A7 - Bracket and Link FEA

Rick Bhattacharya · MEGR 2157
Total time spent: **6 hours**.

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Objective

I used static FEA to test the bracket and link from A5 and A6. I worked through three geometries for each part. The first run showed where the original design needed reinforcement. The second added material around the highly stressed regions. For the third, I used topology results to guide material removal, then ran static studies on the edited CAD.

My goal was to balance strength, stiffness, and mass. I used mesh refinement to assess the results before changing geometry, then compared each revised design with the selected stress, displacement, and safety-factor limits.

Bracket FEA

Preprocessing - material, fixtures, and loading

I kept 6061-T6 aluminum throughout the work. SOLIDWORKS lists a yield strength of 39,885.38 psi. Carrying forward the A5 safety factor of 4 gives an allowable von Mises stress of approximately **9,971 psi**. I also retained the 0.005-in displacement target for this comparison.

Check	Target used
Maximum von Mises stress	At most 9,971 psi
Minimum yield-based safety factor	At least 4
Maximum resultant displacement	At most 0.005 in
Change between successive meshes	At most 7% for stress, displacement, and FOS

For the bracket, I restrained the selected interior mounting faces. This represents a fully fixed interface to the T-beam. It prevents movement on those faces and does not model sliding, clearance, or separation at the actual fit. The original bracket was modeled with separate solid bodies, making the connections between them part of the analysis setup.

I applied a downward force of **600 lbf per selected pin face**. I used Split Line to make the original pin overhangs independently selectable. The setup figures show the selected faces for the force definitions.

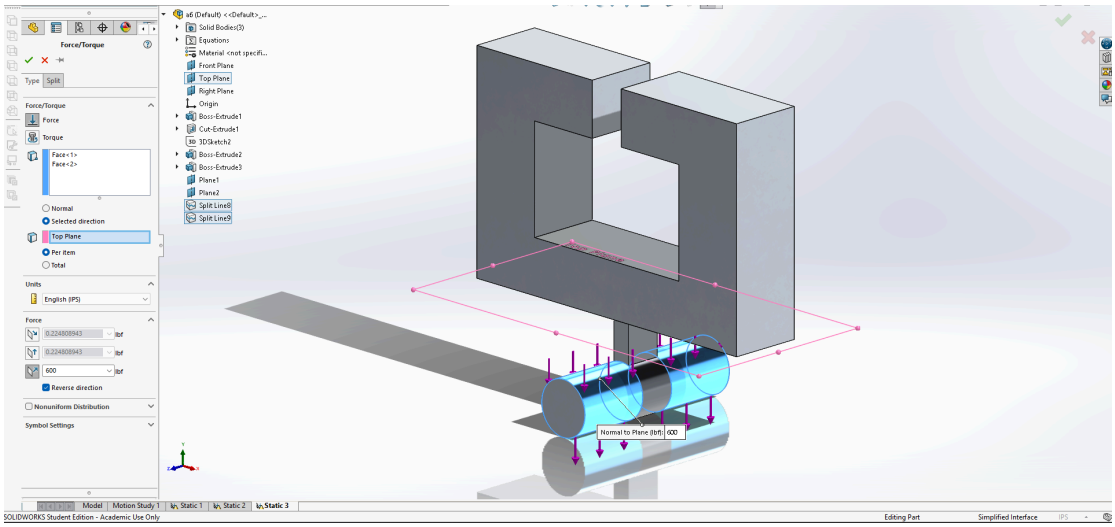


Figure 4. *Baseline force selection: two split pin faces, 600 lbf per item. Two selected faces imply 1,200 lbf total. .*

Mesh selection and validation method

I used high-quality solid meshes with the blended curvature-based mesher shown in the mesh-detail screenshots. Here, "coarse" and "medium" describe relative mesh density. They do not mean that the first run used draft-quality elements.

For each unchanged geometry, I compared a coarse and a refined mesh using:

Change (%) = |medium result - coarse result| / |coarse result| × 100.

The acceptance check uses maximum stress, maximum resultant displacement, and minimum FOS. The assignment allows proceeding with the finest mesh after three attempts if the results still do not converge. I recorded two mesh levels for each design. Mass does not change just because the mesh is refined. A stable mesh result also does not prove that the chosen loading or supports represent the physical part correctly.

Plot deformation is exaggerated for visibility. For example, several Design 2 plots use a scale near 982; the shape on the screen is not the actual deflection magnitude. I used the legend values for the tables.

Design 1 - original connector

The first bracket retained the narrow connector between the body and pin. The medium stress plot reaches 15,580 psi at the connector-to-pin junction. The 0.002004-in displacement is below the comparison limit, but a minimum FOS of 2.561 does not meet the target of 4. That made the lower connector the first redesign target.

Coarse mesh - solve and results

The first recorded mesh has **1,261 elements**. Its maximum stress is 14,830 psi, maximum displacement is 0.00196 in, and minimum FOS is 2.567.

Medium mesh - solve and results

I refined the mesh to **7,173 elements** and solved the same design again. The recorded maximum stress is 15,580 psi, maximum displacement is 0.002004 in, and minimum FOS is 2.561.

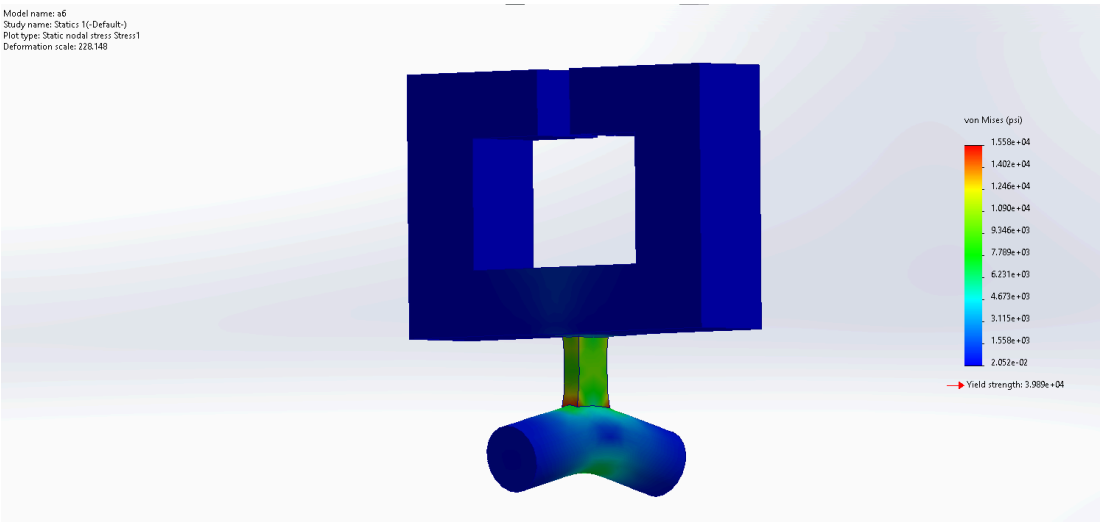


Figure 10. *Bracket Design 1: medium stress. .*

Convergence and interpretation

Result	Coarse	Medium	Change
Max stress (psi)	14,830	15,580	5.06%
Max displacement (in)	0.00196	0.002004	2.24%
Min FOS	2.567	2.561	0.23%
Elements	1,261	7,173	-
Mass (lb, rounded)	1.03	1.03	Unchanged geometry

Locations: stress - connector-to-pin junction; displacement - outer pin ends.

The displayed mesh-to-mesh changes are below 7%. The medium result has a minimum FOS below 4, so I reinforced the lower connector for the next design.

Design 2 - diagonal supports and fillets

I replaced the single narrow connection with two inclined supports and rounded their transitions. The intent was to distribute the load through a wider path rather than leave it concentrated at one neck. I used separate extrusion and fillet operations to develop this shape.

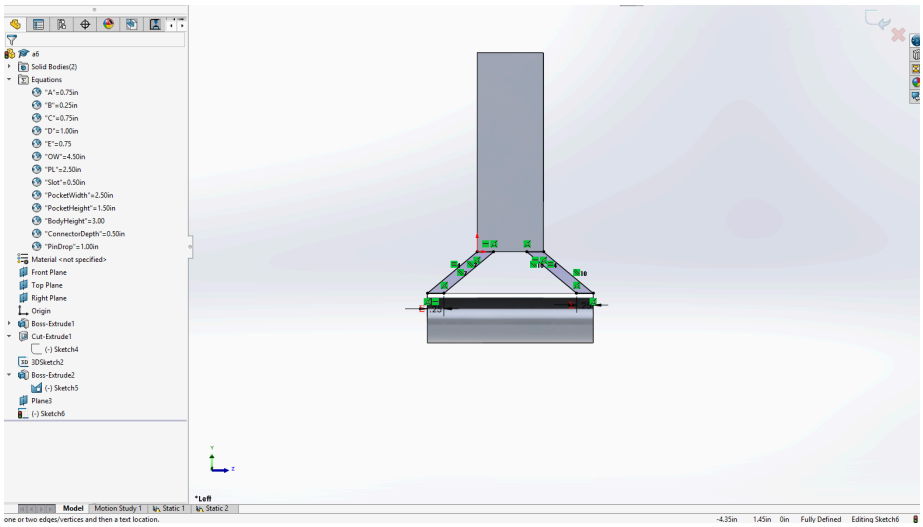


Figure 14. *Diagonal support sketch used to replace the narrow baseline connector. .*

The medium run gives 6,640 psi, 0.0004583 in, and a minimum FOS of 6.007 for this load setup. Its mesh changes are below 7%. The mass is **1.06 lb**, slightly higher than Design 1's 1.03 lb because of the added support material.

Coarse mesh - solve and results

The first recorded mesh has **6,466 elements**. Its maximum stress is 6,299 psi, maximum displacement is 0.0004575 in, and minimum FOS is 6.332.

Medium mesh - solve and results

I refined the mesh to **9,966 elements** and solved the same design again. The recorded maximum stress is 6,640 psi, maximum displacement is 0.0004583 in, and minimum FOS is 6.007.

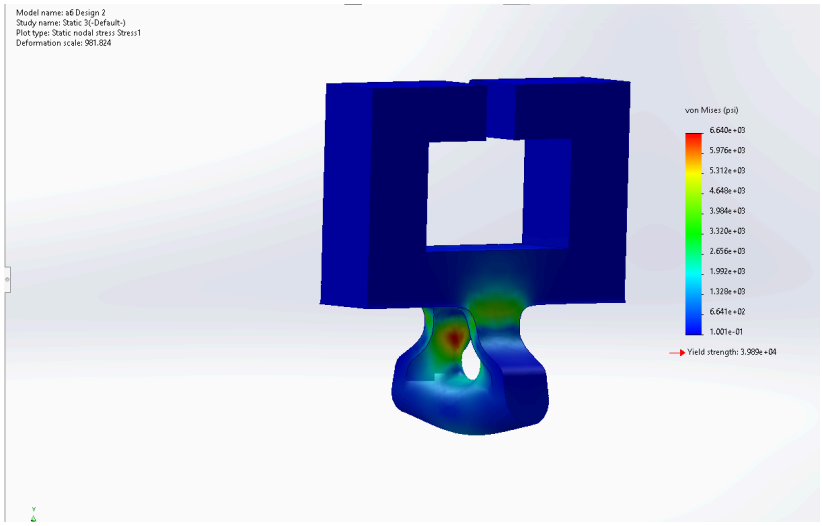


Figure 26. *Bracket Design 2: Design 2 Medium Stress. .*

Convergence and interpretation

Result	Coarse	Medium	Change
Max stress (psi)	6,299	6,640	5.41%
Max displacement (in)	0.0004575	0.0004583	0.17%
Min FOS	6.332	6.007	5.13%
Elements	6,466	9,966	-
Mass (lb, rounded)	1.06	1.06	Unchanged geometry

Locations: stress - inner curved support beside the central opening; displacement - loaded lower pin span.

All three changes are within 7%. The medium results satisfy the strength and displacement targets for this load setup.

Design 3 - topology-guided material removal

I started with the reinforced bracket and used a topology study to look for material that could be removed. My settings were Best Stiffness to Weight Ratio, a 30% mass-reduction target, a displacement limit of 0.005 in, and FOS 4. I added half symmetry, a 0.25-in minimum member-thickness control, and preserved the mounting and pin interfaces.

The first material-mass plot had jagged boundaries. I used the smoothed result as a reference, sketched a cleaner outer profile, and cut material from the upper bracket envelope. The topology target and the later manual CAD mass reduction are separate results; the 30% setting does not mean the final CAD removed exactly 30%.

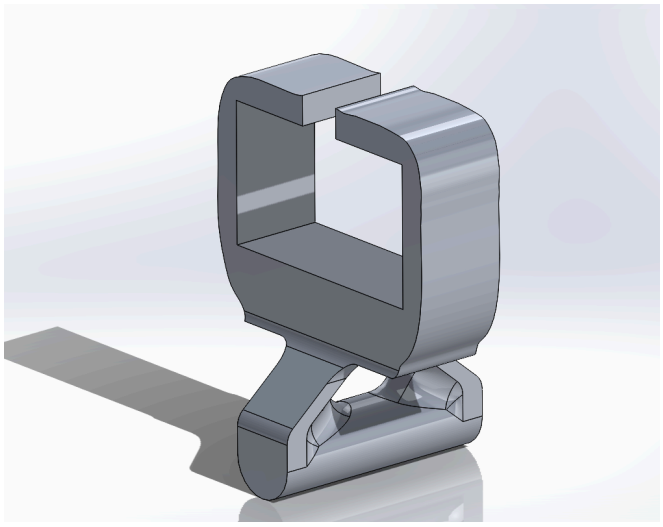


Figure 39. *Remodeled bracket after the outer material-removal cut. .*

The remodeled bracket has a medium static result of 6,613 psi, 0.0004591 in, and FOS 6.032 for its applied loading. Its mass is 0.50 lb, approximately **52.8% below Design 2**, using the rounded mass readings.

Coarse mesh - solve and results

The first recorded mesh has **7,246 elements**. Its maximum stress is 6,377 psi, maximum displacement is 0.0004576 in, and minimum FOS is 6.254.

Medium mesh - solve and results

I refined the mesh to **10,375 elements** and solved the same design again. The recorded maximum stress is 6,613 psi, maximum displacement is 0.0004591 in, and minimum FOS is 6.032.

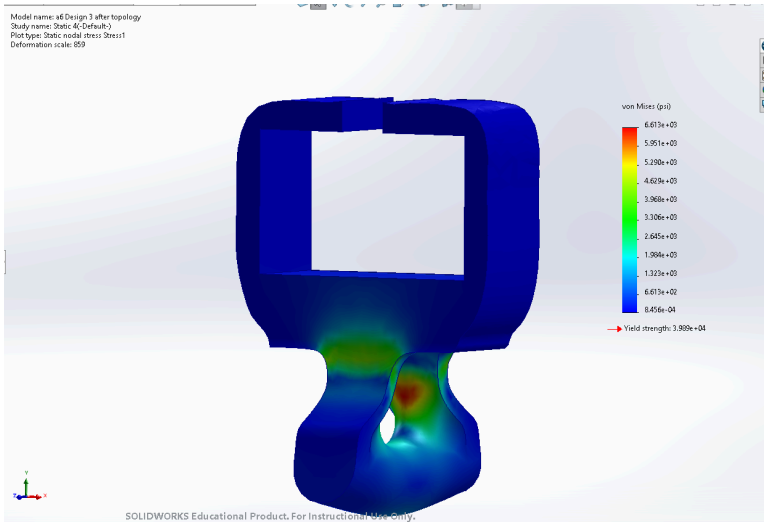


Figure 46. *Bracket Design 3: Design 3 Medium Stress. .*

Convergence and interpretation

Result	Coarse	Medium	Change
Max stress (psi)	6,377	6,613	3.70%
Max displacement (in)	0.0004576	0.0004591	0.33%
Min FOS	6.254	6.032	3.55%
Elements	7,246	10,375	-
Mass (lb, rounded)	0.50	0.50	Unchanged geometry

Locations: stress - inner support transition beside the central opening; displacement - loaded lower pin span.

All three changes are within 7%. The medium stress, displacement, and FOS meet the selected limits for this load setup. Removing material from the upper envelope reduced the mass to 0.50 lb.

Finalized bracket results

Refined result	Design 1	Design 2	Design 3
Maximum stress and location	15,580 psi Connector-to-pin junction	6,640 psi Inner curved support beside the central opening	6,613 psi Inner support transition beside the central opening
Maximum deflection and location	0.002004 in Outer pin ends	0.0004583 in Loaded lower pin span	0.0004591 in Loaded lower pin span
Mass (lb, rounded)	1.03	1.06	0.50
Minimum safety factor	2.561	6.007	6.032
Number of elements	7,173	9,966	10,375
Target comparison	FOS below 4	Within selected limits	Within selected limits

Removing material from the upper envelope reduced mass while retaining the reinforced lower supports. I kept those supports because the stress plots placed the largest stresses near the inner curved transitions beside the central opening.

Link FEA - MEGR 2157

Preprocessing - material, fixtures, and loading

I kept 6061-T6 aluminum for the link. I fixed the inner cylindrical surface of the 0.750-in hole and applied **600 lbf** to the 1.000-in hole, along the line between the hole centers. The fixed-hole model approximates the attachment; I did not model a separate shaft or press-fit contact.

I used the same targets as the bracket: maximum von Mises stress of approximately 9,971 psi, a minimum FOS of 4, and maximum resultant displacement of 0.005 in. Each link geometry was solved with a coarse and a medium high-quality solid mesh. I compared stress, displacement, and FOS with the same 7% change criterion before moving to the next design.

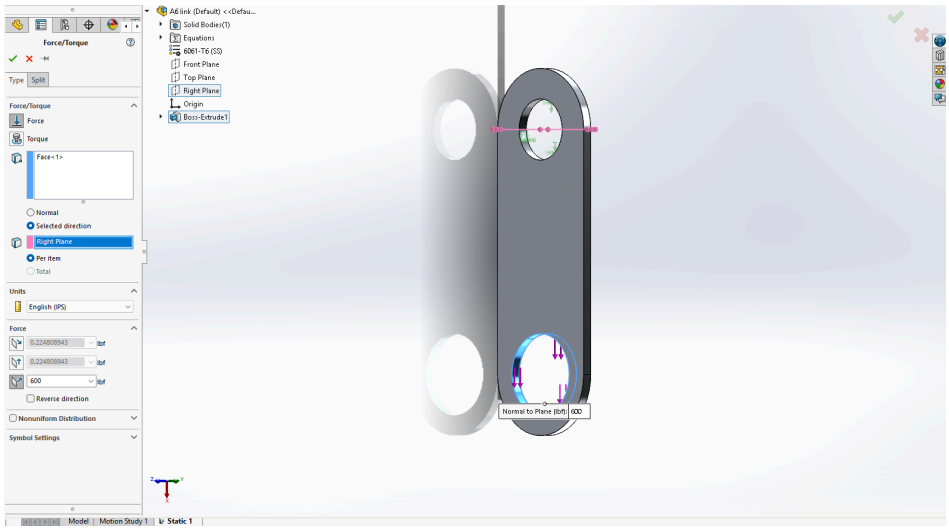


Figure 51. *Link force definition: 600 lbf applied to the larger-hole face along the link centerline. .*

Design 1 - original plate

The original link was 1.50 in wide and 0.125 in thick, with 3.00 in between hole centers. The medium run reaches 14,870 psi around the larger loaded hole. Displacement is below 0.005 in, but FOS 2.681 falls short of 4. The local stress at the hole was more demanding than the nominal net-section calculation alone suggested.

Coarse mesh - solve and results

The first recorded mesh has **1,305 elements**. Its maximum stress is 14,710 psi, maximum displacement is 0.002331 in, and minimum FOS is 2.711.

Medium mesh - solve and results

I refined the mesh to **7,886 elements** and solved the same design again. The recorded maximum stress is 14,870 psi, maximum displacement is 0.002339 in, and minimum FOS is 2.681.

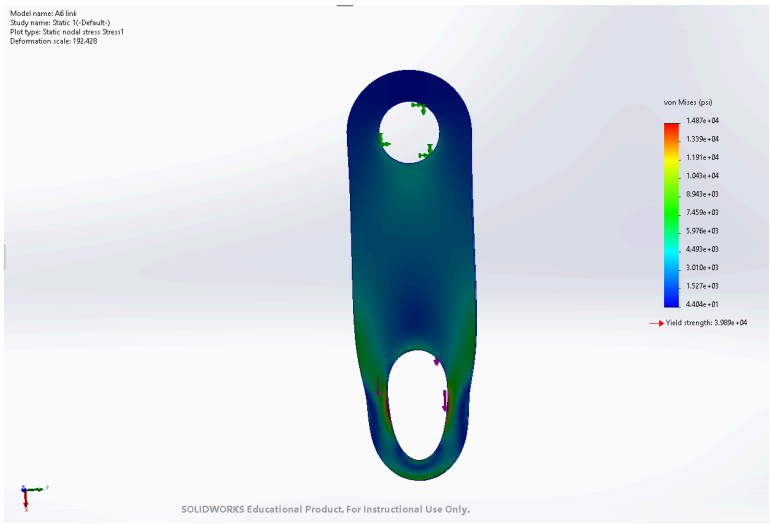


Figure 57. *Link Design 1: Design 1 Medium Stress. .*

Convergence and interpretation

Result	Coarse	Medium	Change
Max stress (psi)	14,710	14,870	1.09%
Max displacement (in)	0.002331	0.002339	0.34%
Min FOS	2.711	2.681	1.11%
Elements	1,305	7,886	-
Mass (lb, rounded)	0.06	0.06	Unchanged geometry

Locations: stress - side ligaments around the loaded 1-in hole; displacement - far end of the loaded large-hole region.

All three changes are below 7%. Displacement meets the target, but the minimum FOS of 2.681 is below 4. The baseline therefore needed a strength change.

Design 2 - wider plate

I increased the link width to **2.00 in**, keeping the 0.125-in thickness and 3.00-in hole-center spacing. This added material around the holes and increased the minimum ligament area.

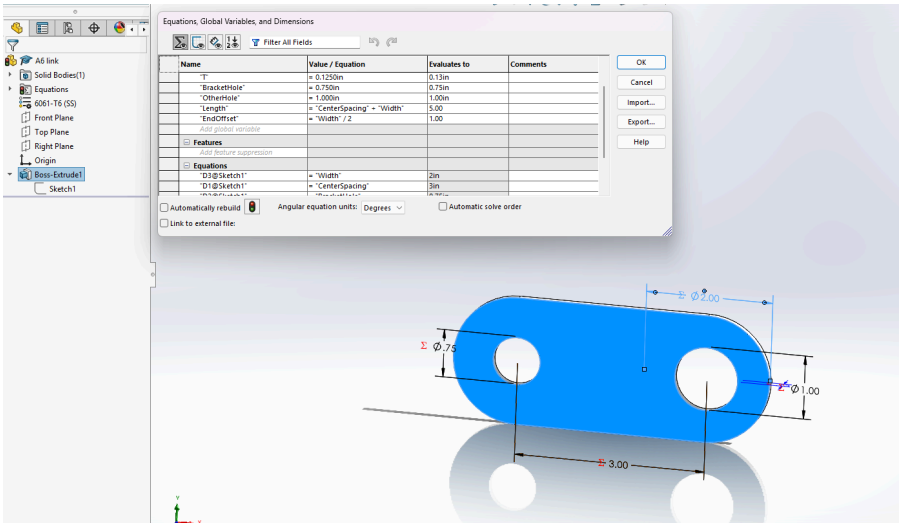


Figure 61. *Link width increased from 1.50 to 2.00 in; thickness remains 0.125 in and hole-center spacing remains 3.00 in. .*

The medium results are 7,403 psi, 0.001278 in, and FOS 5.388. All three mesh changes are below 7%. The mass rose from 0.06 to 0.10 lb, so the wider plate solved the recorded strength issue at a weight cost.

Coarse mesh - solve and results

The first recorded mesh has **1,470 elements**. Its maximum stress is 7,489 psi, maximum displacement is 0.001277 in, and minimum FOS is 5.326.

Medium mesh - solve and results

I refined the mesh to **10,097 elements** and solved the same design again. The recorded maximum stress is 7,403 psi, maximum displacement is 0.001278 in, and minimum FOS is 5.388.

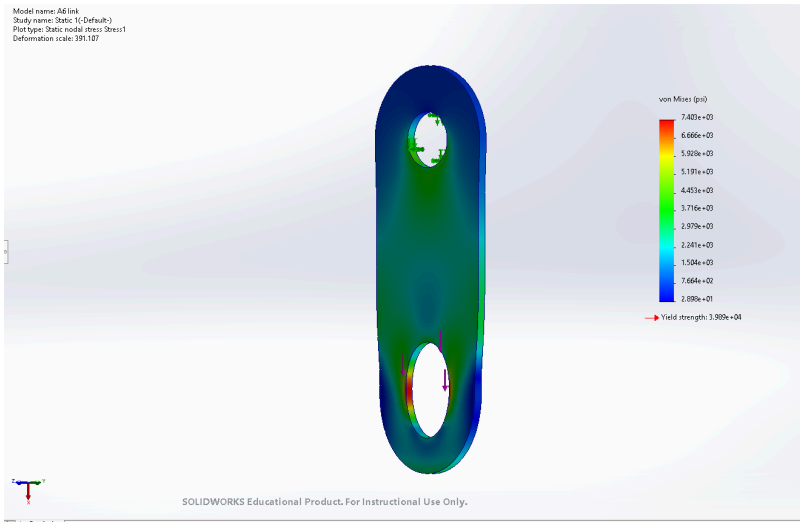


Figure 67. *Link Design 2: Design 2 Medium Stress. .*

Convergence and interpretation

Result	Coarse	Medium	Change
Max stress (psi)	7,489	7,403	1.15%
Max displacement (in)	0.001277	0.001278	0.08%
Min FOS	5.326	5.388	1.16%
Elements	1,470	10,097	-
Mass (lb, rounded)	0.10	0.10	Unchanged geometry

Locations: stress - ligaments around the loaded large hole; displacement - far end of the loaded large-hole region.

All three changes are below 7%. The wider link satisfies both strength and displacement limits, with a mass increase over Design 1.

Design 3 - topology-guided cut

I used another topology study to guide trimming the wider link. The setup retained the two hole interfaces and used a 30% mass-reduction target, 0.005-in displacement constraint, FOS 4, and a 0.125-in minimum member-thickness control. I smoothed the topology display before drawing the manual cut.

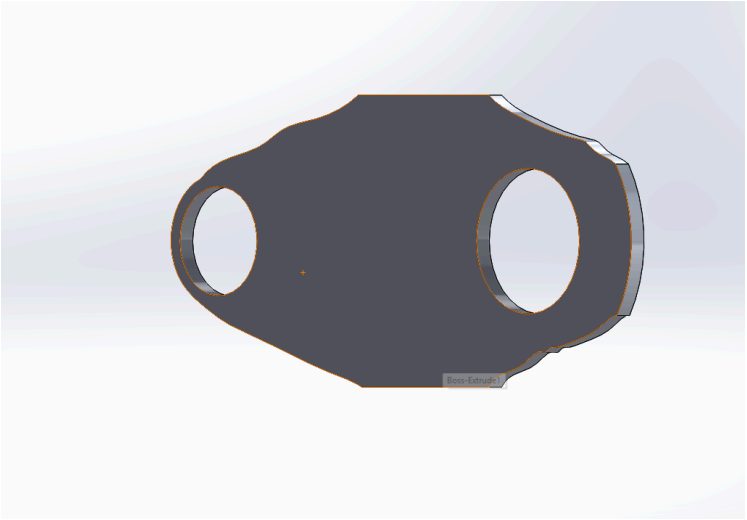


Figure 77. *Link after the manual material-removal operation. .*

The resulting link has a recorded mass of 0.07 lb. Its medium static result is 8,093 psi, 0.001355 in, and FOS 4.928. The coarse-to-medium changes are below 7%. It is approximately **30% lighter than Design 2**, using the rounded readings, but remains heavier than the original 0.06-lb link. That is a useful tradeoff: the original link was lighter but did not meet the FOS target.

I used a **0.10-in blind cut** for the manual material-removal operation. I then meshed and solved the edited part rather than using the topology result alone as the final strength assessment.

Coarse mesh - solve and results

The first recorded mesh has **609 elements**. Its maximum stress is 7,751 psi, maximum displacement is 0.00135 in, and minimum FOS is 5.146.

Medium mesh - solve and results

I refined the mesh to **1,338 elements** and solved the same design again. The recorded maximum stress is 8,093 psi, maximum displacement is 0.001355 in, and minimum FOS is 4.928.

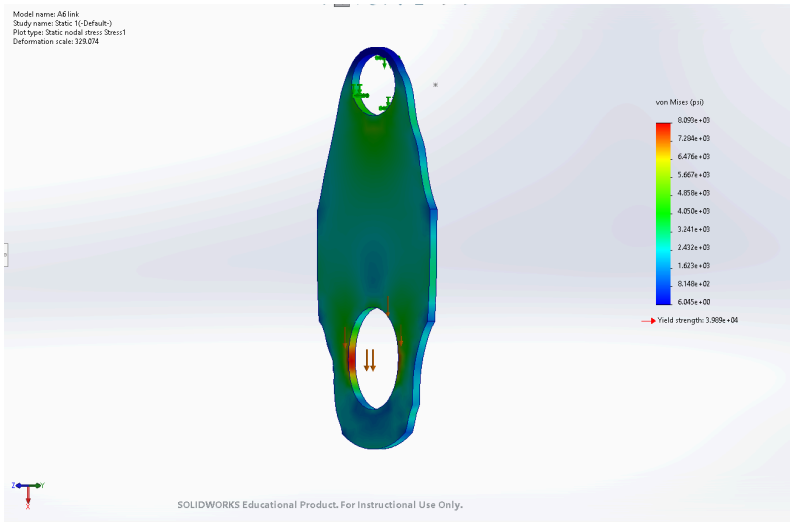


Figure 83. *Link Design 3: Medium Stress. .*

Convergence and interpretation

Result	Coarse	Medium	Change
Max stress (psi)	7,751	8,093	4.41%
Max displacement (in)	0.00135	0.001355	0.37%
Min FOS	5.146	4.928	4.24%
Elements	609	1,338	-
Mass (lb, rounded)	0.07	0.07	Unchanged geometry

Locations: stress - trimmed ligament beside the loaded large hole; displacement - far end of the loaded large-hole region.

All three changes are below 7%. The medium FOS is 4.928 and the displacement is 0.001355 in. Mass is approximately 30% below Design 2, using the rounded mass readings.

Finalized link results

Refined result	Design 1	Design 2	Design 3
Maximum stress and location	14,870 psi Side ligaments around the loaded 1-in hole	7,403 psi Ligaments around the loaded large hole	8,093 psi Trimmed ligament beside the loaded large hole
Maximum deflection and location	0.002339 in Far end of the loaded large-hole region	0.001278 in Far end of the loaded large-hole region	0.001355 in Far end of the loaded large-hole region
Mass (lb, rounded)	0.06	0.10	0.07
Minimum safety factor	2.681	5.388	4.928
Number of elements	7,886	10,097	1,338
Target comparison	FOS below 4	Within selected limits	Within selected limits

The final recorded link study meets the selected stress, displacement, and FOS targets. The stress maximum remains near the larger loaded hole, so that interface needs enough material even when the outer profile is trimmed.

Lessons learned and mistakes

Convergence and strength answer different questions. My original bracket and link could give relatively stable mesh results while still missing the required FOS. Refining the mesh improves confidence in a numerical result; it does not reinforce the part. I used the stress location to decide where to change geometry after the mesh comparison.

A color plot needs its legend. The FOS plots were largely red even when the minimum was above 4 because the automatically scaled maximum was very large. I read the minimum number rather than using red as a pass/fail signal. I also compared stress in psi consistently after initially viewing some results in N/m².

Load definitions depend on face selection. Split Line let me select the original pin overhangs independently. Rebuilding the connection changed the available faces. I learned that a force entered "per item" applies to each selected face, so both the magnitude and the face count matter when setting up a study.

Topology output needed interpretation. I went beyond the required geometry iterations by trying topology optimization on both parts. The raw result looked rough because it followed the analysis mesh. Smoothing made the suggested boundary easier to follow, but it did not by itself produce a finished parametric model. I used sketches, cuts, and fillets to create the revised shapes, then ran static studies on the edited CAD. This took more work than simply increasing the fillets, but it helped me decide where to remove material.

Lower mass involved a tradeoff. The wider link increased FOS but also added mass. Trimming it brought the mass back down while reducing the strength margin. I compared the final FOS with 4 rather than trying to maximize FOS without considering weight.

The support assumption matters. Fixing the interior mounting faces prevents sliding and separation. The low stress in much of that restrained region does not prove that it would carry little stress with a more realistic contact model. I kept this simplification explicit when interpreting the topology result.

File history matters too. Saving later work into the same part file made the design history harder to organize. Separate saves or named configurations would make it easier to return to each geometry and compare its studies. The screenshots helped me keep track of the sequence.

Actual time

Total time spent: 6 hours. This included the bracket and link studies, mesh comparisons, geometry changes, topology setup, screenshots, and corrections.

CAD and source downloads

Download	Contents
Complete report PDF	Written results, convergence tables, and all 86 screenshots
CAD models ZIP	Native parts, baseline folder, and file manifest
Saved static results	Saved static-study CWR files
Saved topology results	Available topology-study CWR files
Original screenshot ZIP	Original names and folders, unchanged
Recorded results CSV	Both mesh results for all six designs
a6 Design 2.SLDPRT	Bracket Design 2 - saved model
a6 Design 3 after topology.SLDPRT	Bracket Design 3 - final named model
a6 Design 3.SLDPRT	Bracket Design 3 - earlier working file
A6 link.SLDPRT	Link - final saved model
a6.SLDPRT	Bracket - working model

The model ZIP contains the native SOLIDWORKS parts. The static and topology result ZIPs contain the saved CWR files; extract them beside the matching parts when opening the studies in SOLIDWORKS. The original A6 models are in the baseline folder.

References

- Course assignment: *Bracket FEA*, including the additional MEGR 2157 link process, the three design iterations, and the 7% mesh-change criterion.
- My A5 calculations and A6 CAD work.
- SOLIDWORKS Simulation results, material properties, and topology settings recorded in my screenshots.

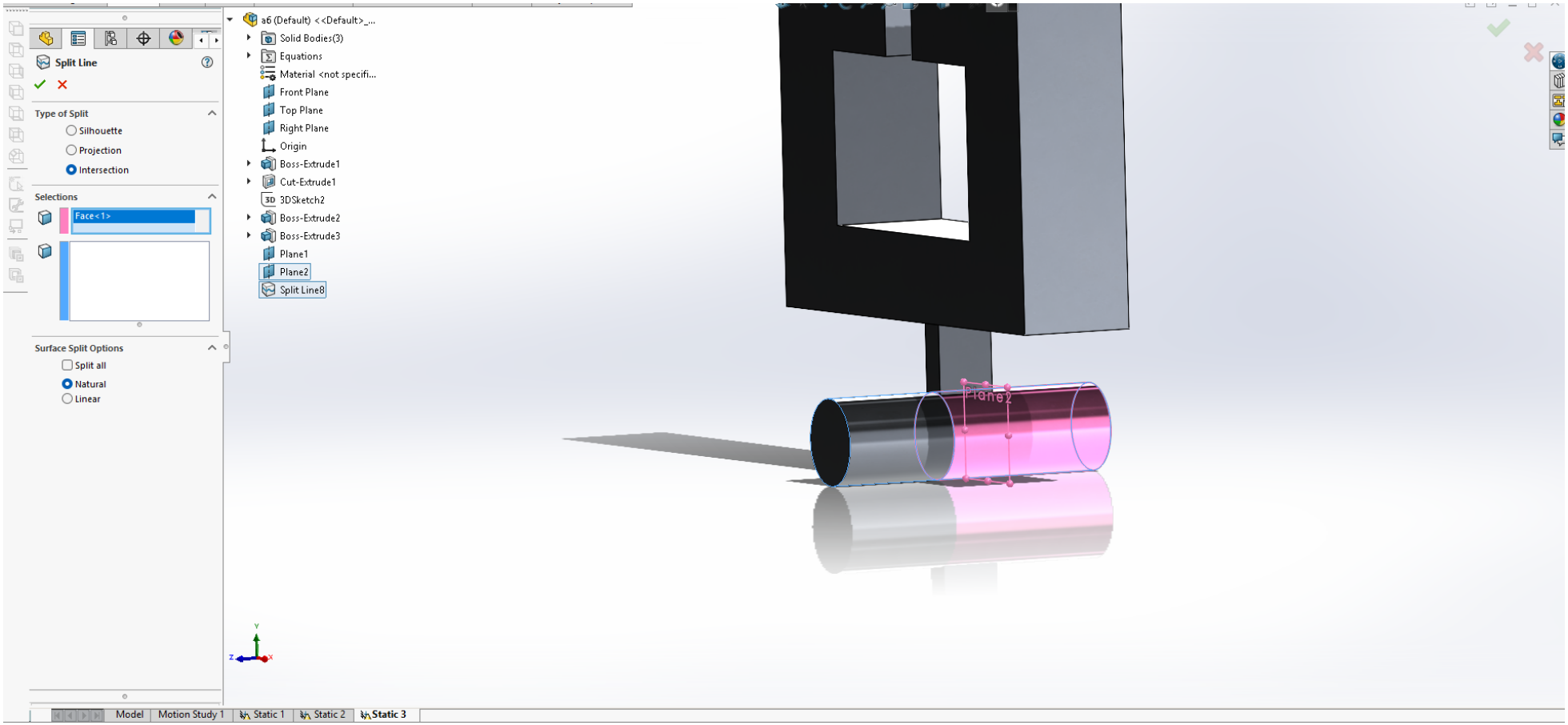
A7 - Figure 1

6061-T6 material properties. The library yield strength is 39,885.38 psi.

Property	Value	Units
Elastic Modulus	10007604	psi
Poisson's Ratio	0.33	N/A
Shear Modulus	3770981.198	psi
Mass Density	0.0975436609	lb/in^3
Tensile Strength	44961.69898	psi
Compressive Strength		psi
Yield Strength	39885.37799	psi
Thermal Expansion Coefficient	1.333333333e-05	/°F
Thermal Conductivity	0.00223225	Btu/(in·sec·°F)
Specific Heat	0.214006	Btu/(lb·°F)
Material Damping Ratio		N/A

A7 - Figure 2

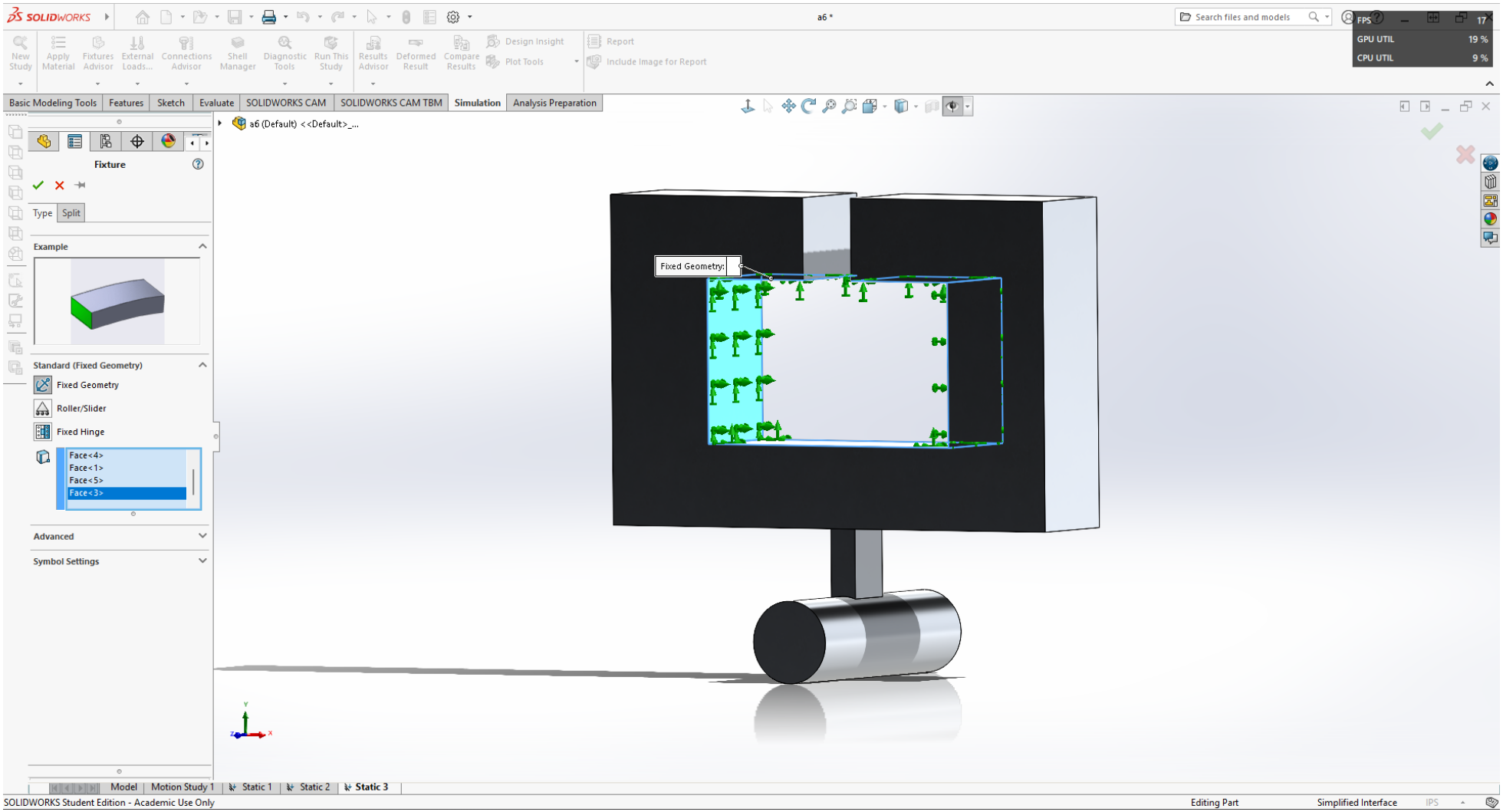
Split Line face division used to select the two original pin overhangs independently.



Source: split line.png

A7 - Figure 3

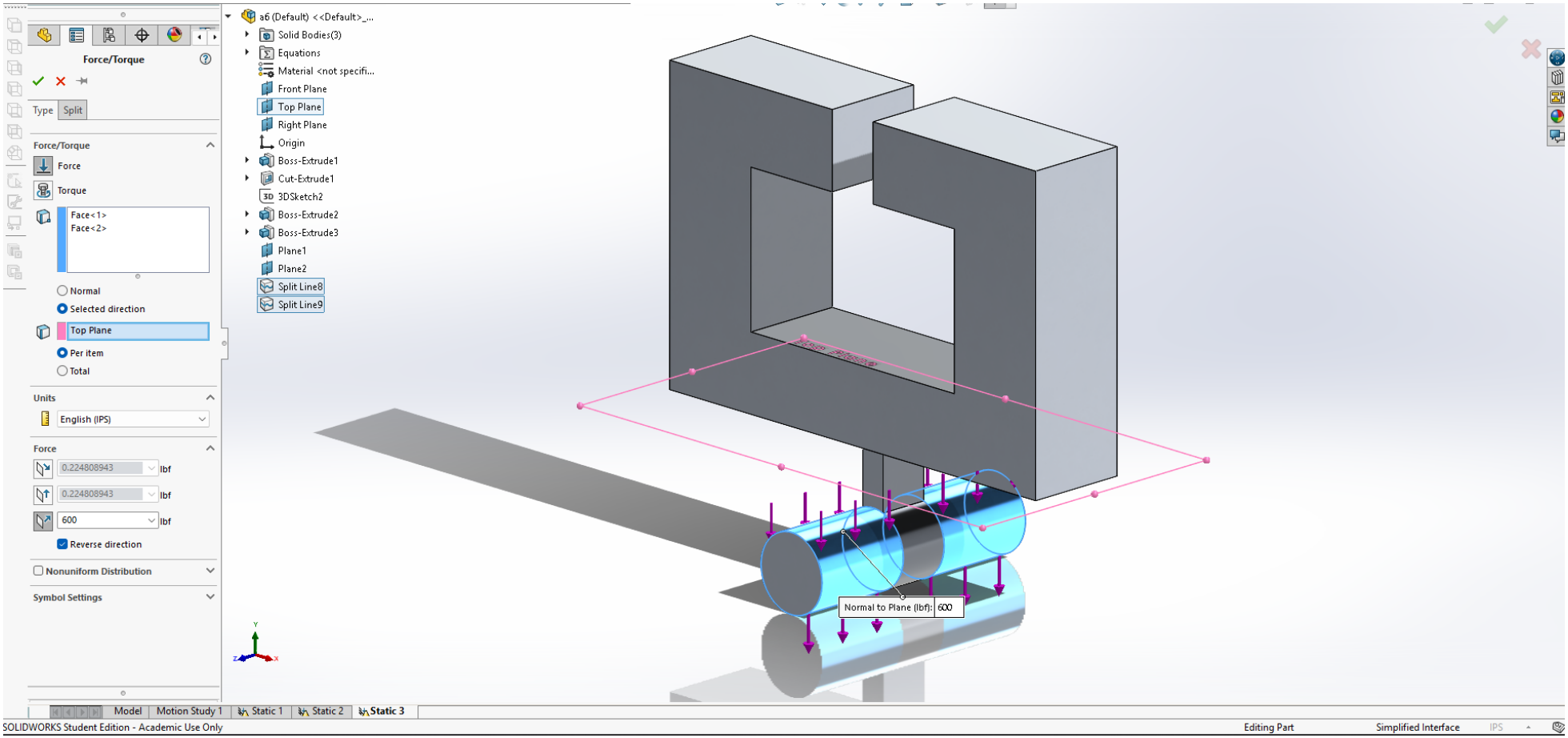
Baseline fixed geometry on the interior mounting surfaces. This approximates a fully restrained T-beam interface.



Source: fixed geometry.png

A7 - Figure 4

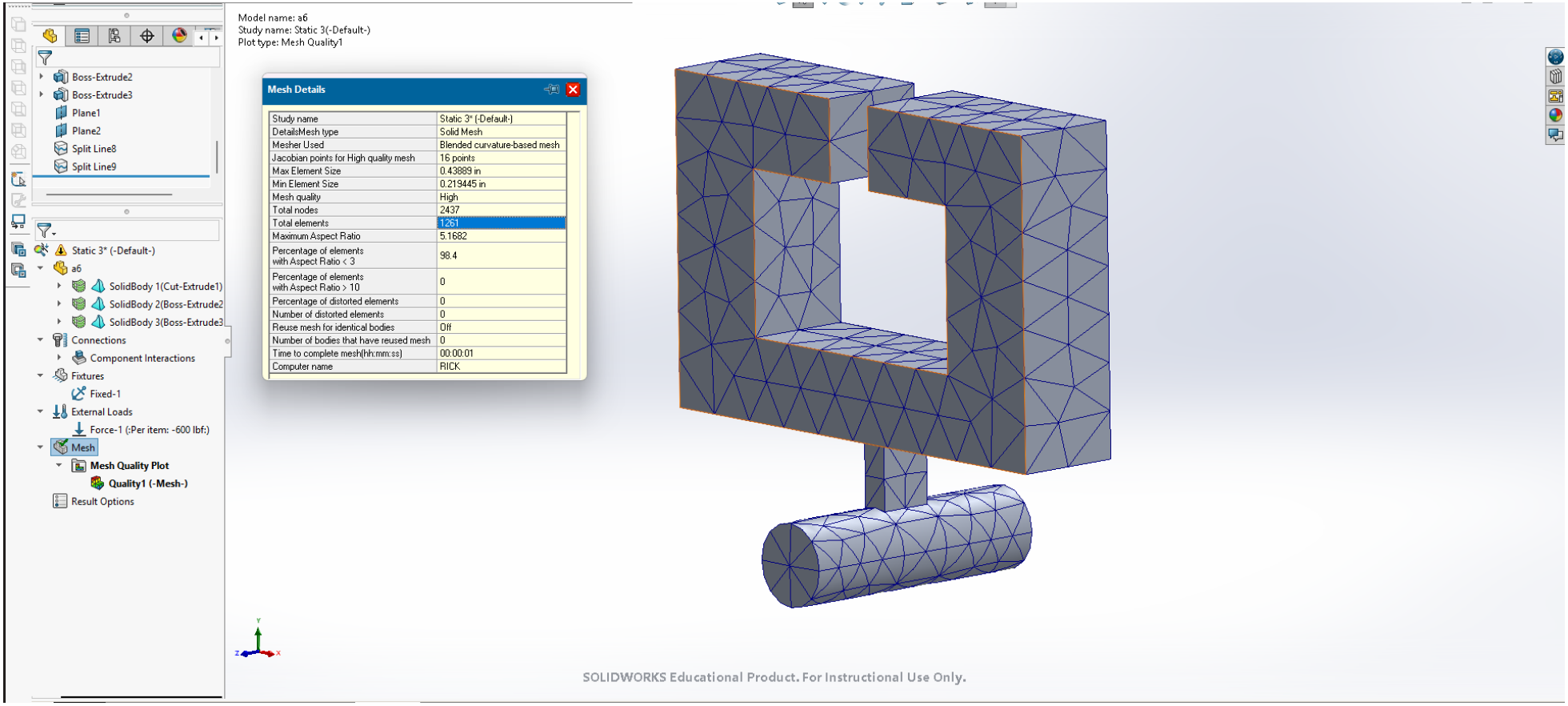
Baseline force selection: two split pin faces, 600 lbf per item. Two selected faces imply 1,200 lbf total.



Source: Force selection.png

A7 - Figure 5

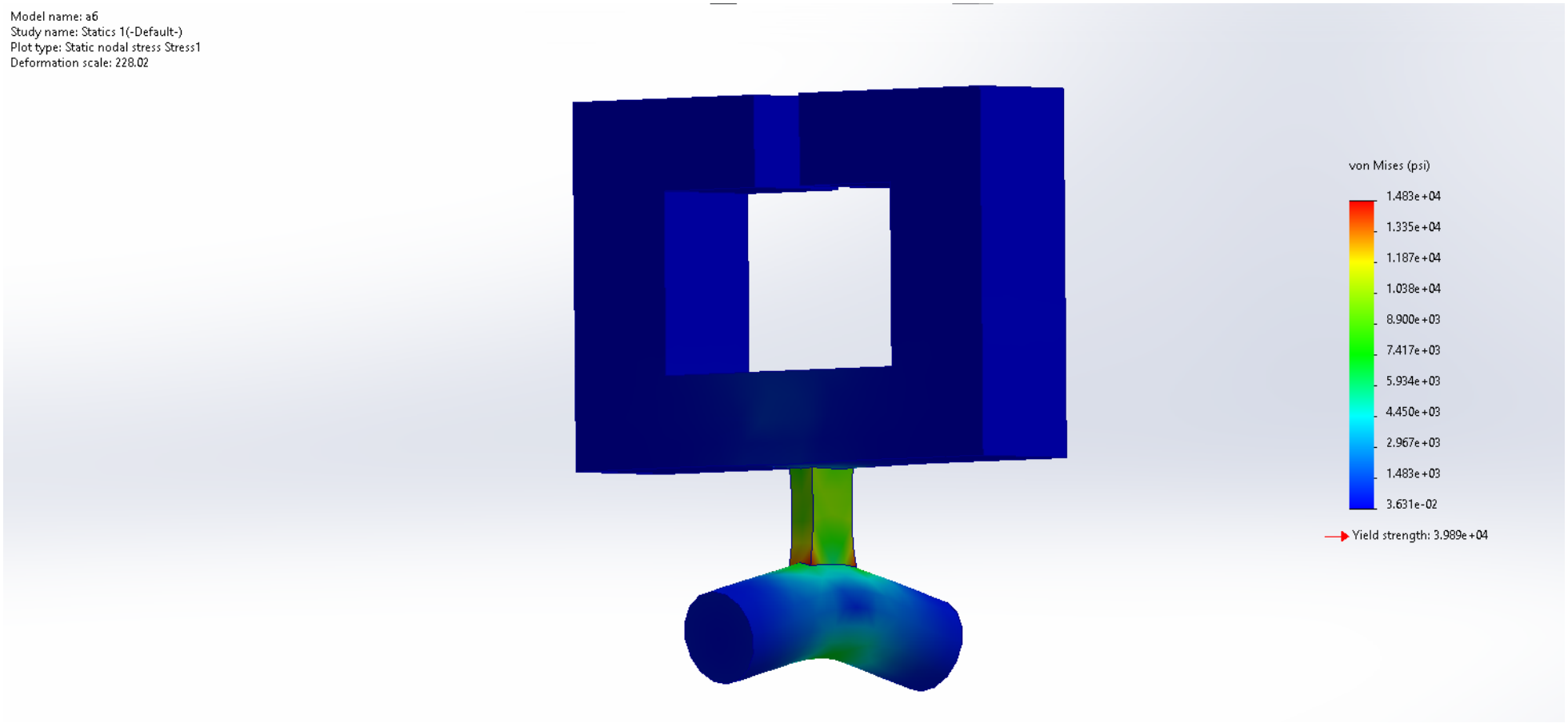
Bracket Design 1: Coarse mesh image.



Source: Coarse mesh image.png

A7 - Figure 6

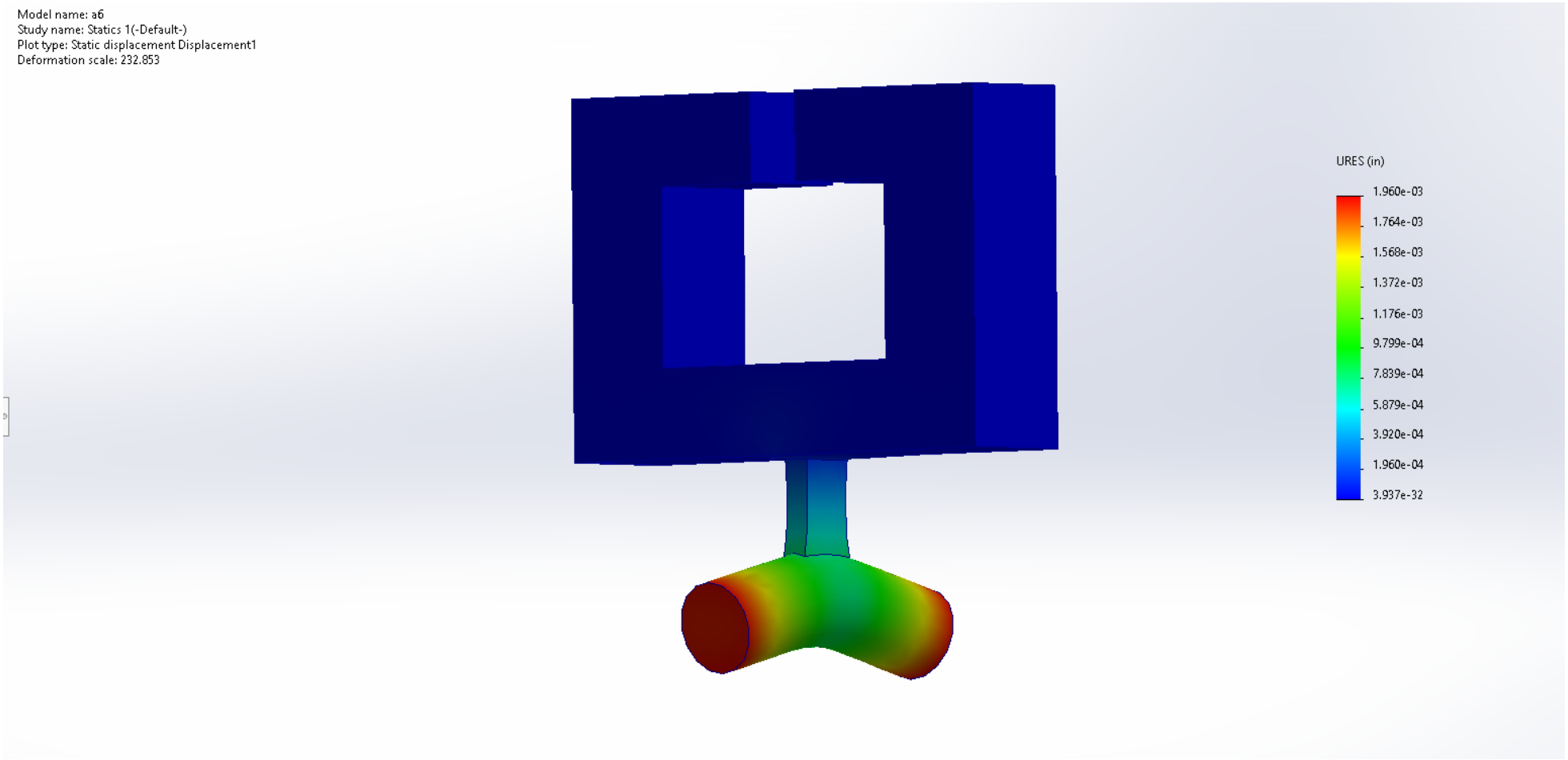
Bracket Design 1: Coarse stress.



Source: Coarse stress.png

A7 - Figure 7

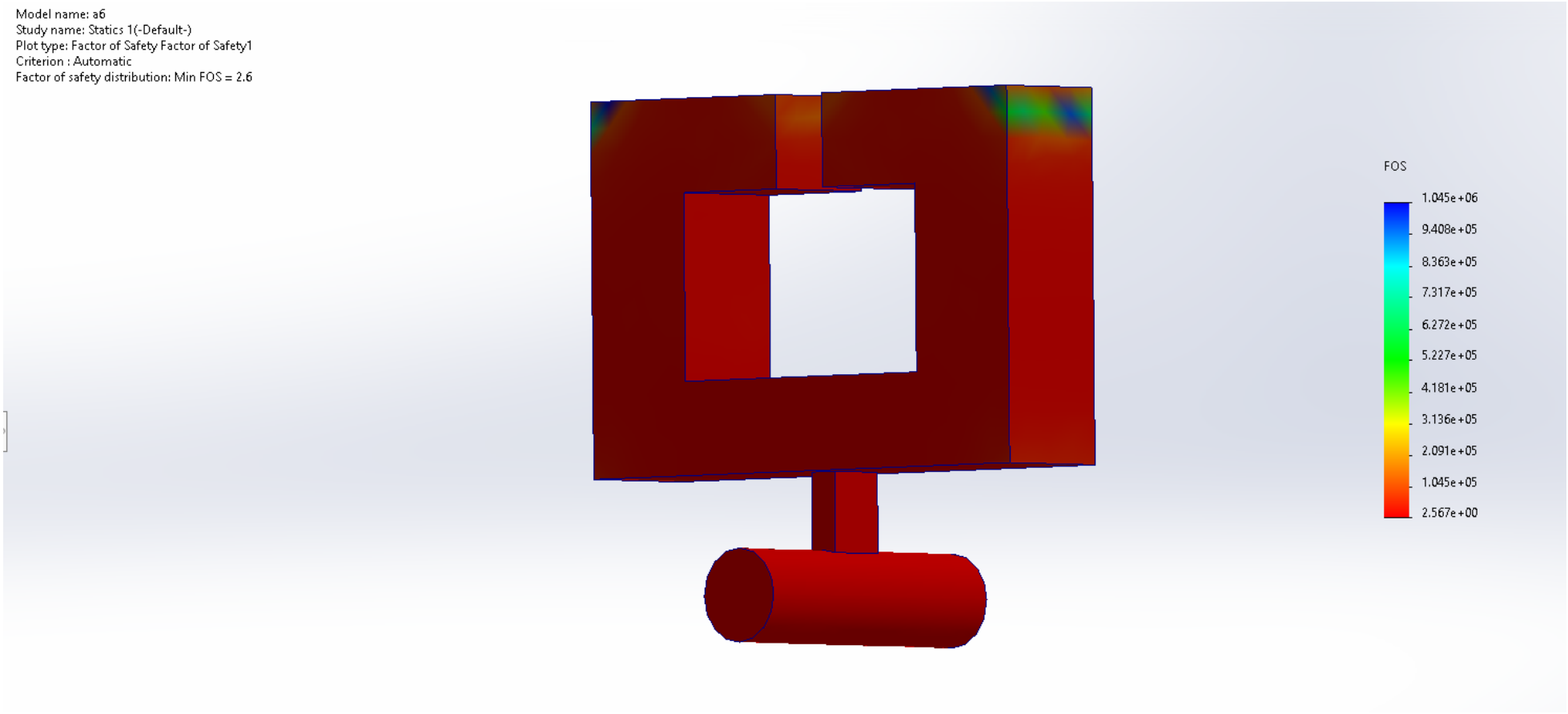
Bracket Design 1: Coarse Displacement.



Source: Coarse Displacement.png

A7 - Figure 8

Bracket Design 1: Coarse FOS.



Source: Coarse FOS.png

A7 - Figure 9

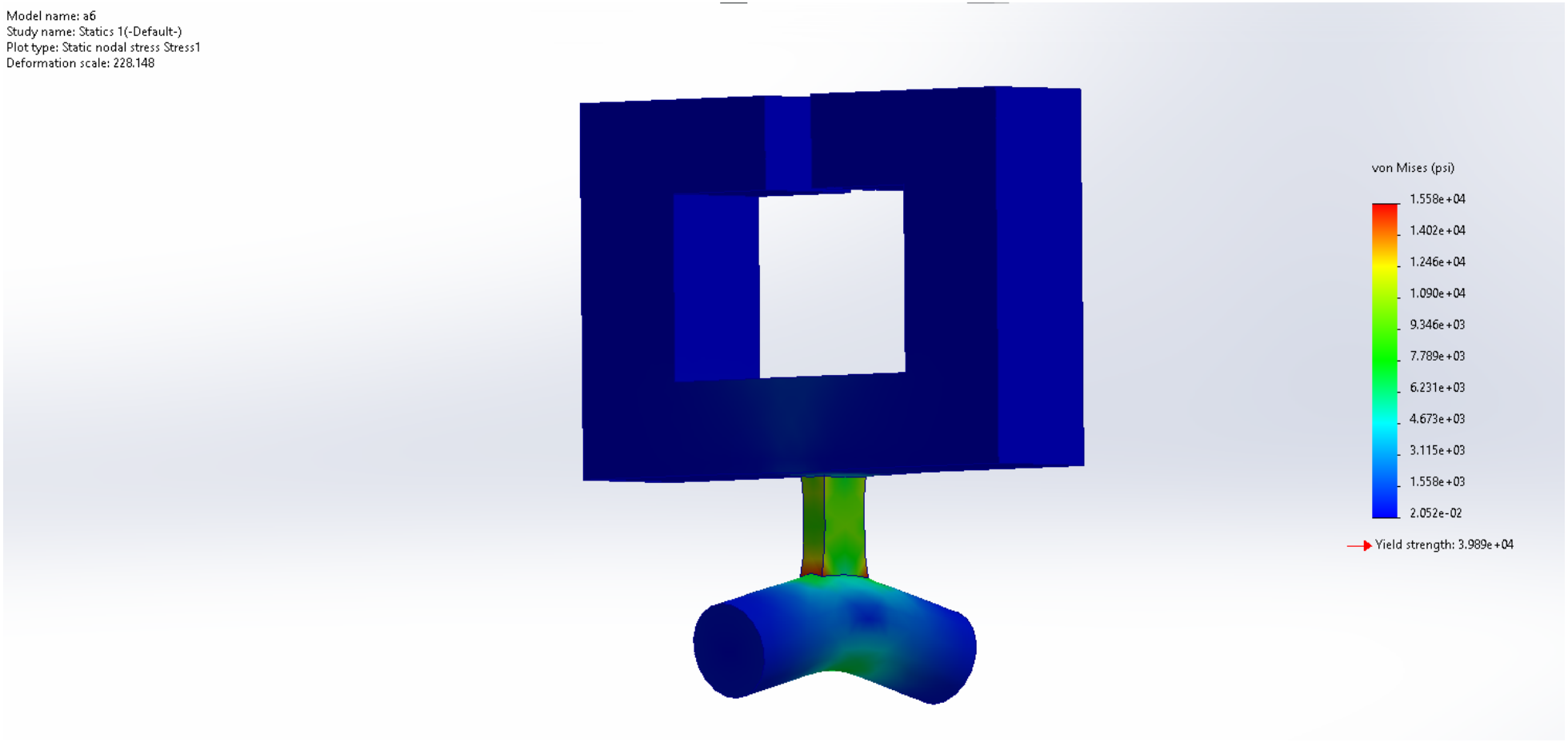
Bracket Design 1: medium mesh elements.

Mesh Details

Study name	Statics 1* (-Default-)
DetailsMesh type	Solid Mesh
Mesher Used	Blended curvature-based mesh
Jacobian points for High quality mesh	16 points
Max Element Size	0.235903 in
Min Element Size	0.235903 in
Mesh quality	High
Total nodes	11743
Total elements	7173
Maximum Aspect Ratio	3.6978
Percentage of elements with Aspect Ratio < 3	99.8
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Number of distorted elements	0
Reuse mesh for identical bodies	Off
Number of bodies that have reused mesh	0
Time to complete mesh(hh:mm:ss)	00:00:01
Computer name	RICK

A7 - Figure 10

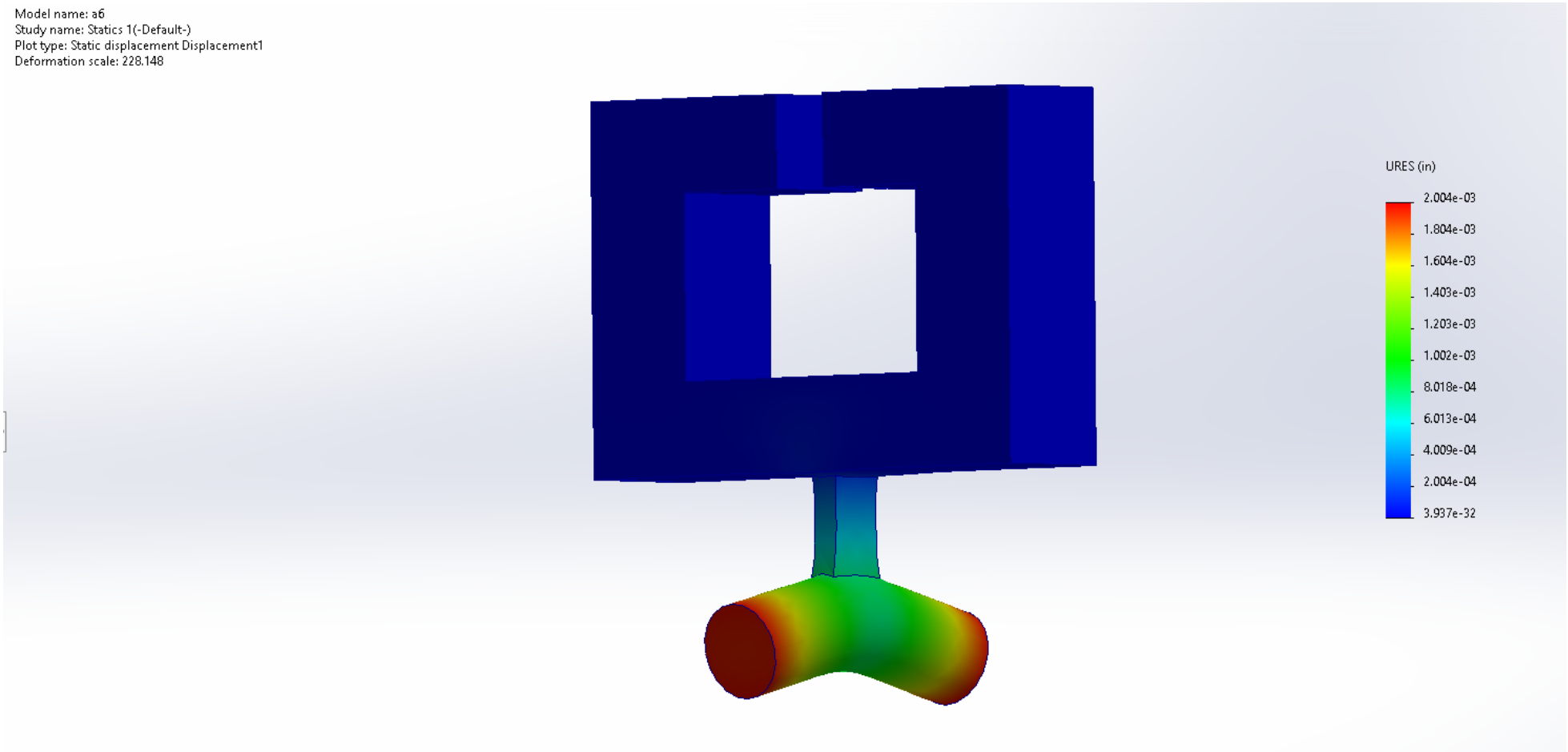
Bracket Design 1: medium stress.



Source: medium stress.png

A7 - Figure 11

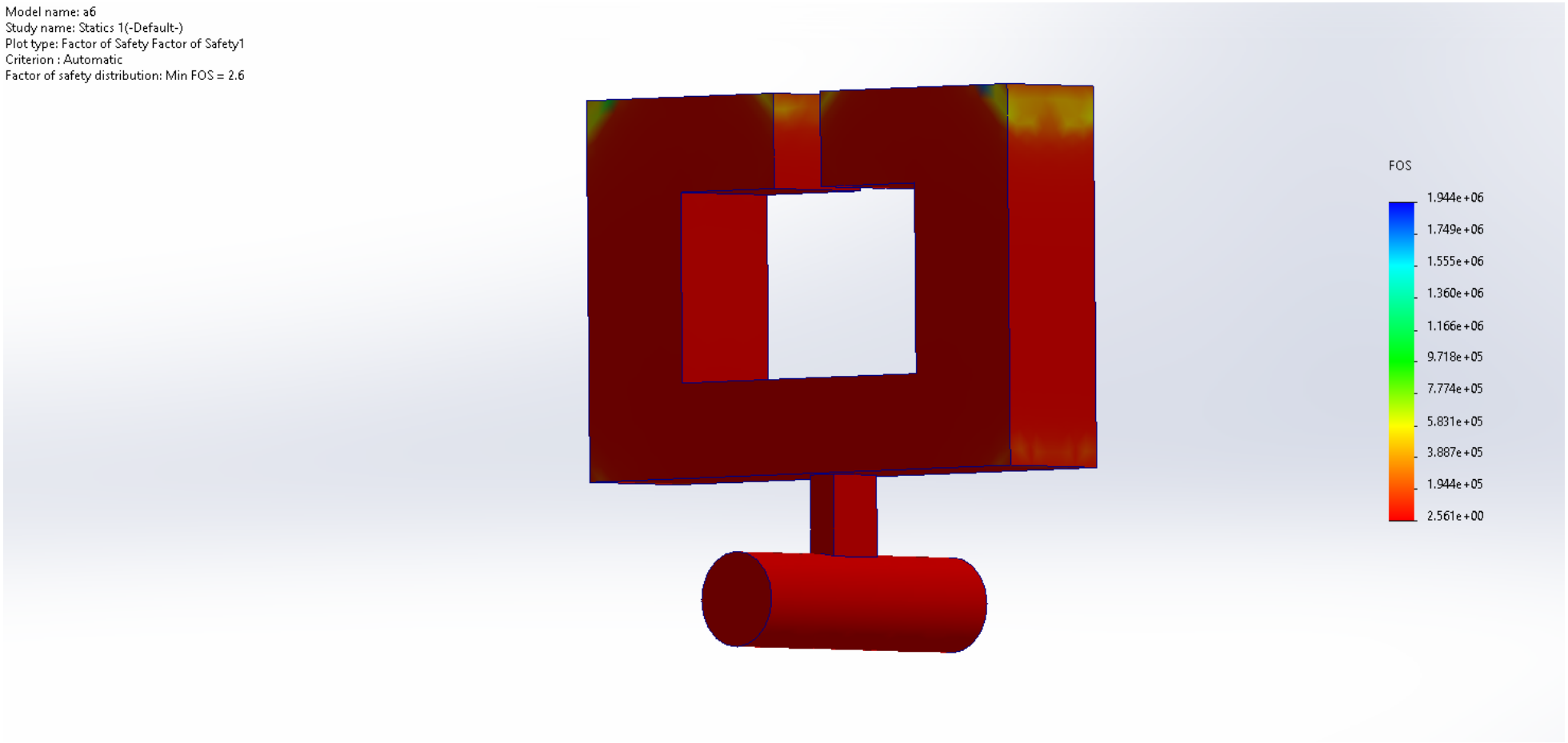
Bracket Design 1: medium displacement.



Source: medium displacement.png

A7 - Figure 12

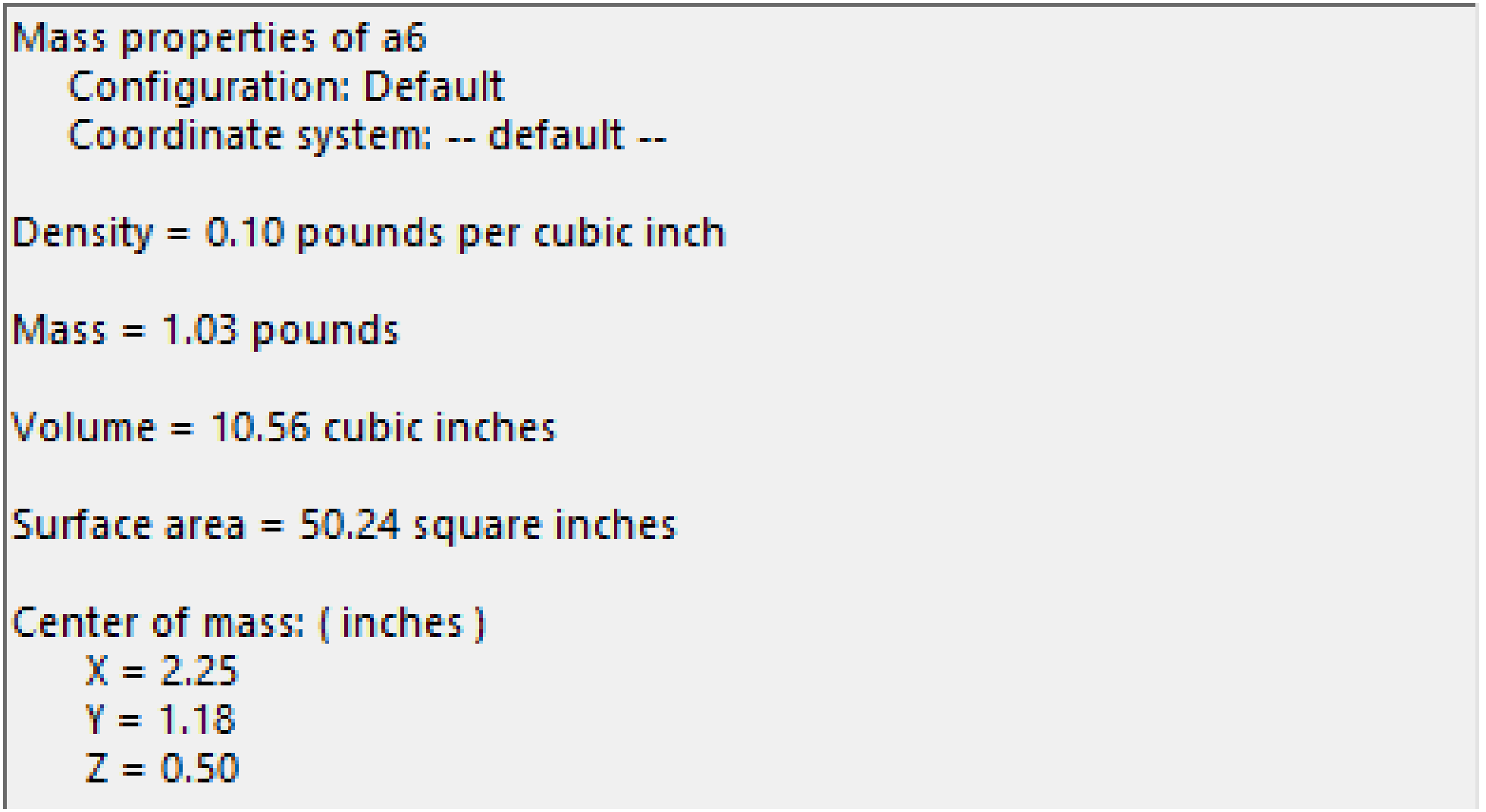
Bracket Design 1: medium FOS.



Source: medium FOS.png

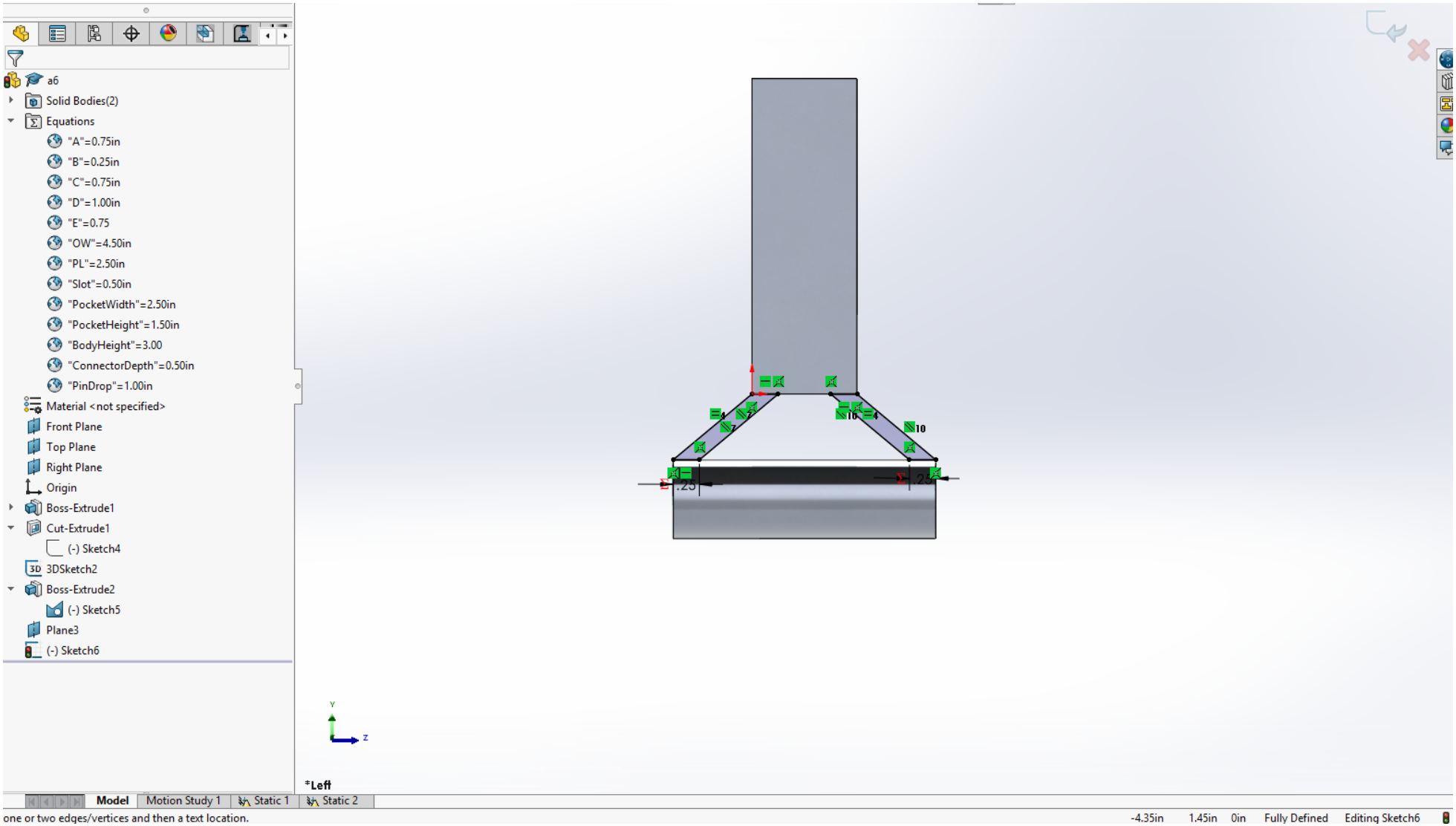
A7 - Figure 13

Bracket Design 1: mass.



A7 - Figure 14

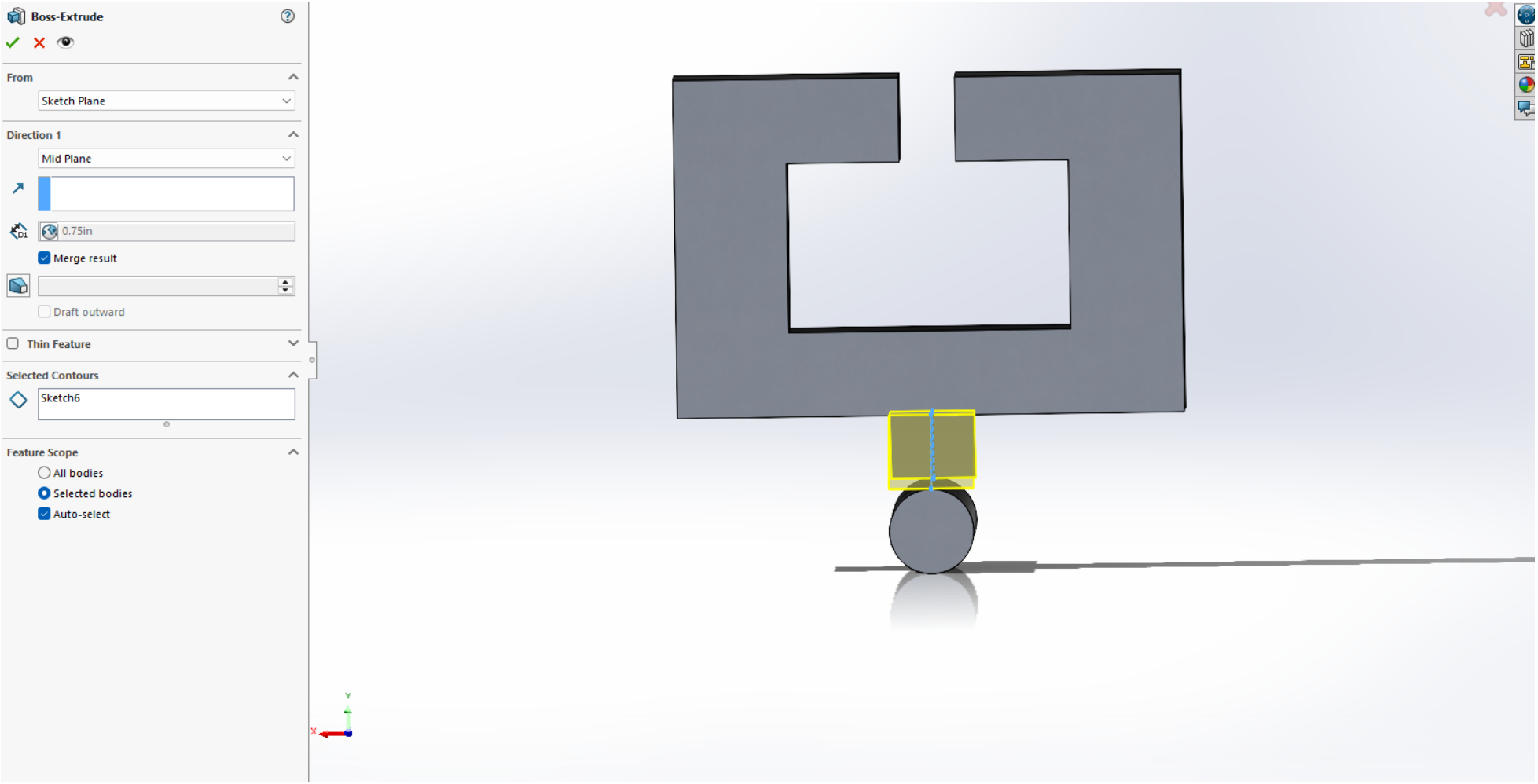
Diagonal support sketch used to replace the narrow baseline connector.



Source: Geometry Change.png

A7 - Figure 15

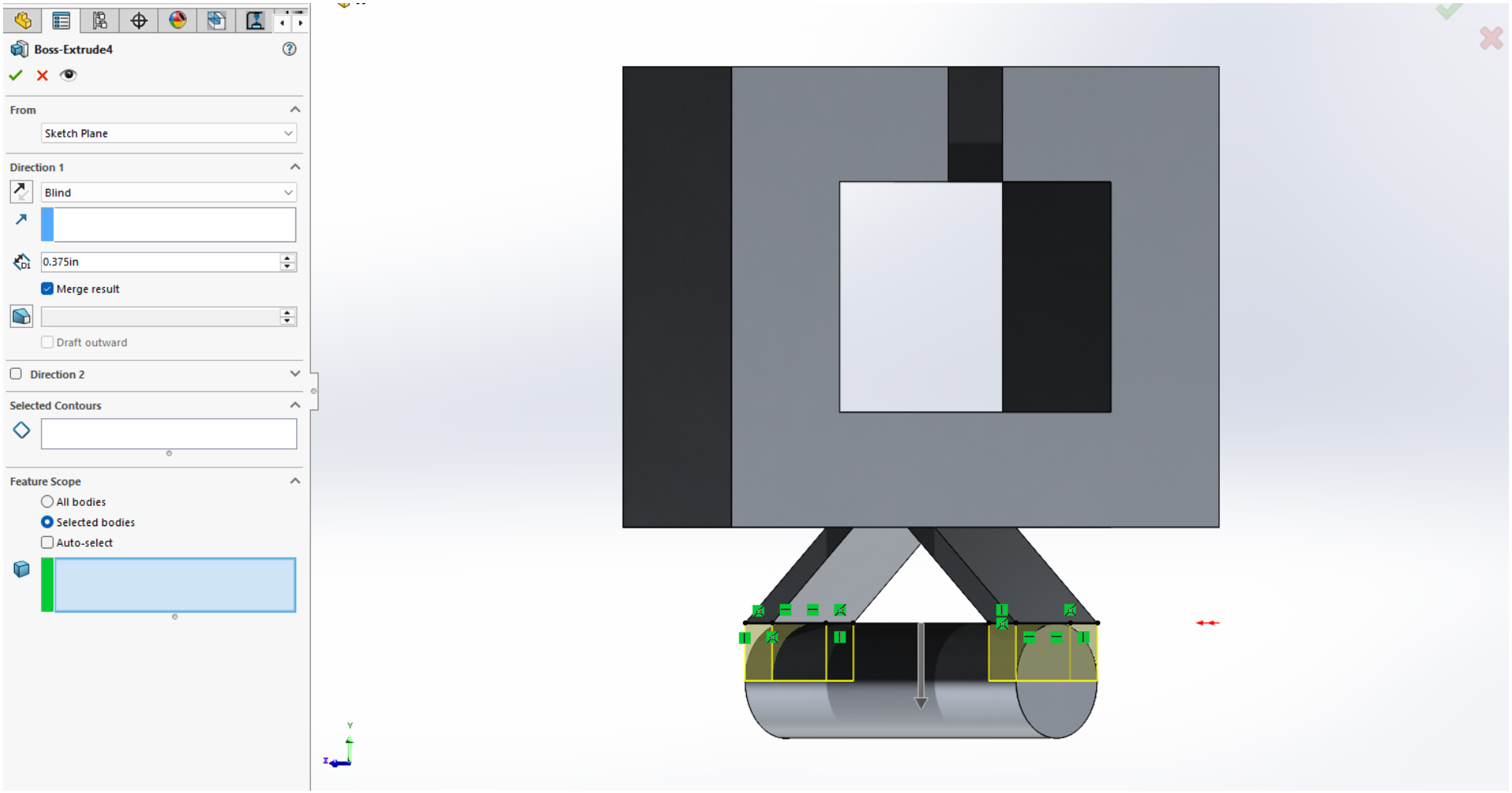
Extrusion used to develop the revised lower support.



Source: extrude.png

A7 - Figure 16

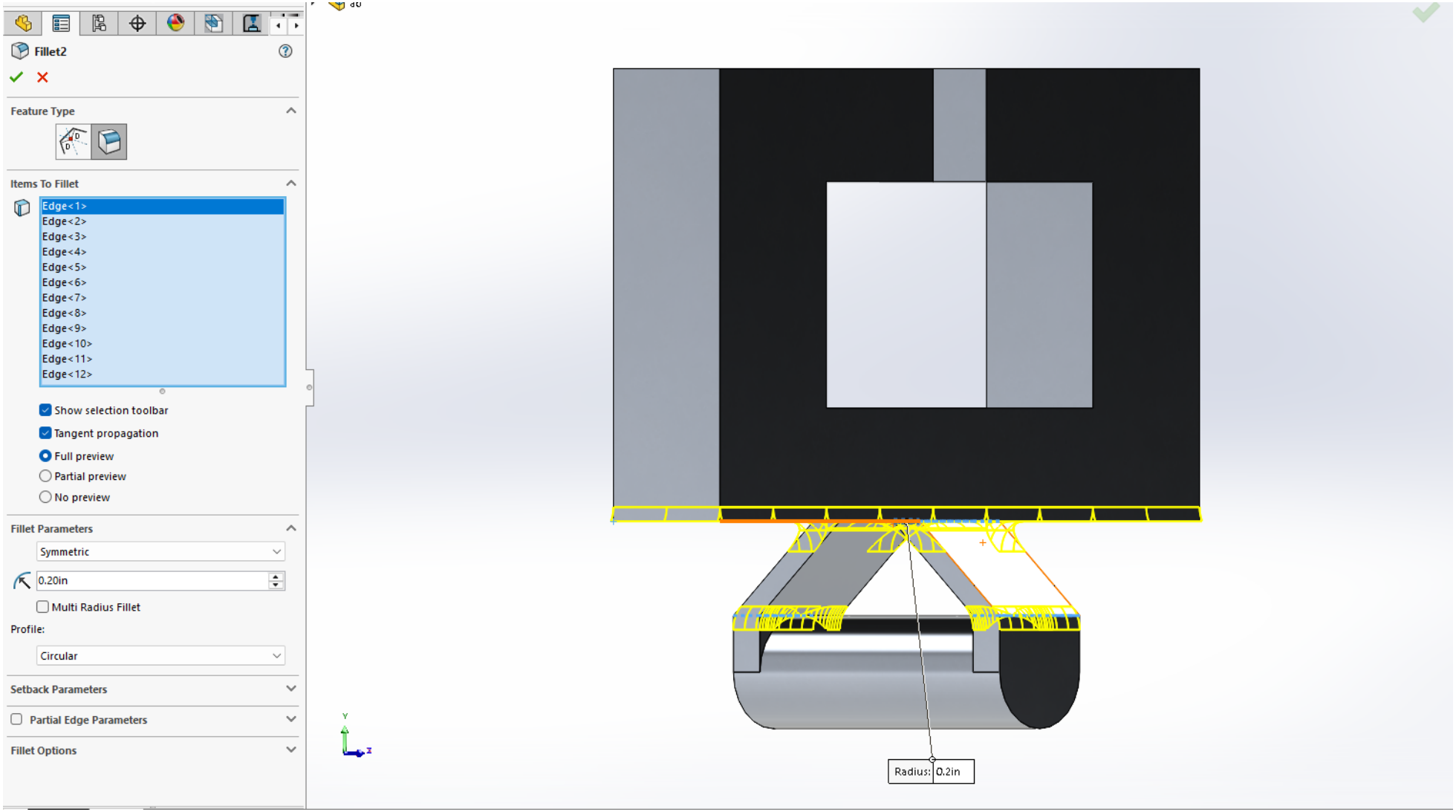
Joining the new diagonal supports to the pin region.



Source: connection with pin.png

A7 - Figure 17

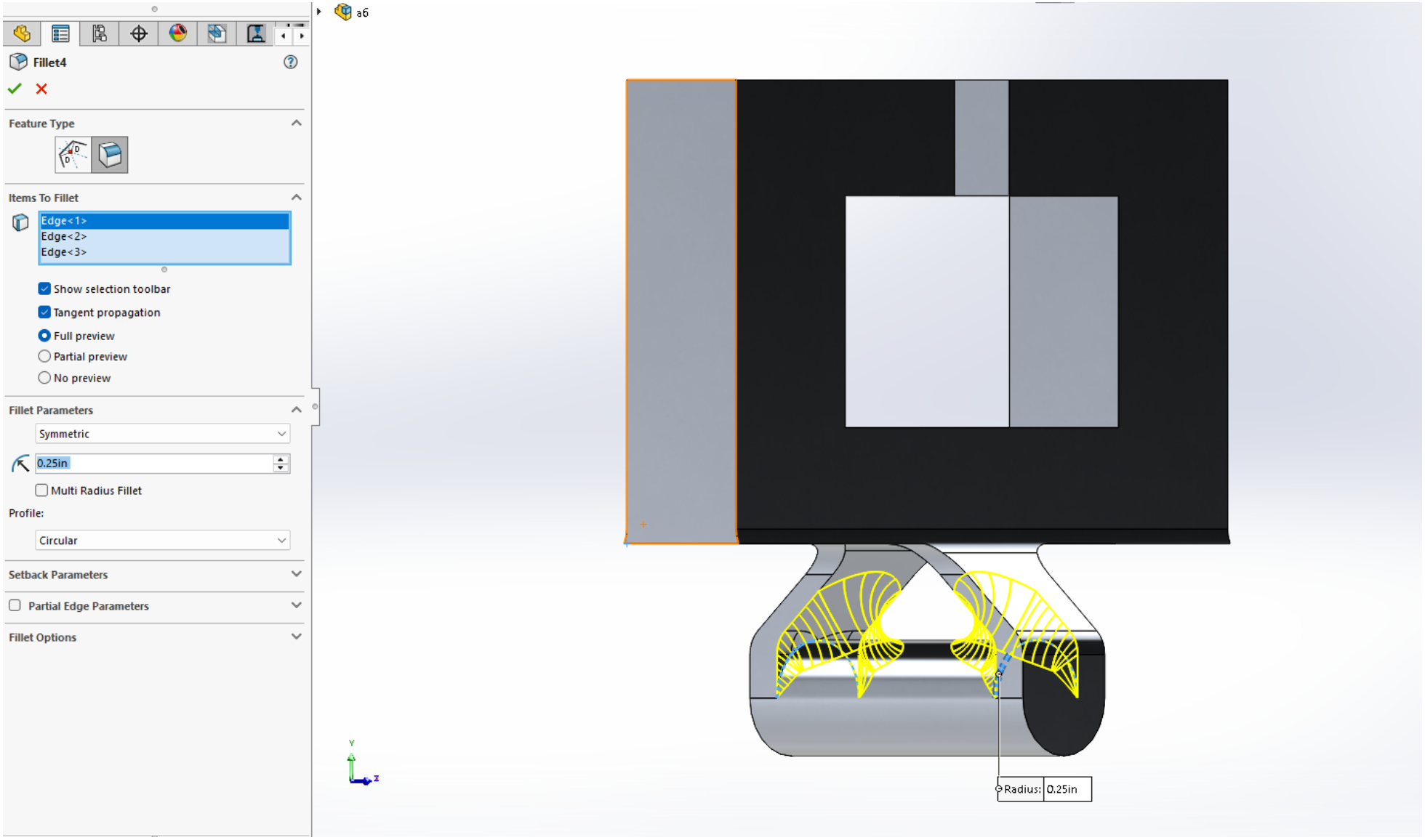
Fillet preview at the support and pin transitions.



Source: fillets.png

A7 - Figure 18

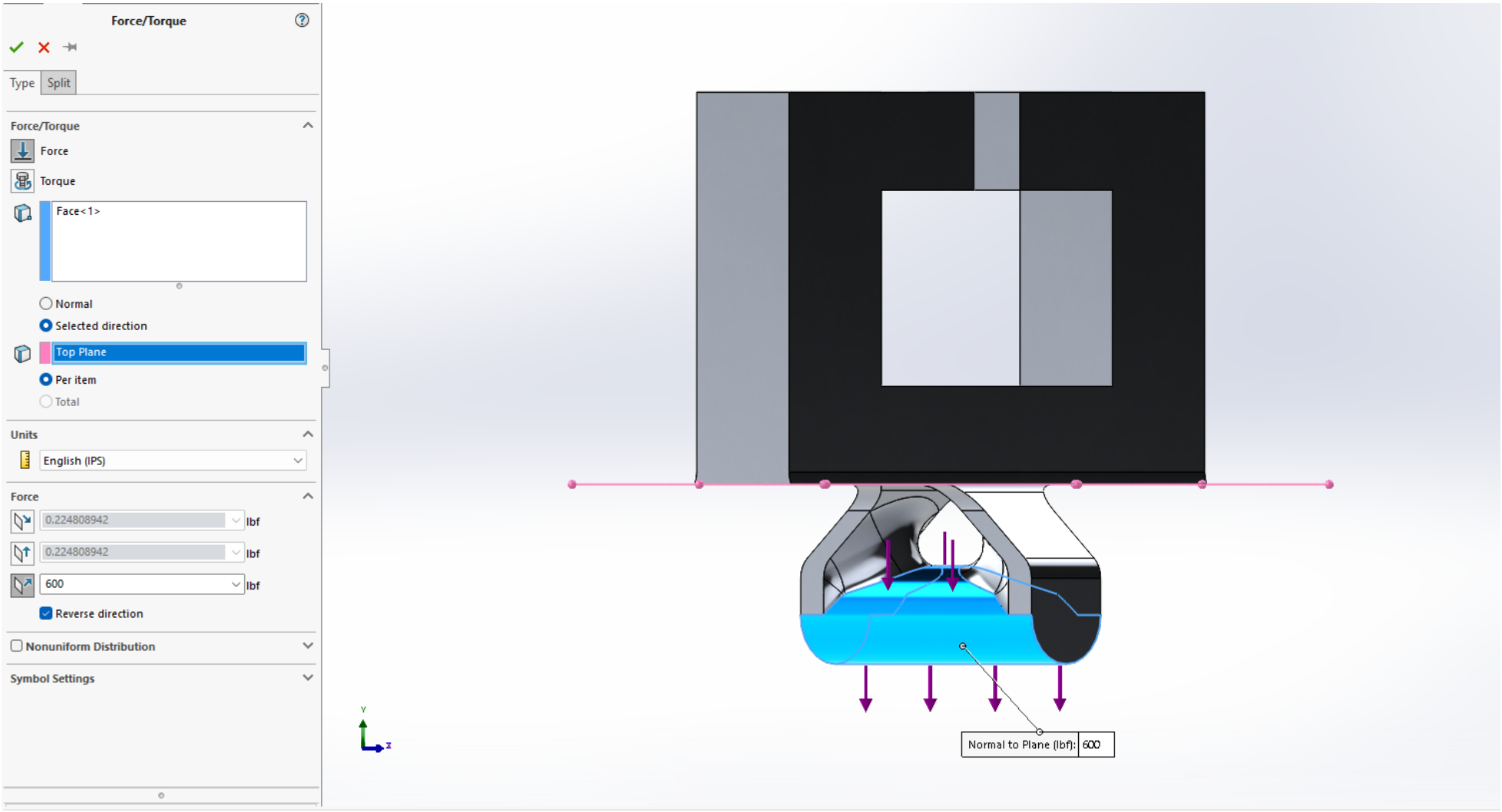
Second fillet operation on the revised support transitions.



Source: fillets 2.png

A7 - Figure 19

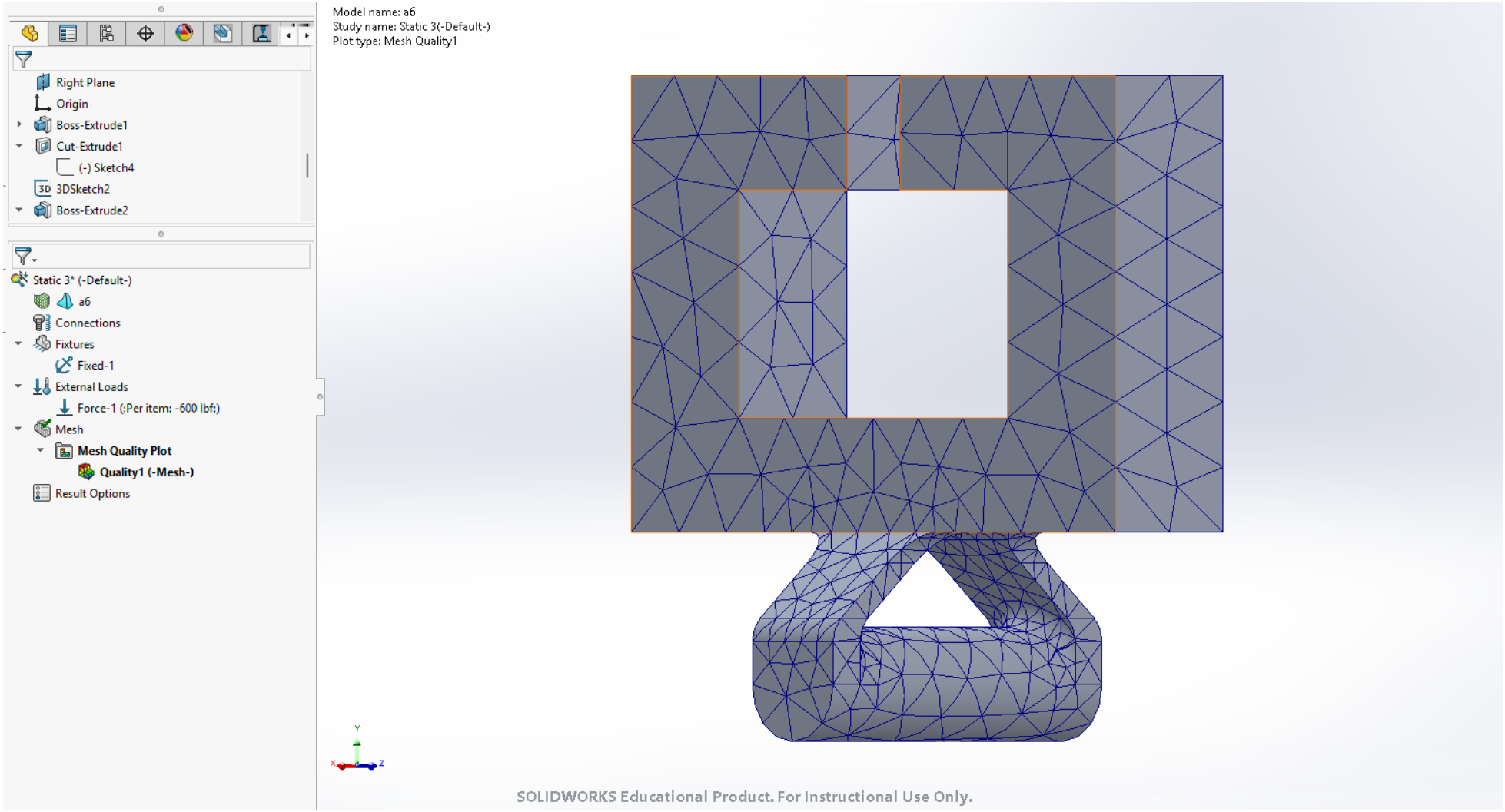
Design 2 force definition: 600 lbf per item on the selected pin face.



Source: Design 2/Design 2 Force.png

A7 - Figure 20

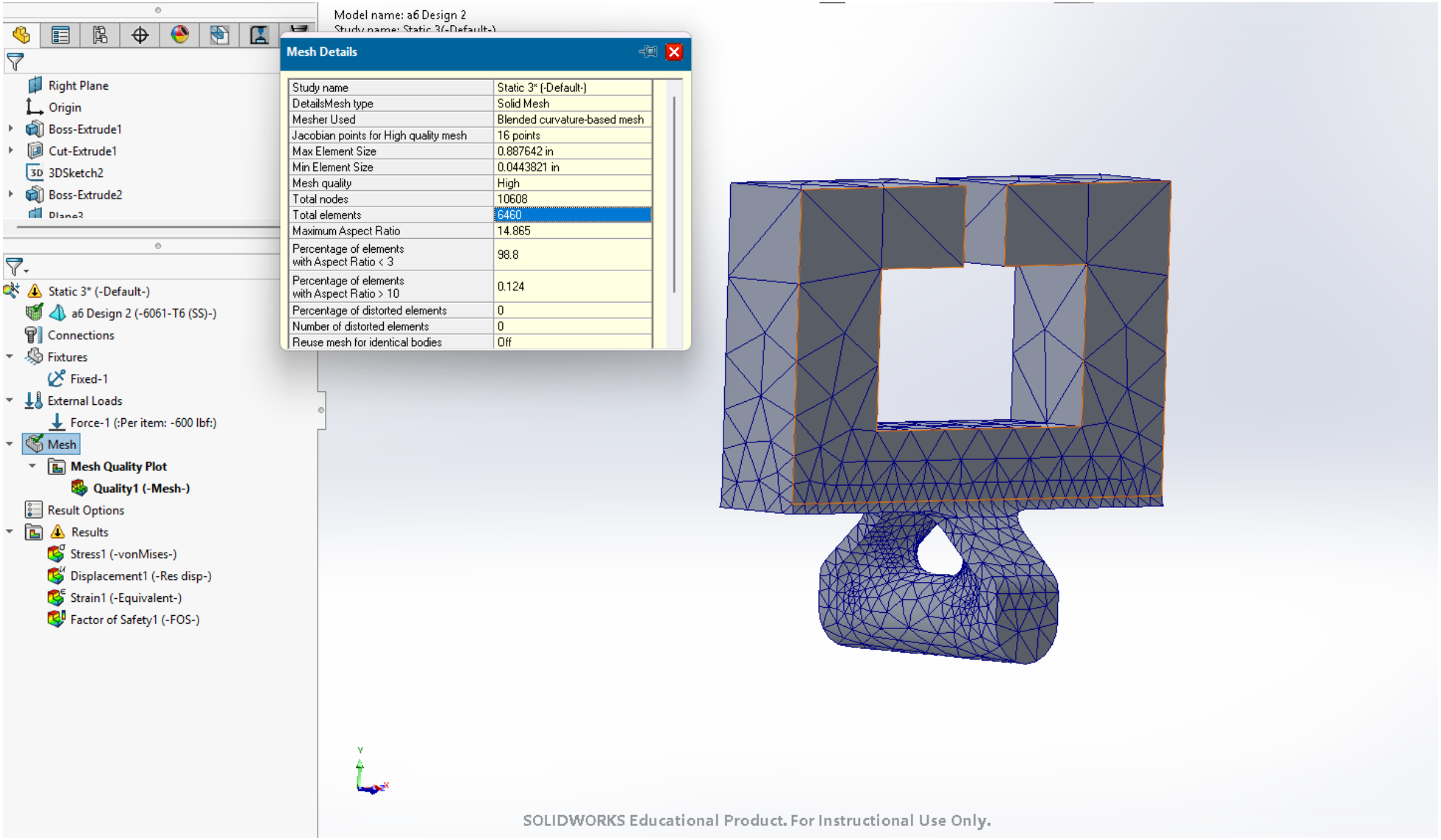
Bracket Design 2: coarse Design 2 mesh.



Source: Design 2/coarse Design 2 mesh.png

A7 - Figure 21

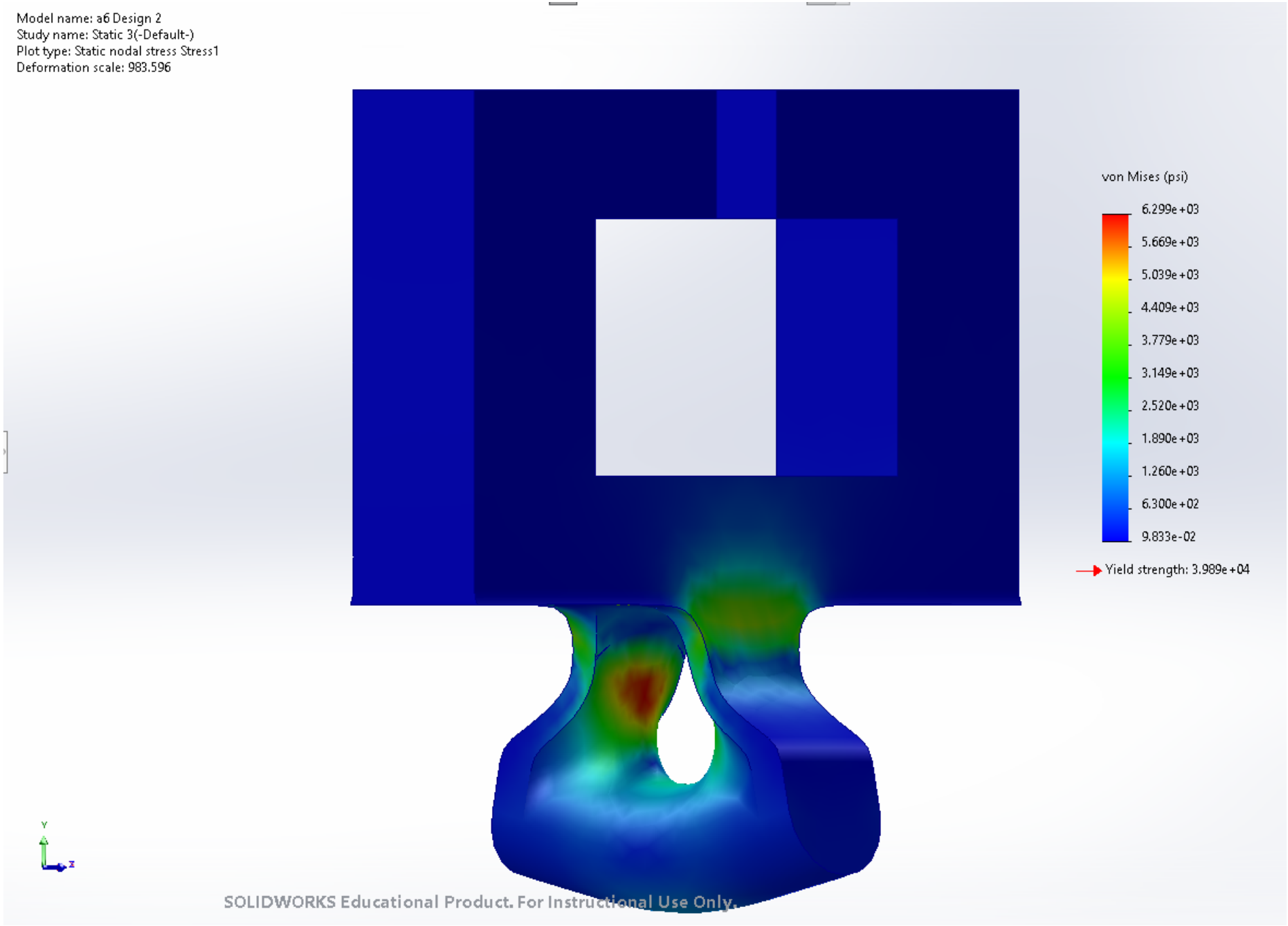
Bracket Design 2: coarse mesh details 2.



Source: Design 2/coarse mesh details 2.png

A7 - Figure 22

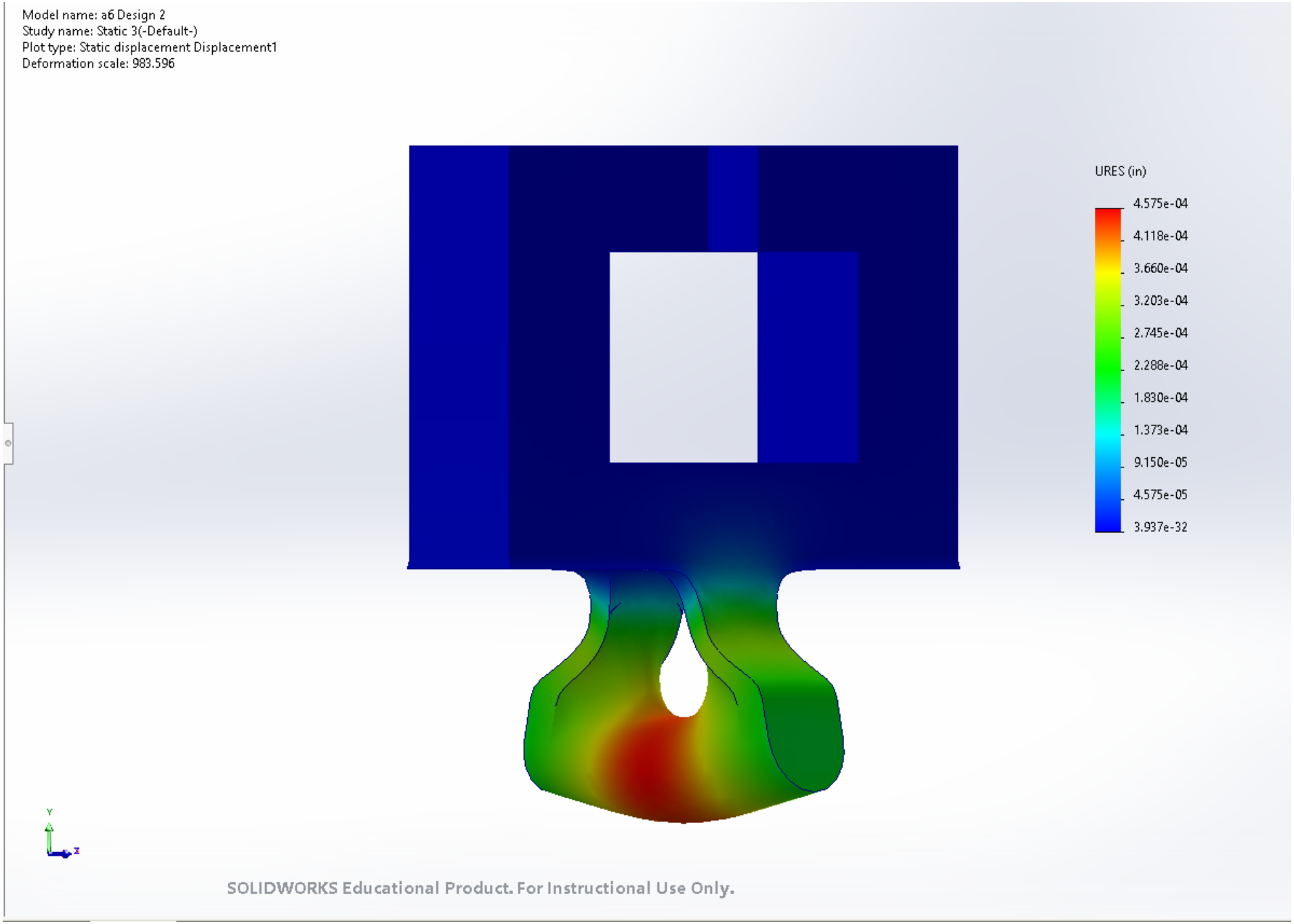
Bracket Design 2: design 2 Coarse Stress.



Source: Design 2/design 2 Coarse Stress.png

A7 - Figure 23

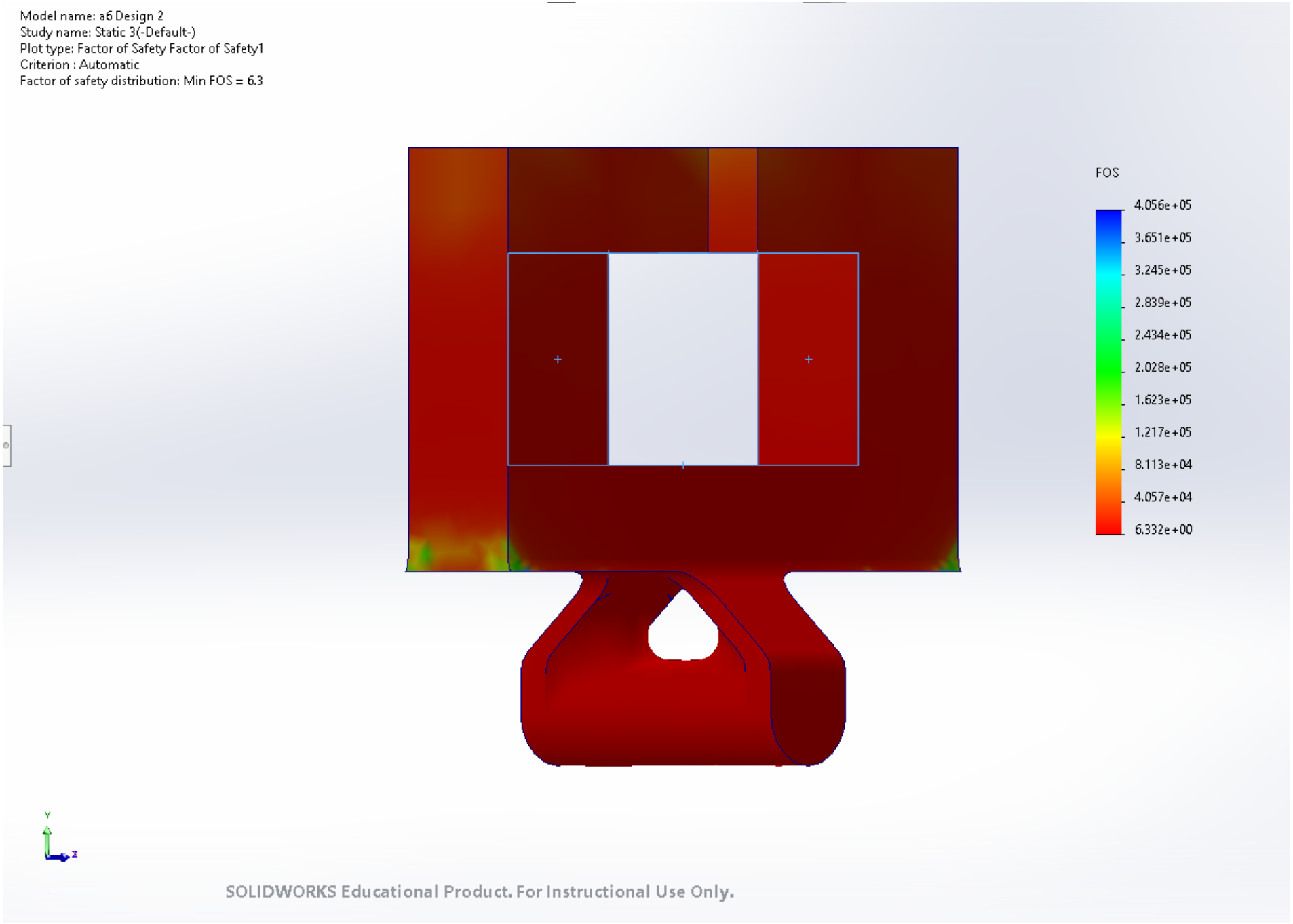
Bracket Design 2: Design 2 Coarse Displacement.



Source: Design 2/Design 2 Coarse Displacement.png

A7 - Figure 24

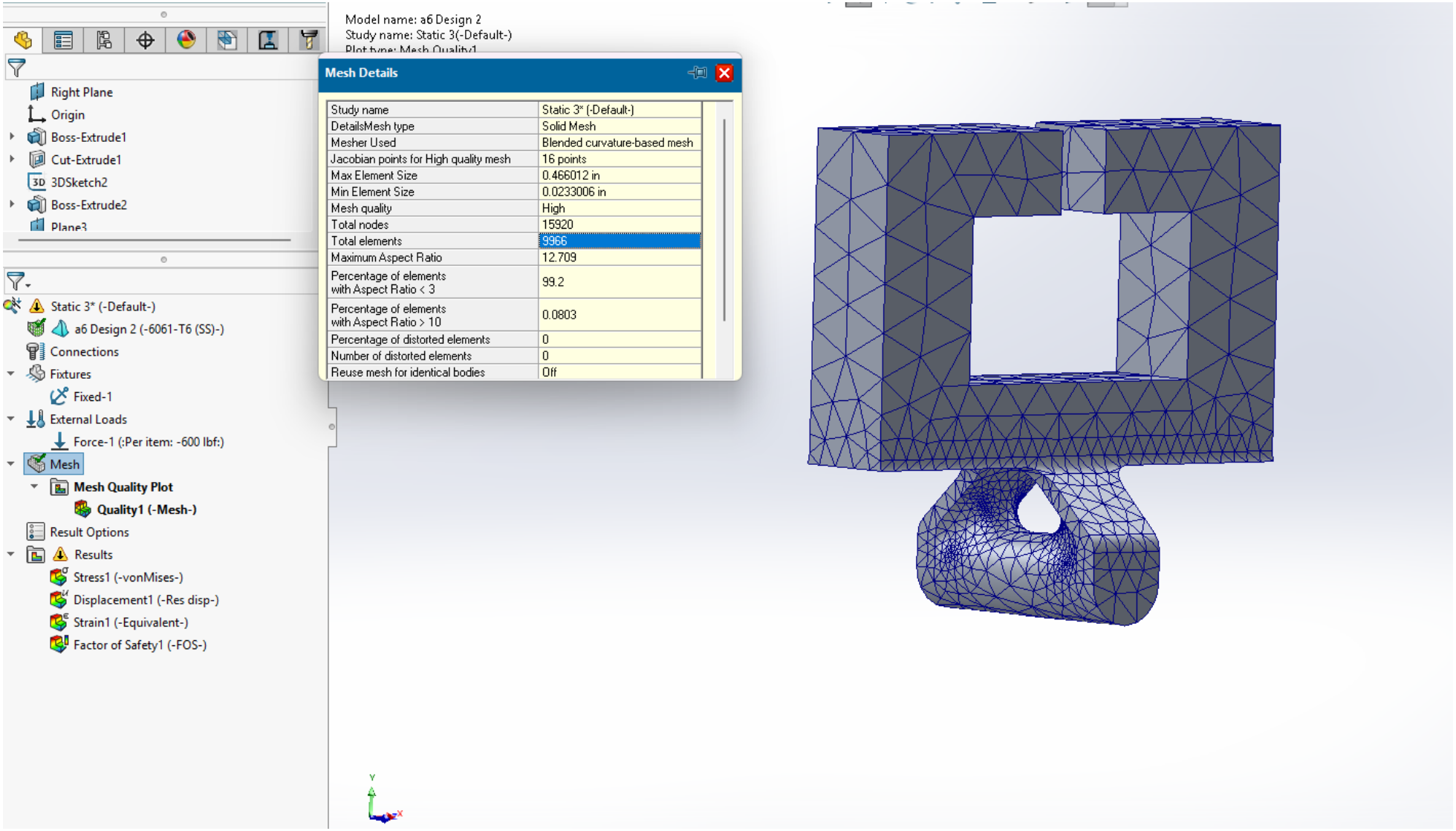
Bracket Design 2: Design 2 Coarse FOS.



Source: Design 2/Design 2 Coarse FOS.png

A7 - Figure 25

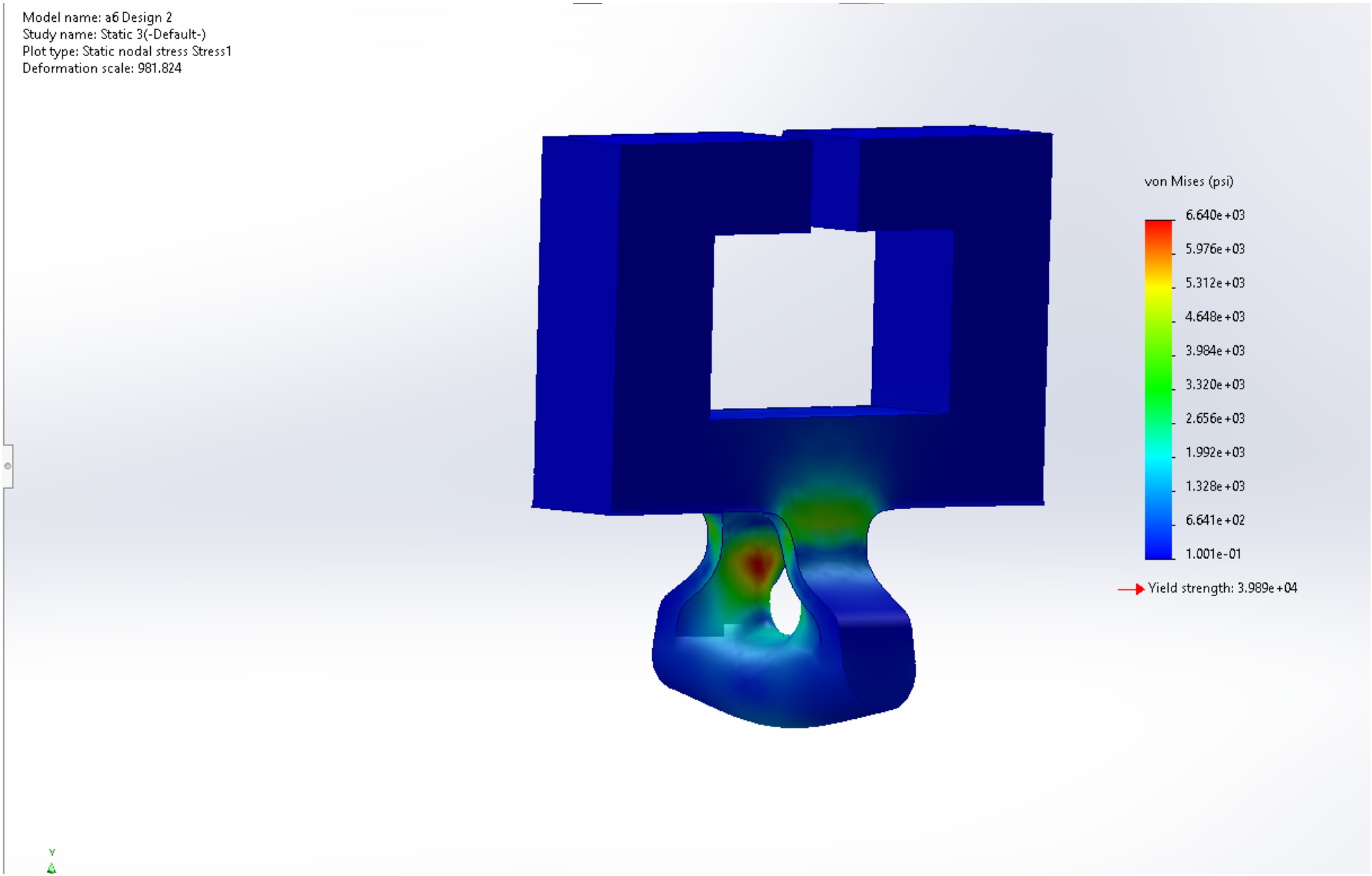
Bracket Design 2: Design 2 Medium mesh.



Source: Design 2/Design 2 Medium mesh.png

A7 - Figure 26

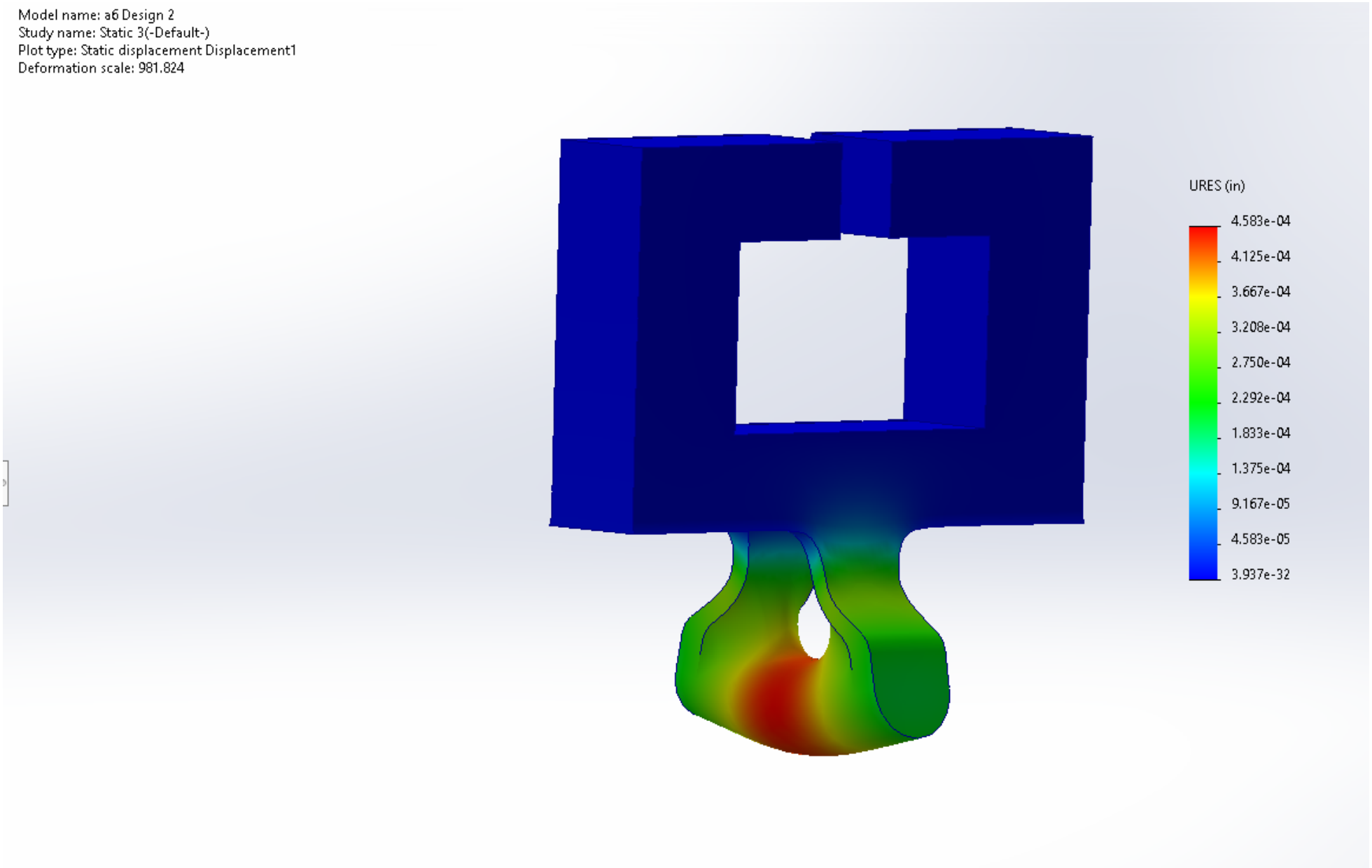
Bracket Design 2: Design 2 Medium Stress.



Source: Design 2/Design 2 Medium Stress.png

A7 - Figure 27

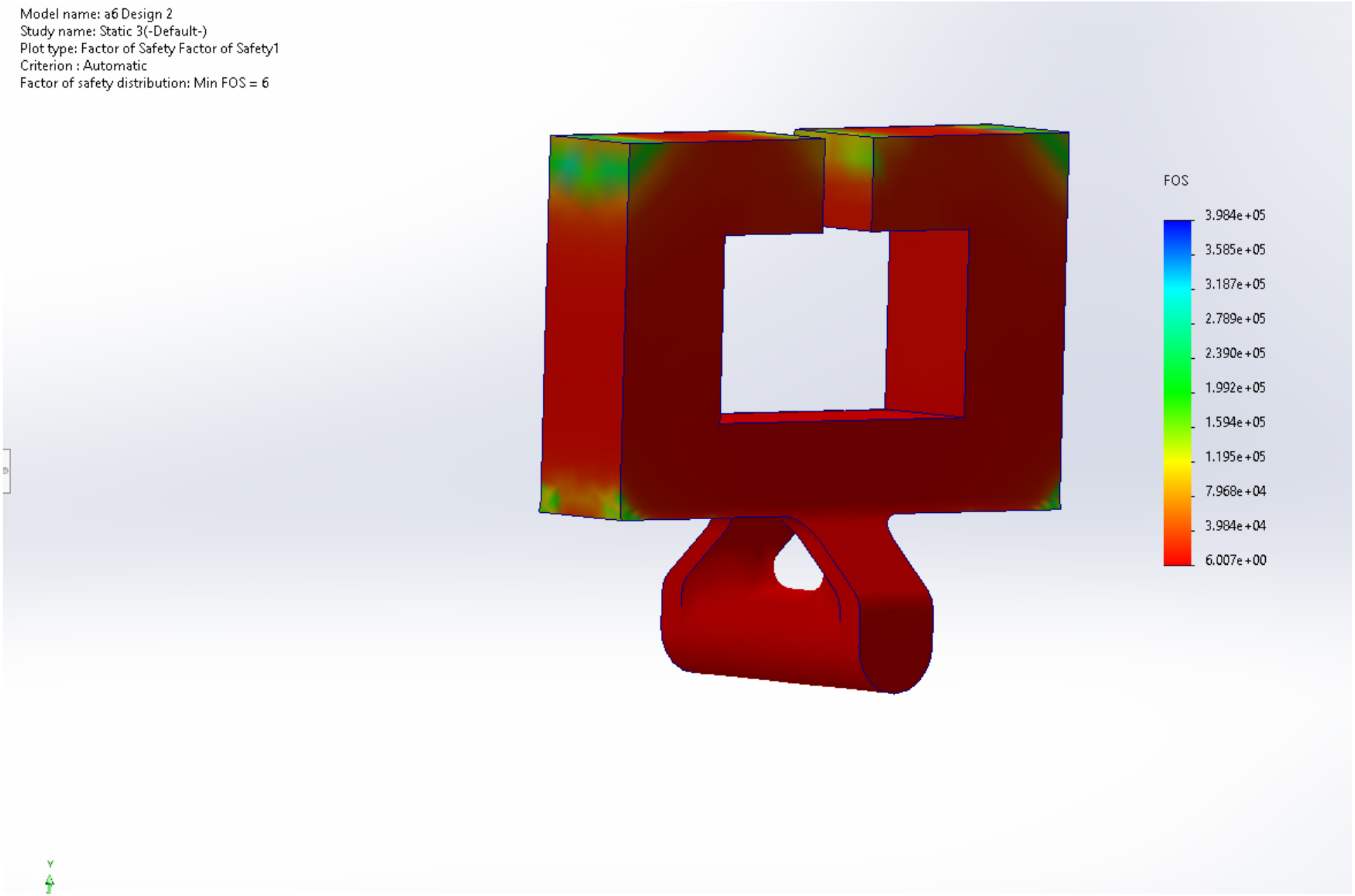
Bracket Design 2: Design 2 Medium Displacement.



Source: Design 2/Design 2 Medium Displacement.png

A7 - Figure 28

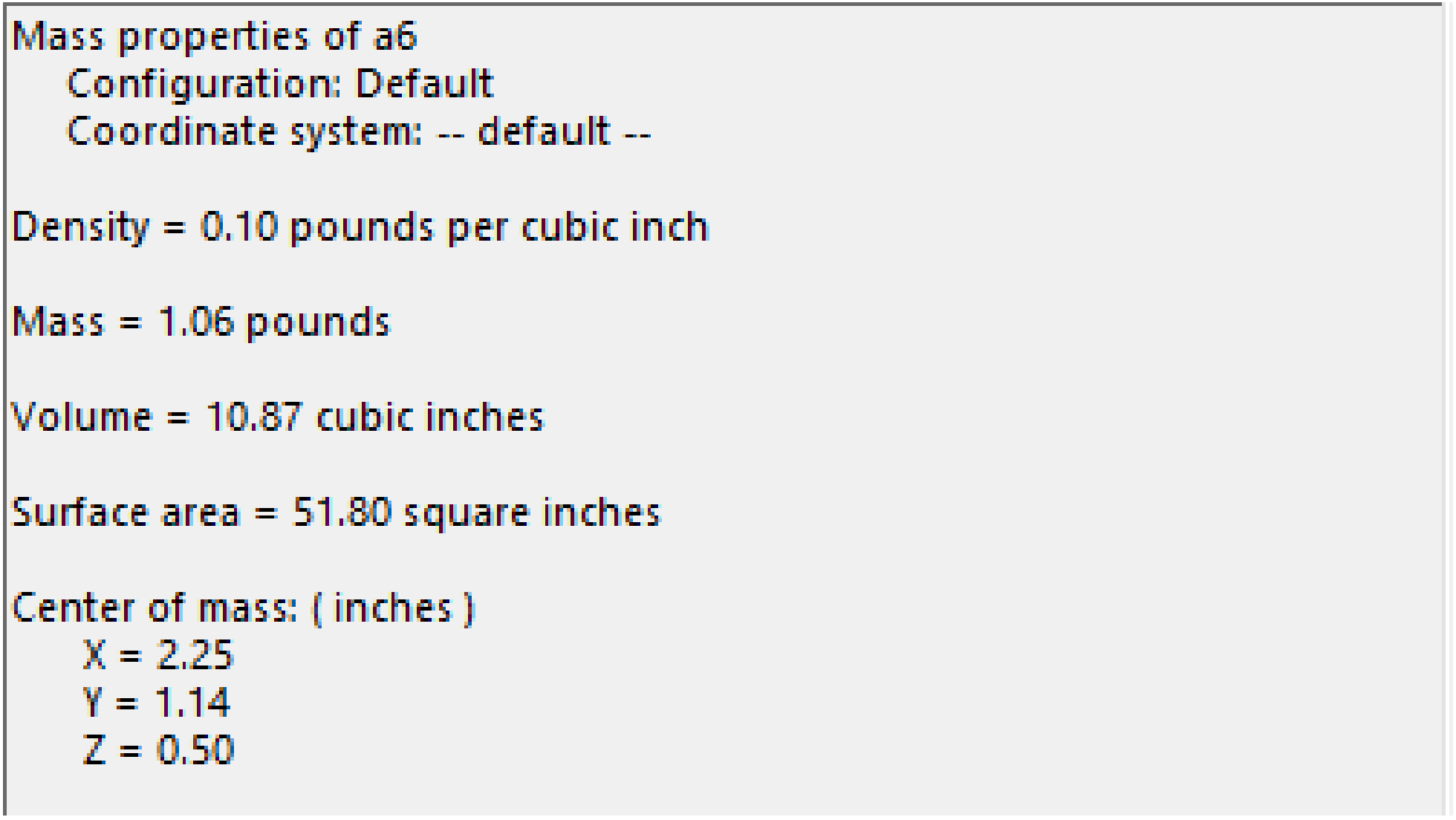
Bracket Design 2: Design 2 Medium FOS.



Source: Design 2/Design 2 Medium FOS.png

A7 - Figure 29

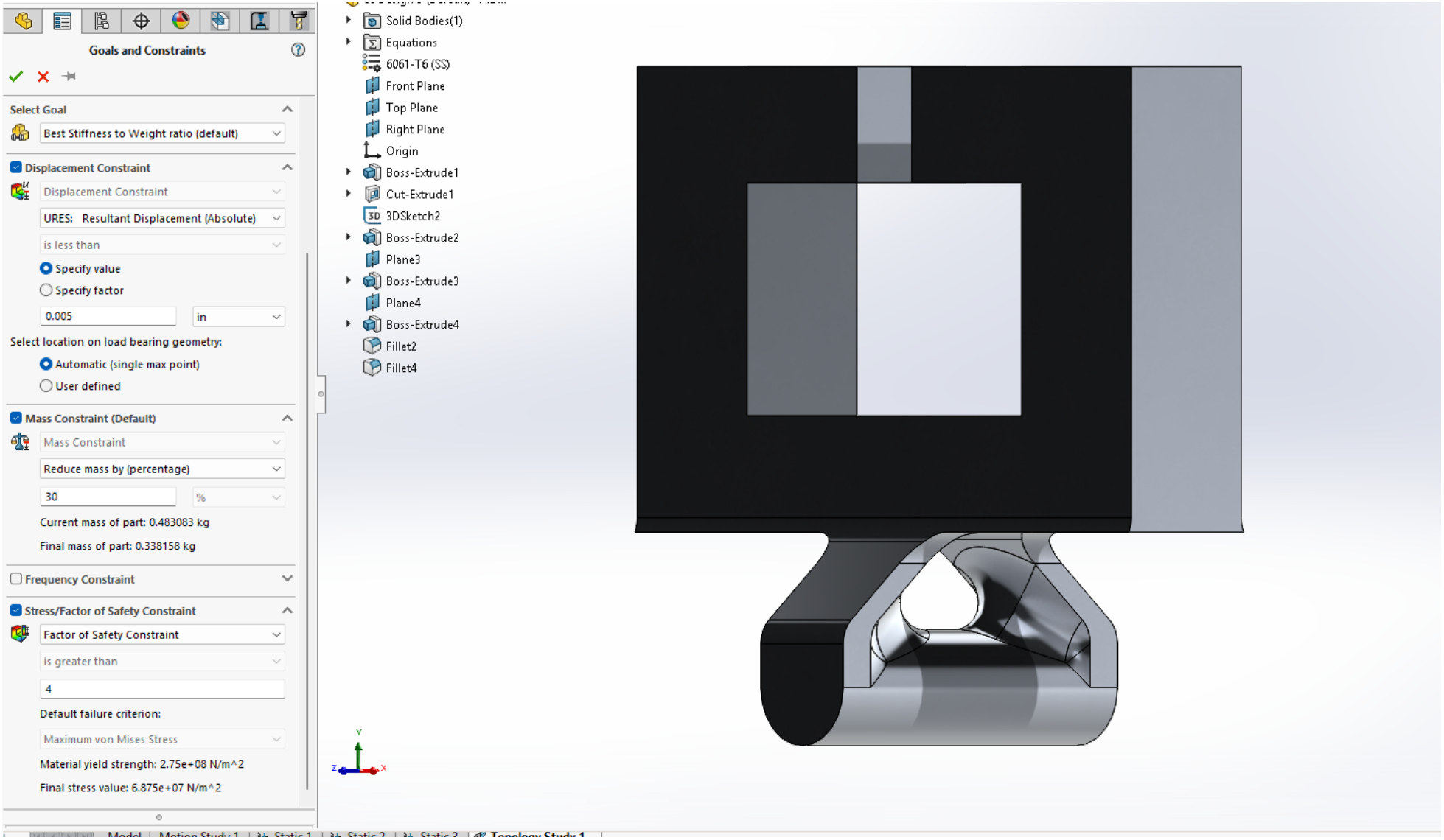
Bracket Design 2: Design 2 mass.



Source: Design 2/Design 2 mass.png

A7 - Figure 30

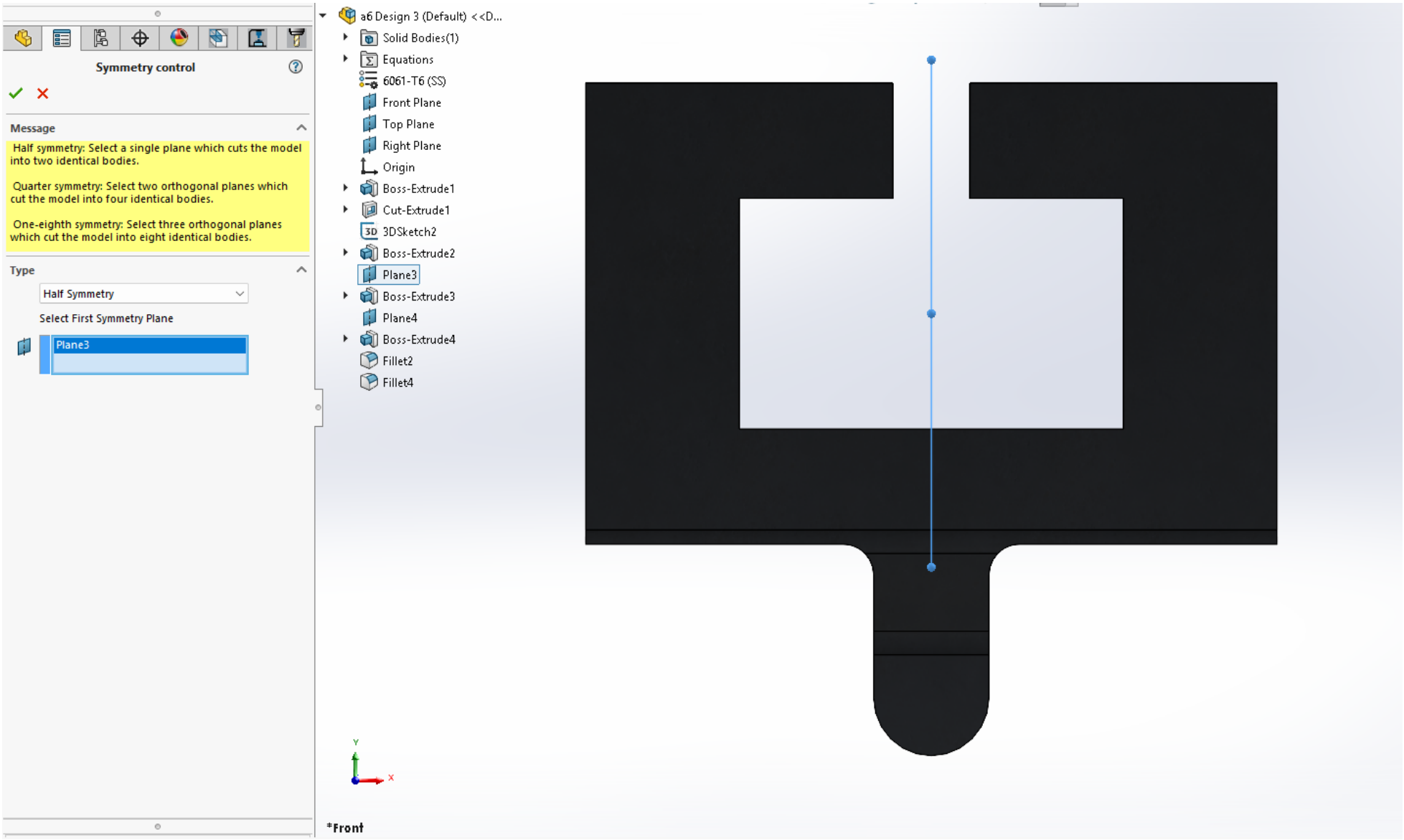
Bracket topology goals: stiffness-to-weight, 30% mass reduction, 0.005-in displacement limit, and FOS 4.



Source: Design 3/Topology Study Design 3.png

A7 - Figure 31

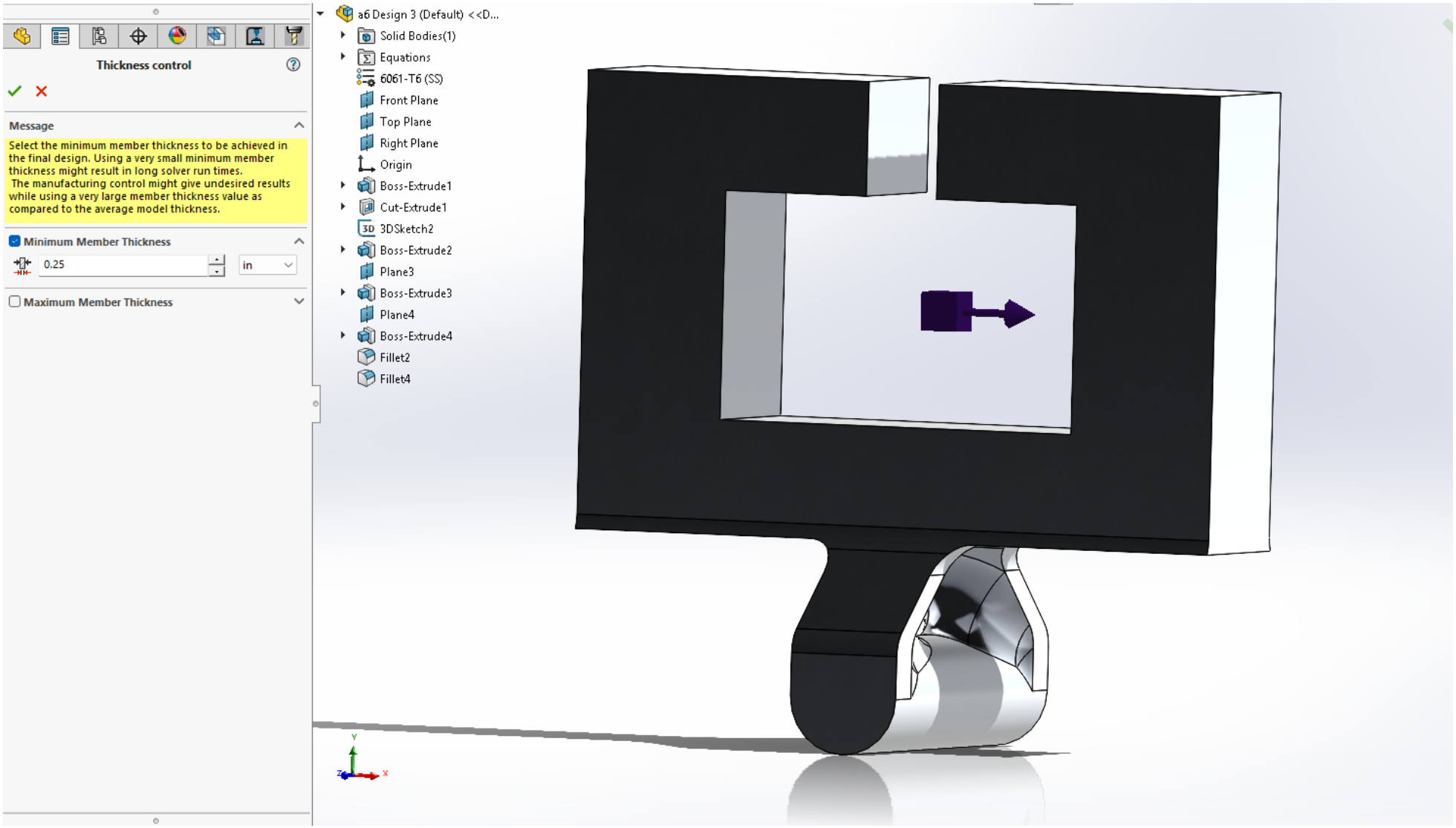
Half-symmetry plane through the bracket center.



Source: Design 3/Symmetry plane.png

A7 - Figure 32

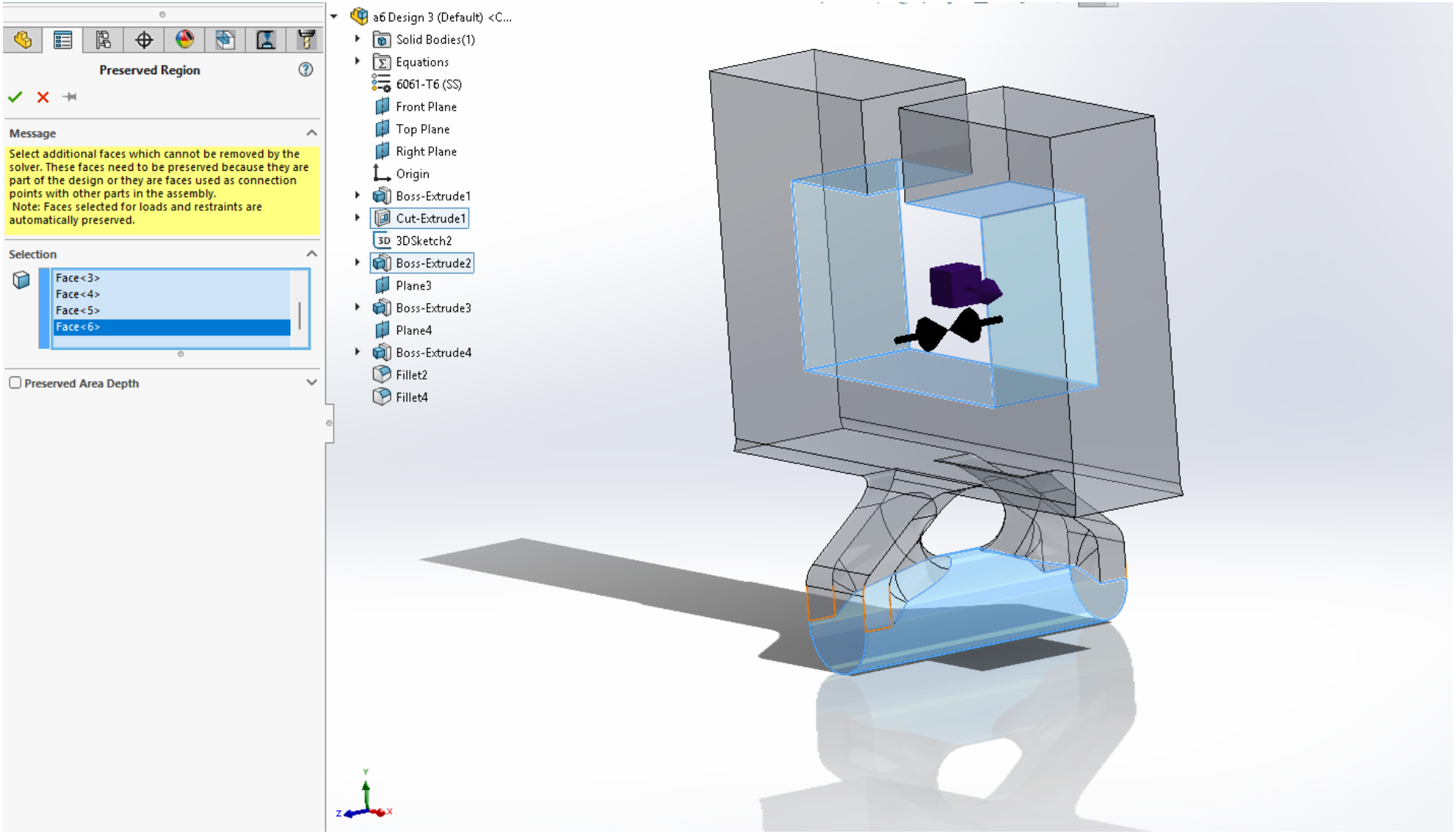
Bracket topology minimum member-thickness setting: 0.25 in.



Source: Design 3/Thickness.png

A7 - Figure 33

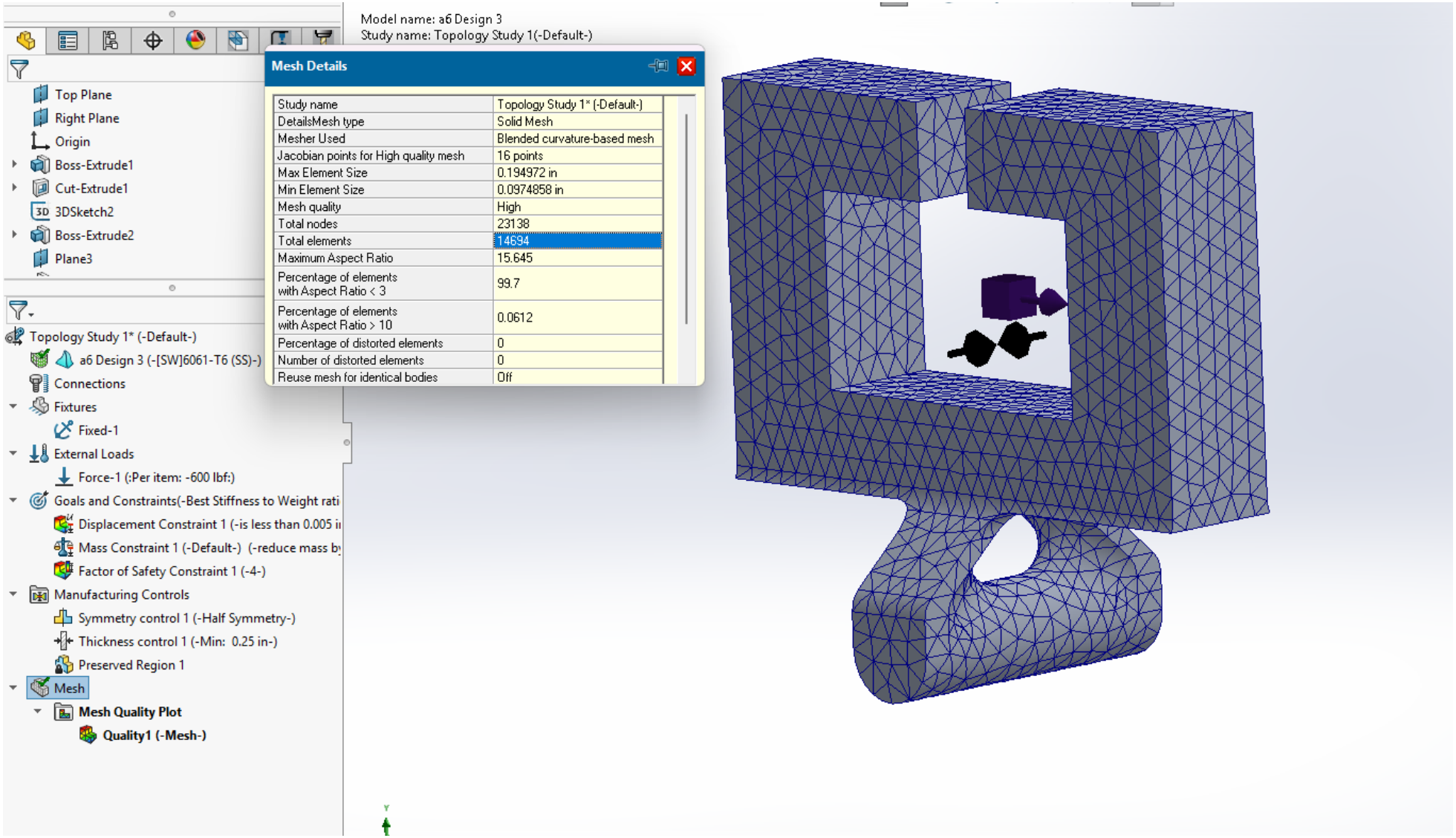
Preserved mounting and pin regions in the bracket topology study.



Source: Design 3/Preserved Surfaces.png

A7 - Figure 34

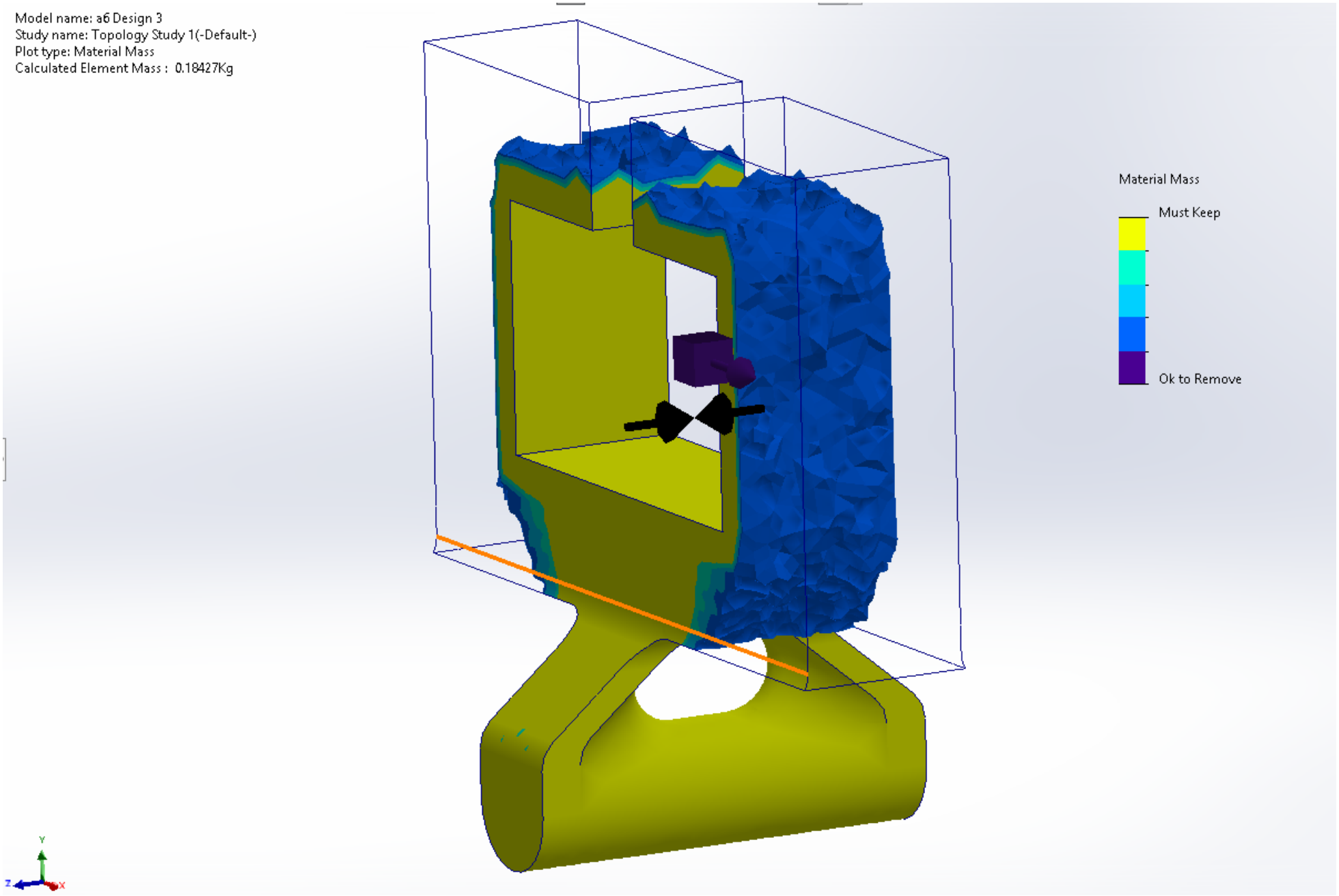
Topology mesh details; this is separate from the final static-study meshes.



Source: Design 3/Topology Coarse mesh.png

A7 - Figure 35

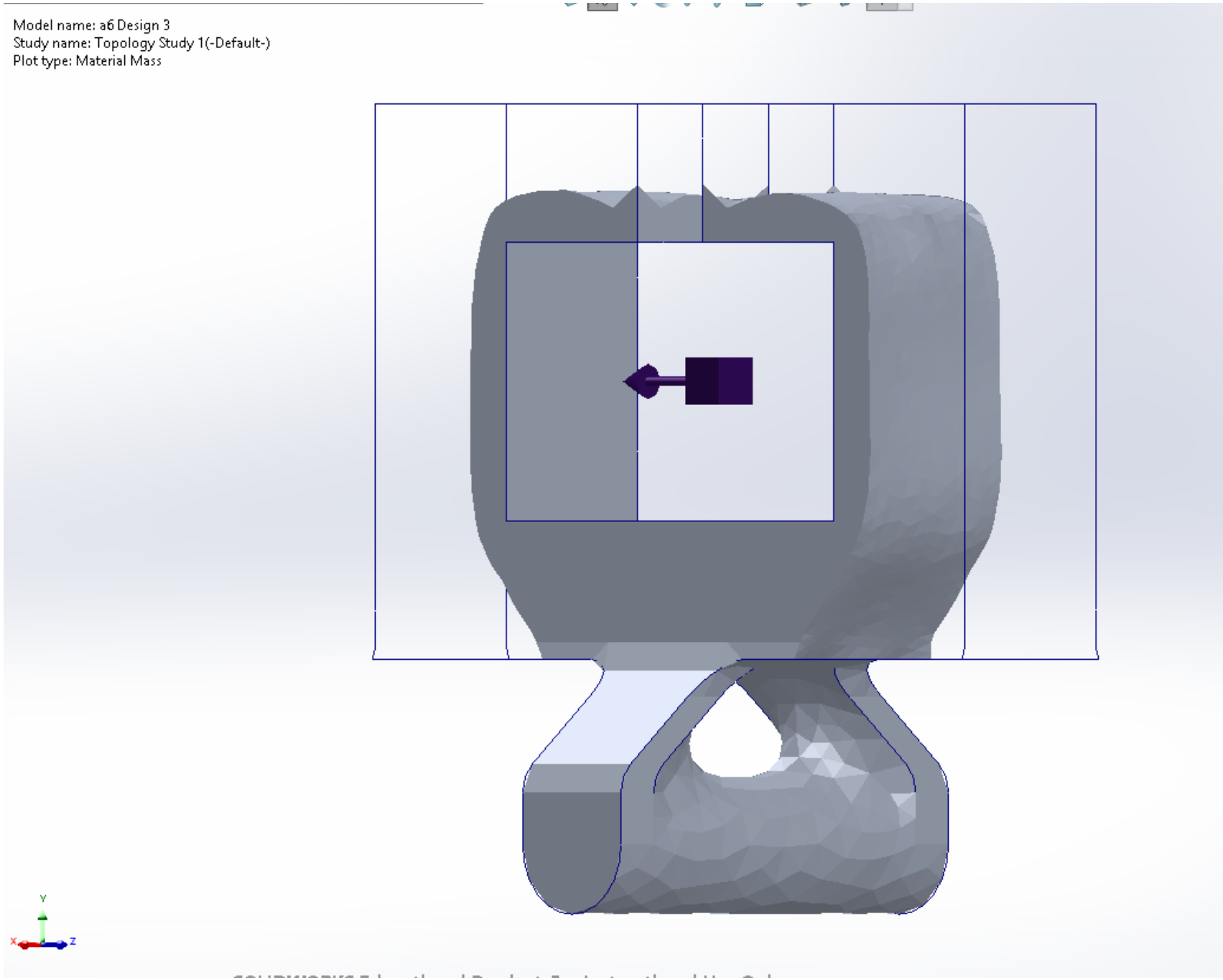
Raw bracket material-mass plot. Its jagged surface prompted the smoothing step.



Source: Design 3/topology.png

A7 - Figure 36

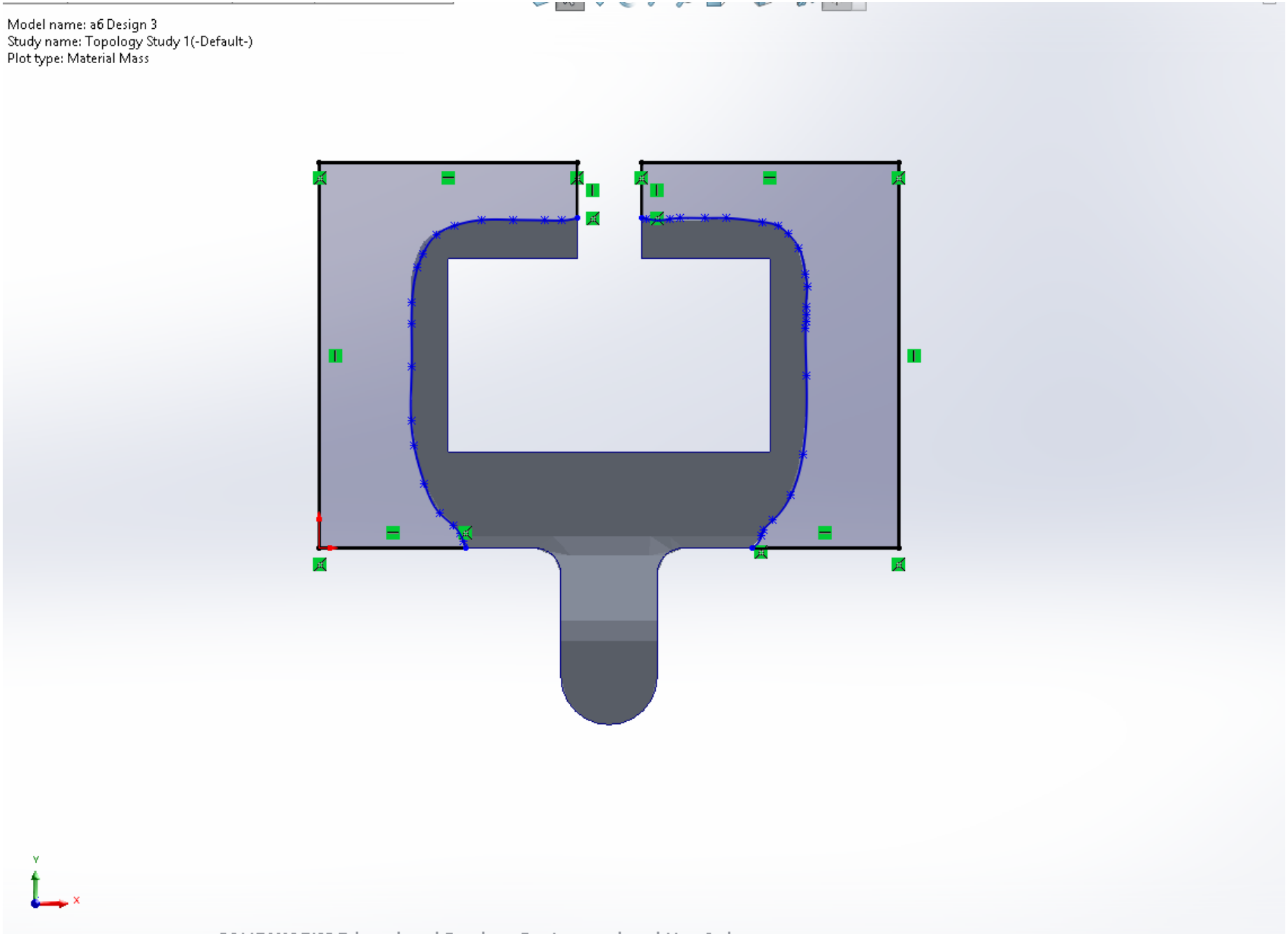
Smoothed bracket topology result used as a visual modeling guide.



Source: Design 3/Topology Smoothining.png

A7 - Figure 37

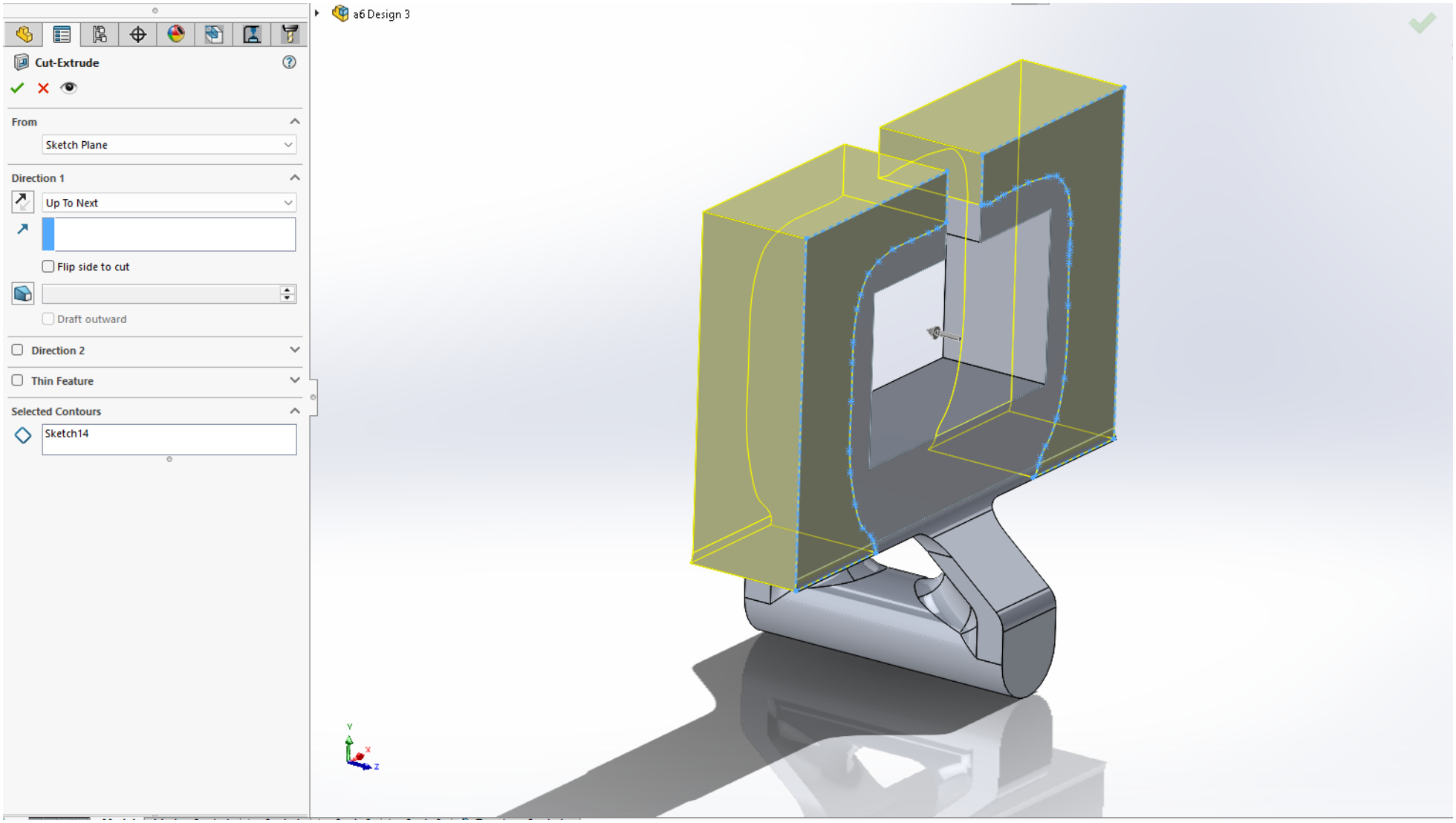
Sketching a smoother outer profile from the topology result.



Source: Design 3/Weight Optimization.png

A7 - Figure 38

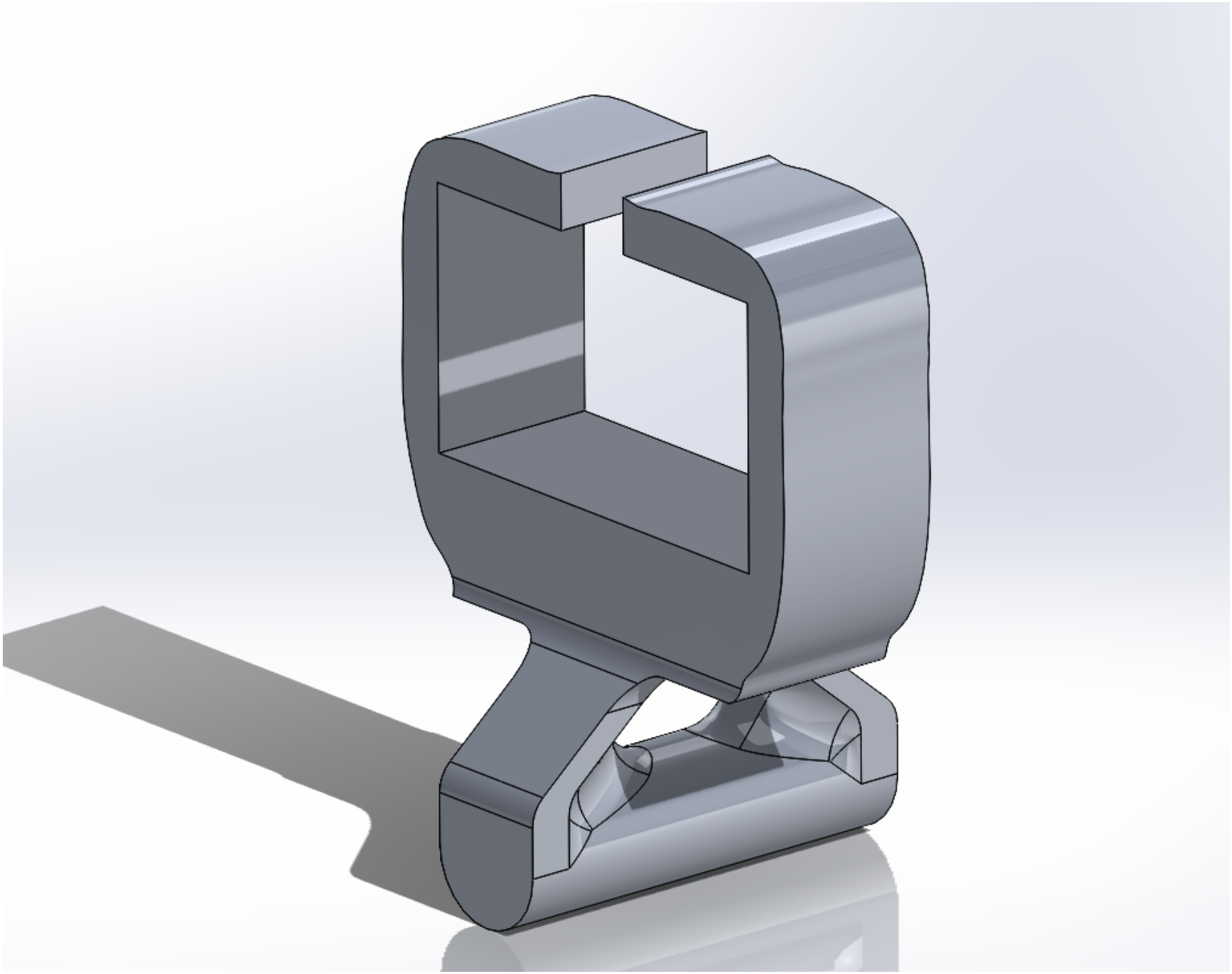
Cut feature used to remove material from the outer bracket envelope.



Source: Design 3/Cut.png

A7 - Figure 39

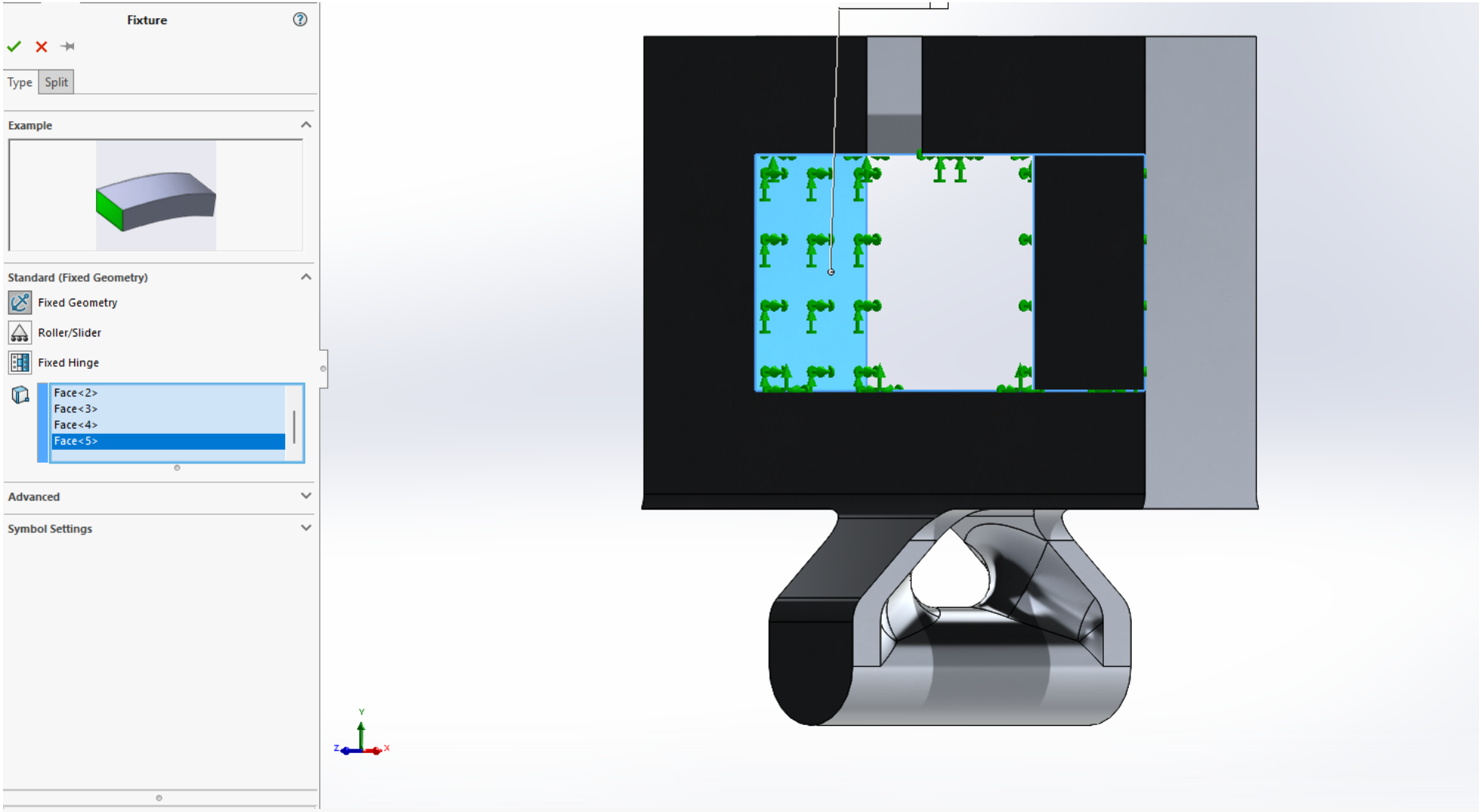
Remodeled bracket after the outer material-removal cut.



Source: Design 3/weight optimized part.png

A7 - Figure 40

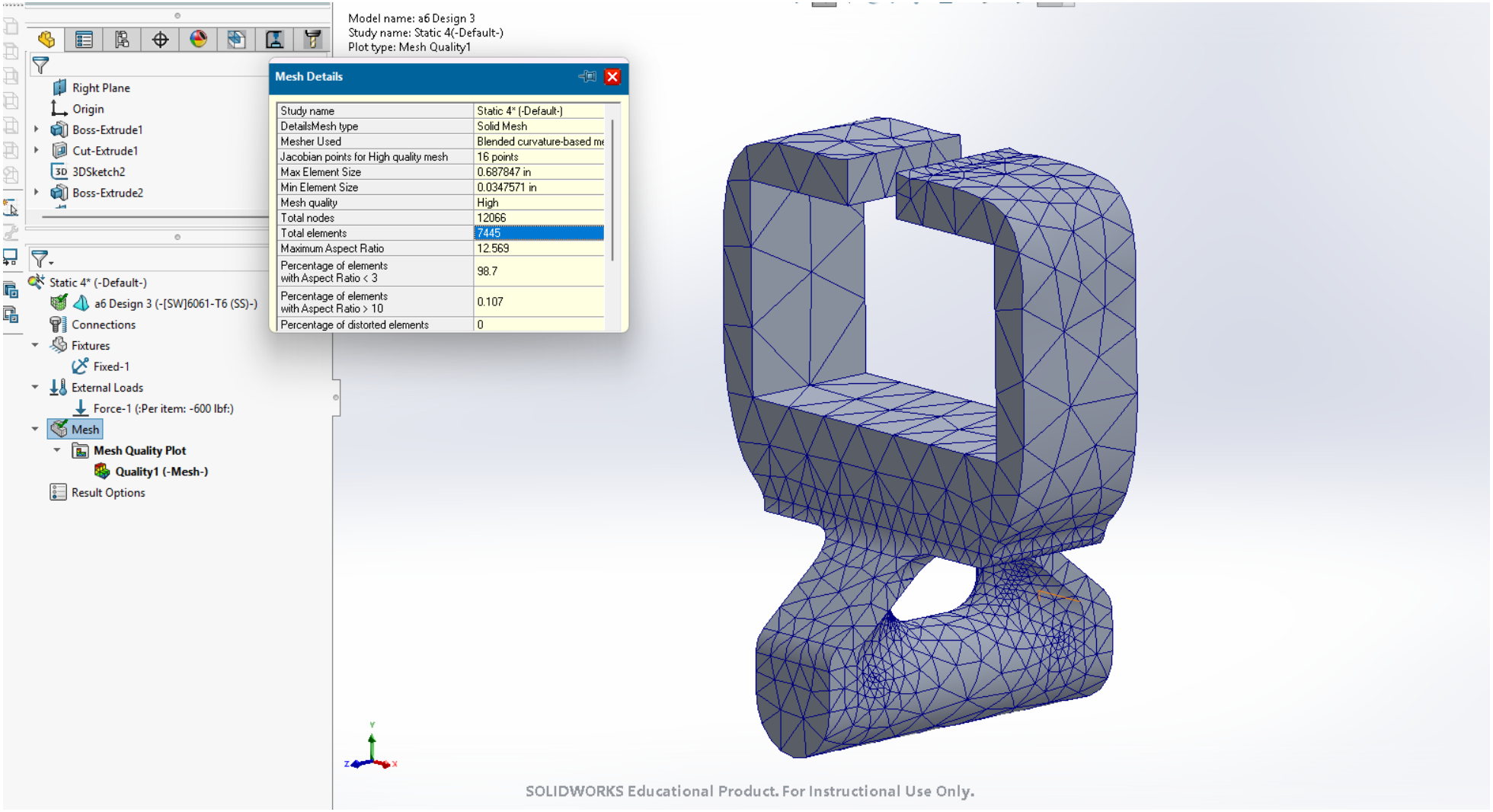
Design 3 fixed-geometry selection on the internal mounting faces.



Source: Design 3/Design 3 Fixtures.png

A7 - Figure 41

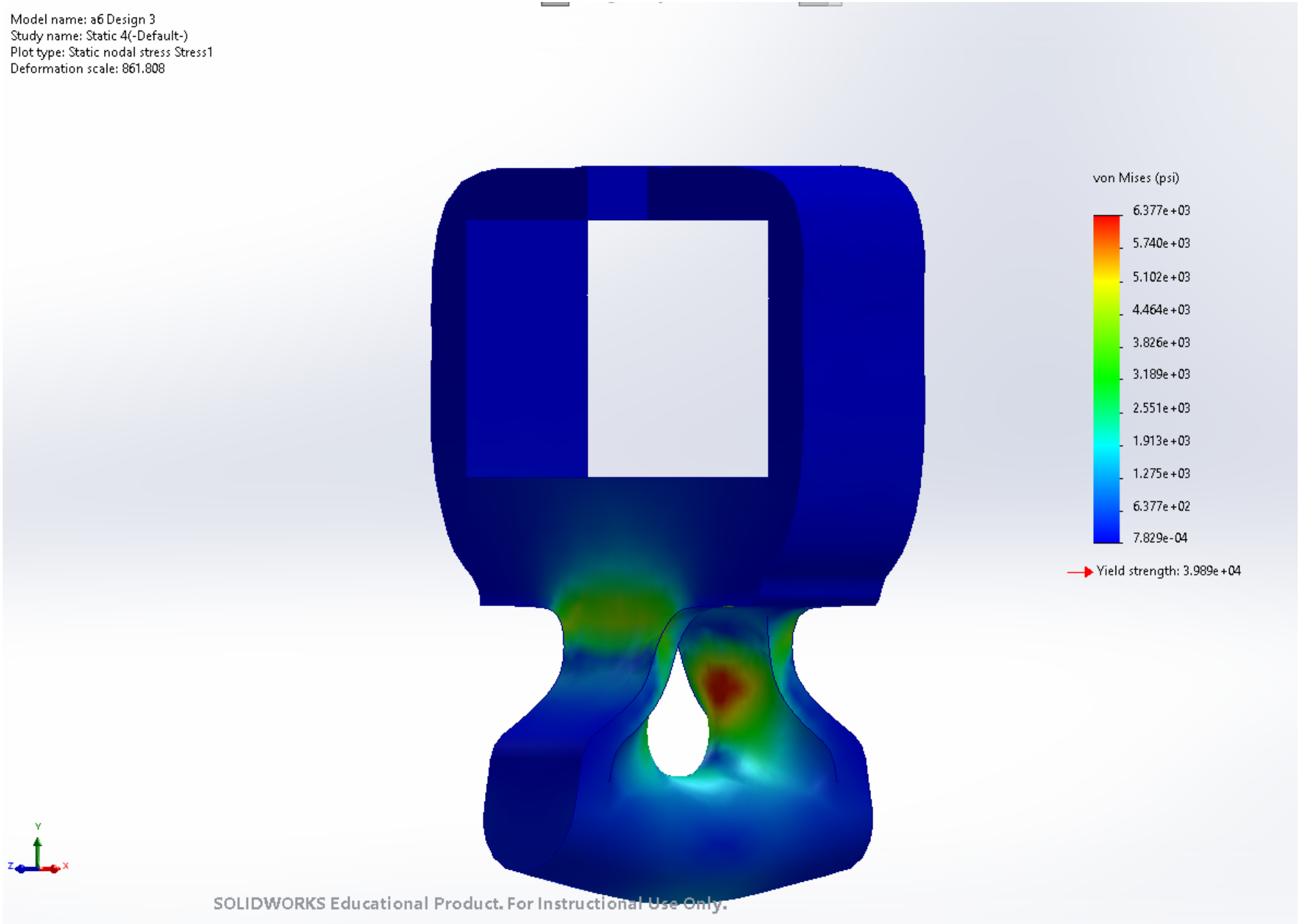
Bracket Design 3: Coarse Mesh Design 3.



Source: Design 3/Coarse Mesh Design 3.png

A7 - Figure 42

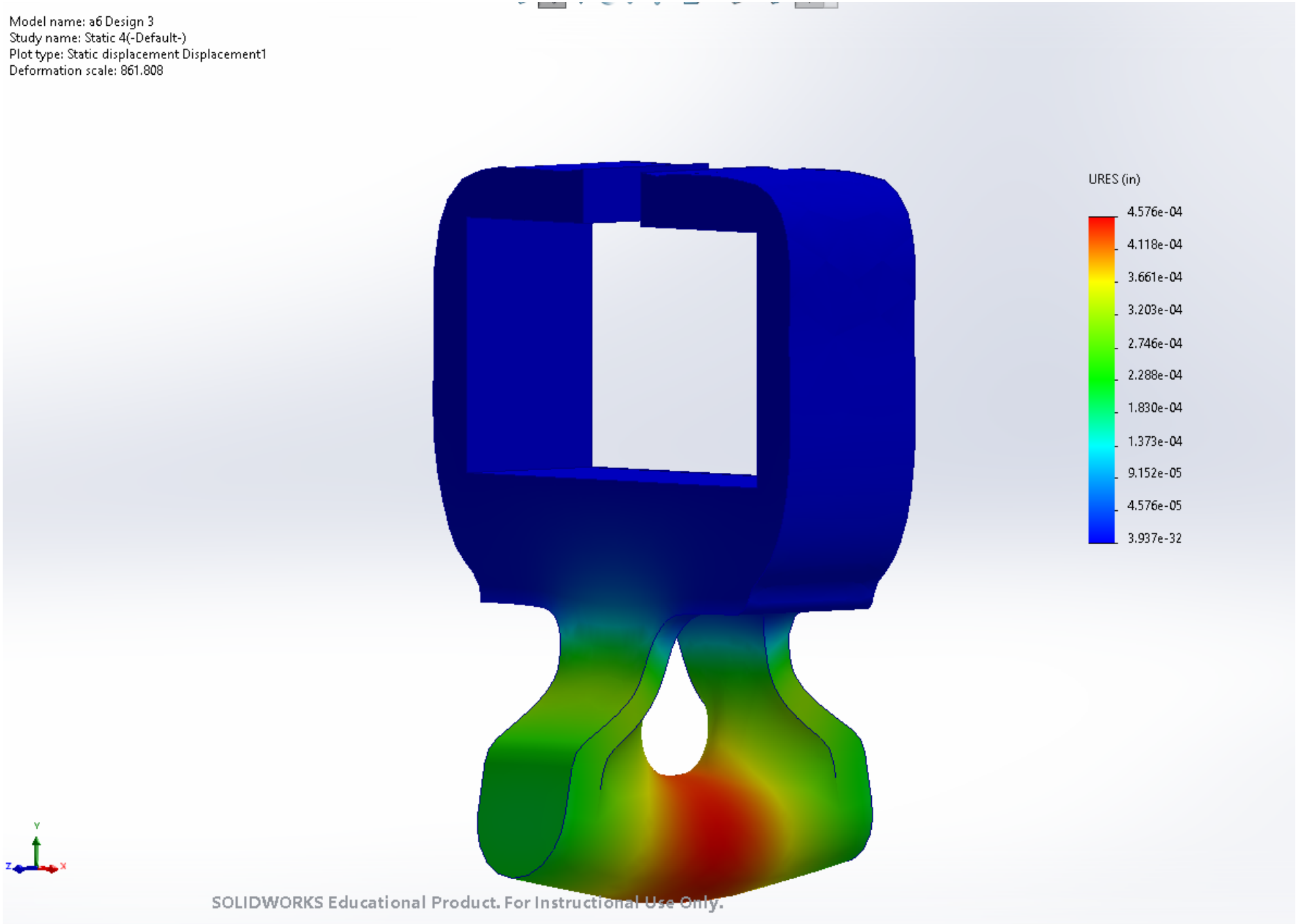
Bracket Design 3: Design 3 Coarse Stress.



Source: Design 3/Design 3 Coarse Stress.png

A7 - Figure 43

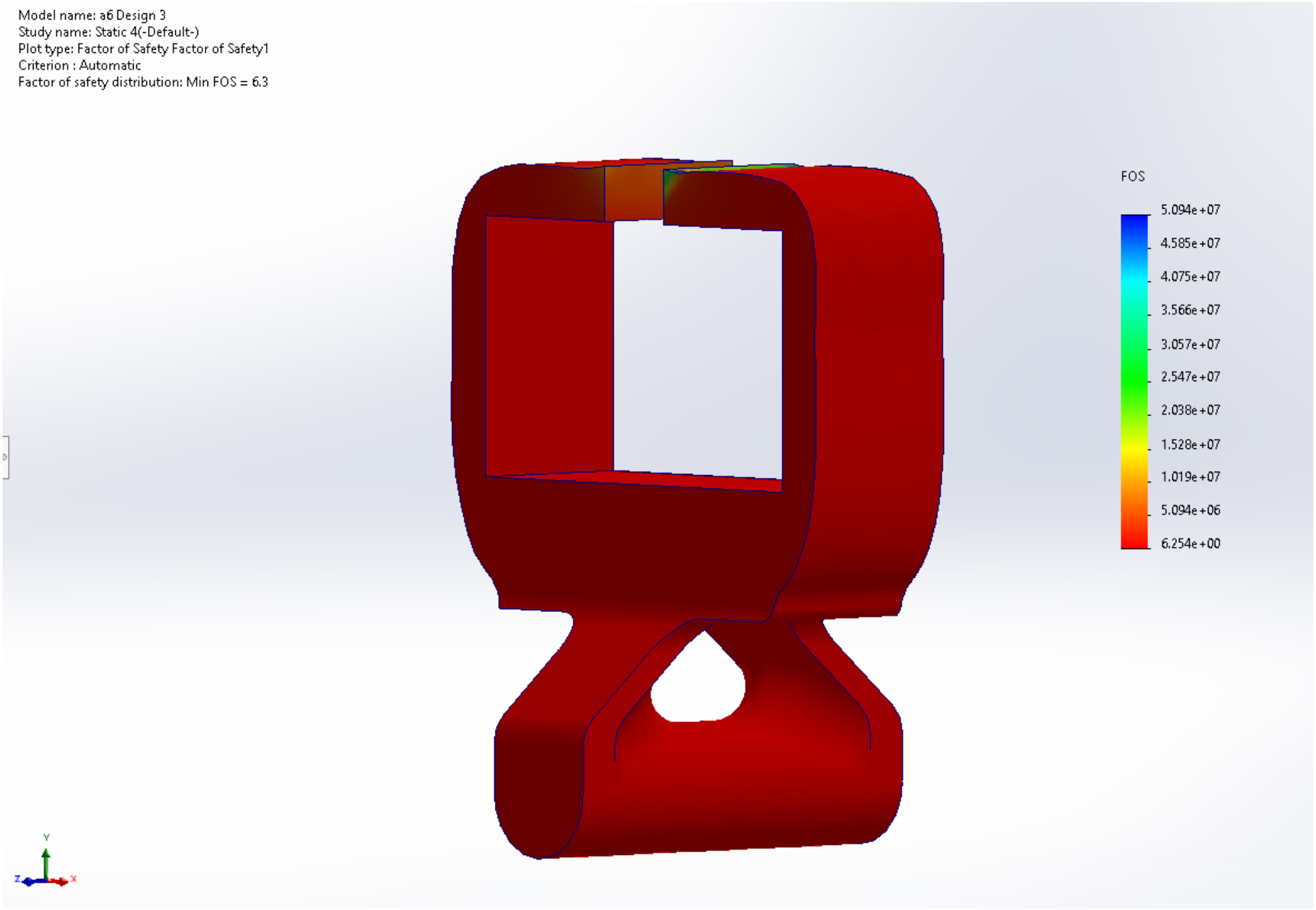
Bracket Design 3: Design 3 Coarse Displacement.



Source: Design 3/Design 3 Coarse Displacement.png

A7 - Figure 44

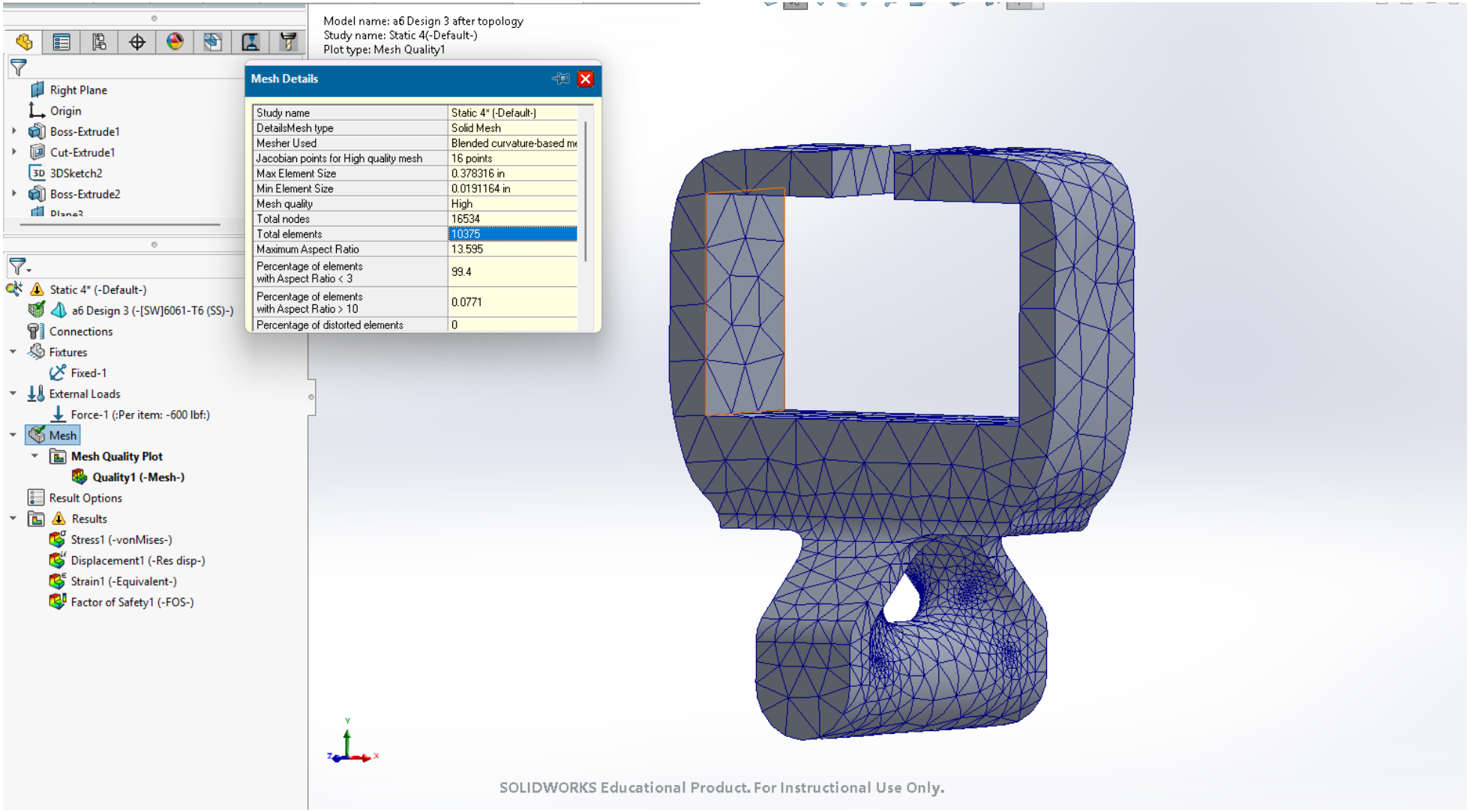
Bracket Design 3: Design 3 Coarse FOS.



Source: Design 3/Design 3 Coarse FOS.png

A7 - Figure 45

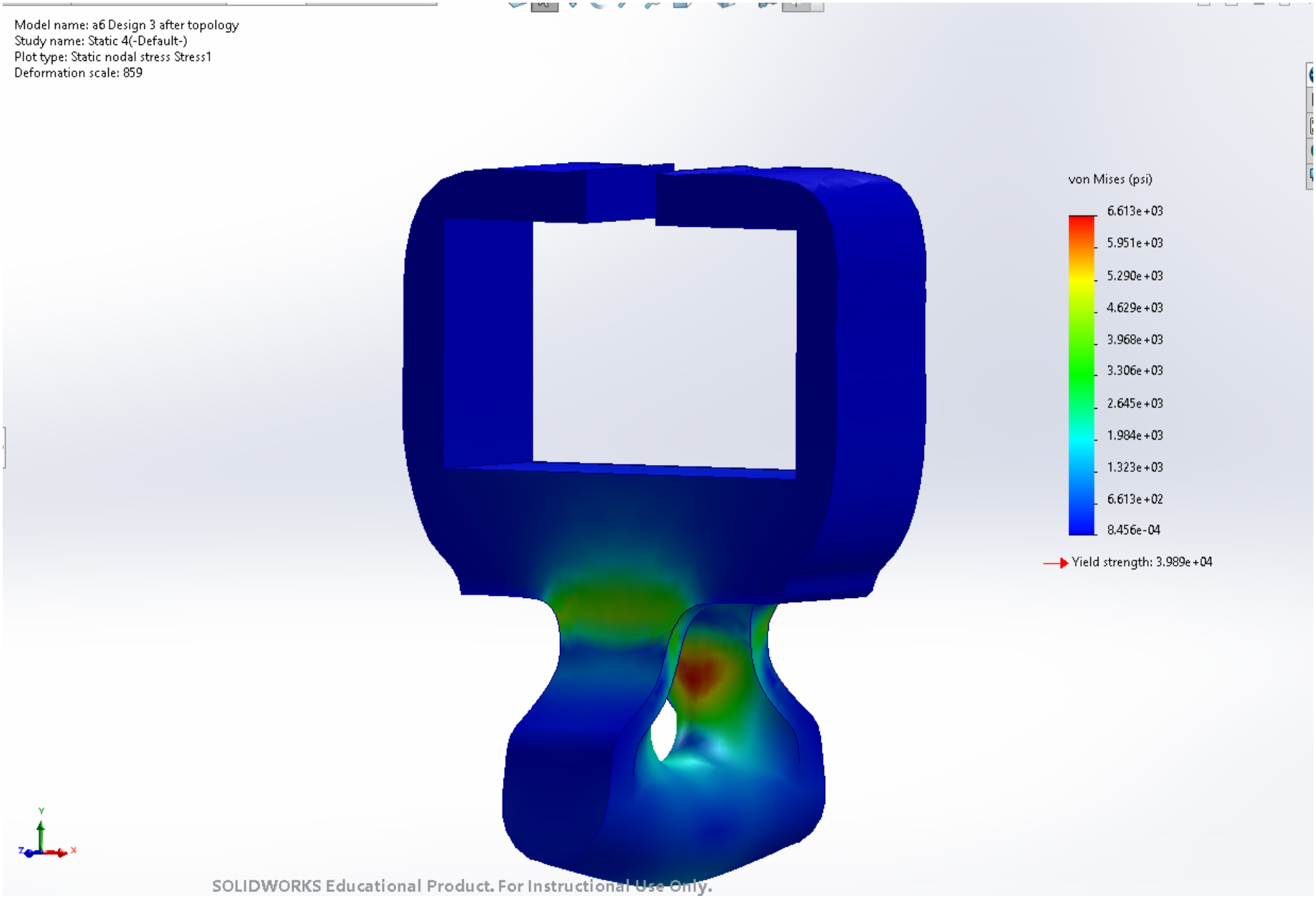
Bracket Design 3: Design 3 Medium mesh.



Source: Design 3/Design 3 Medium mesh.png

A7 - Figure 46

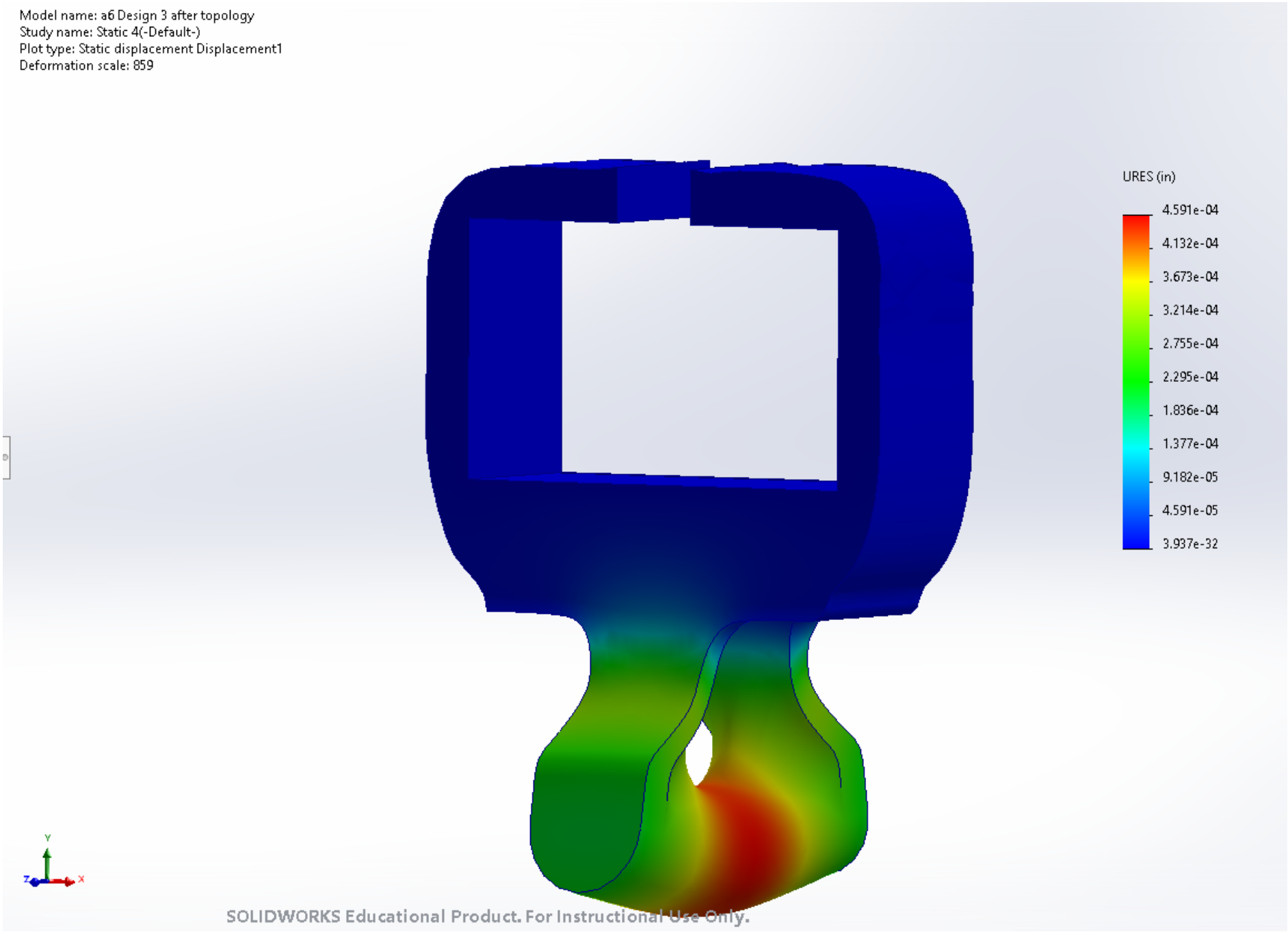
Bracket Design 3: Design 3 Medium Stress.



Source: Design 3/Design 3 Medium Stress.png

A7 - Figure 47

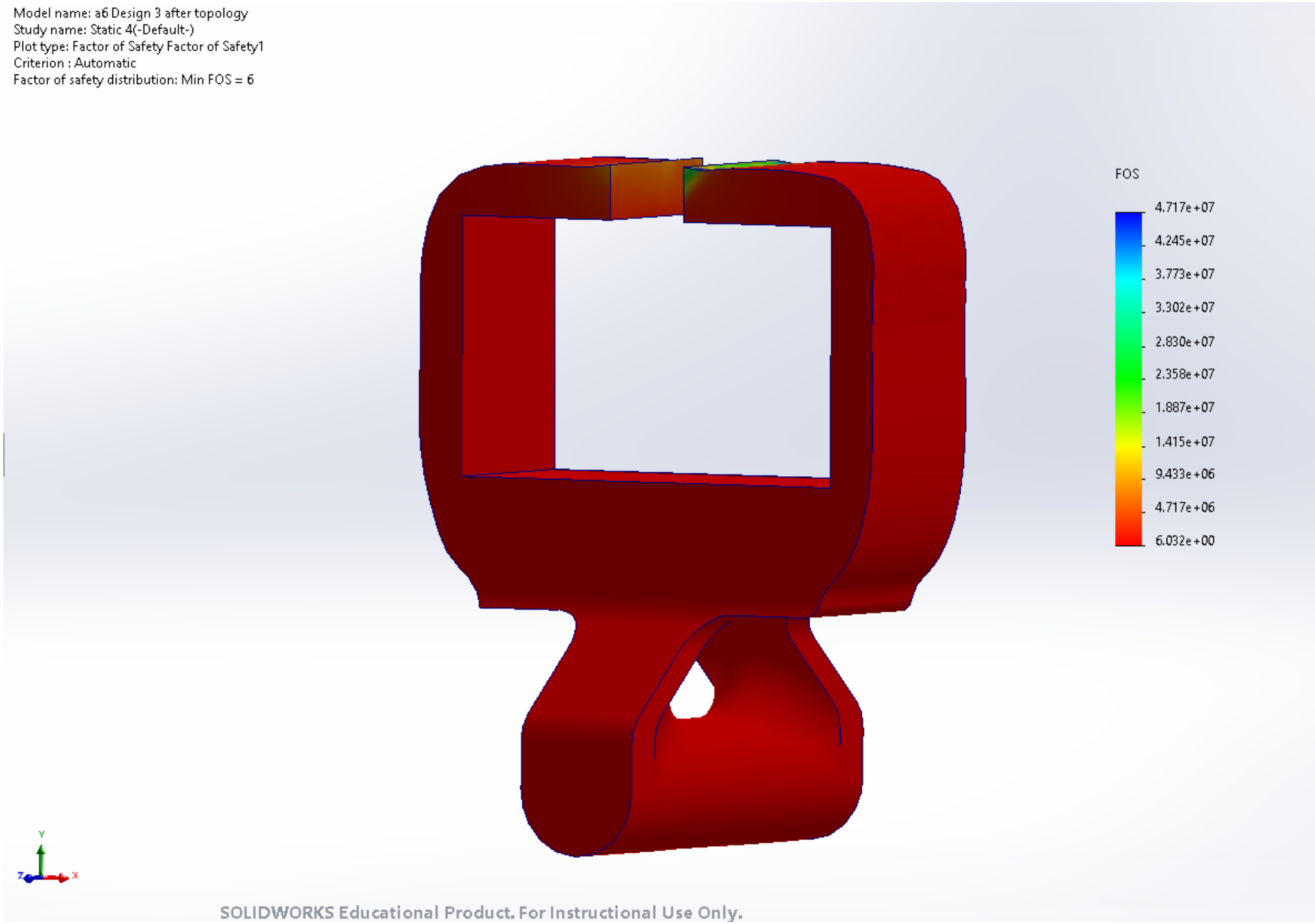
Bracket Design 3: Design 3 Medium Displacement.



Source: Design 3/Design 3 Medium Displacement.png

A7 - Figure 48

Bracket Design 3: Design 3 Medium FOS.



Source: Design 3/Design 3 Medium FOS.png

Bracket Design 3: Mass after optimization.

Mass properties of a6 Design 3 after topology
Configuration: Default
Coordinate system: -- default --

Density = 0.10 pounds per cubic inch

Mass = 0.50 pounds

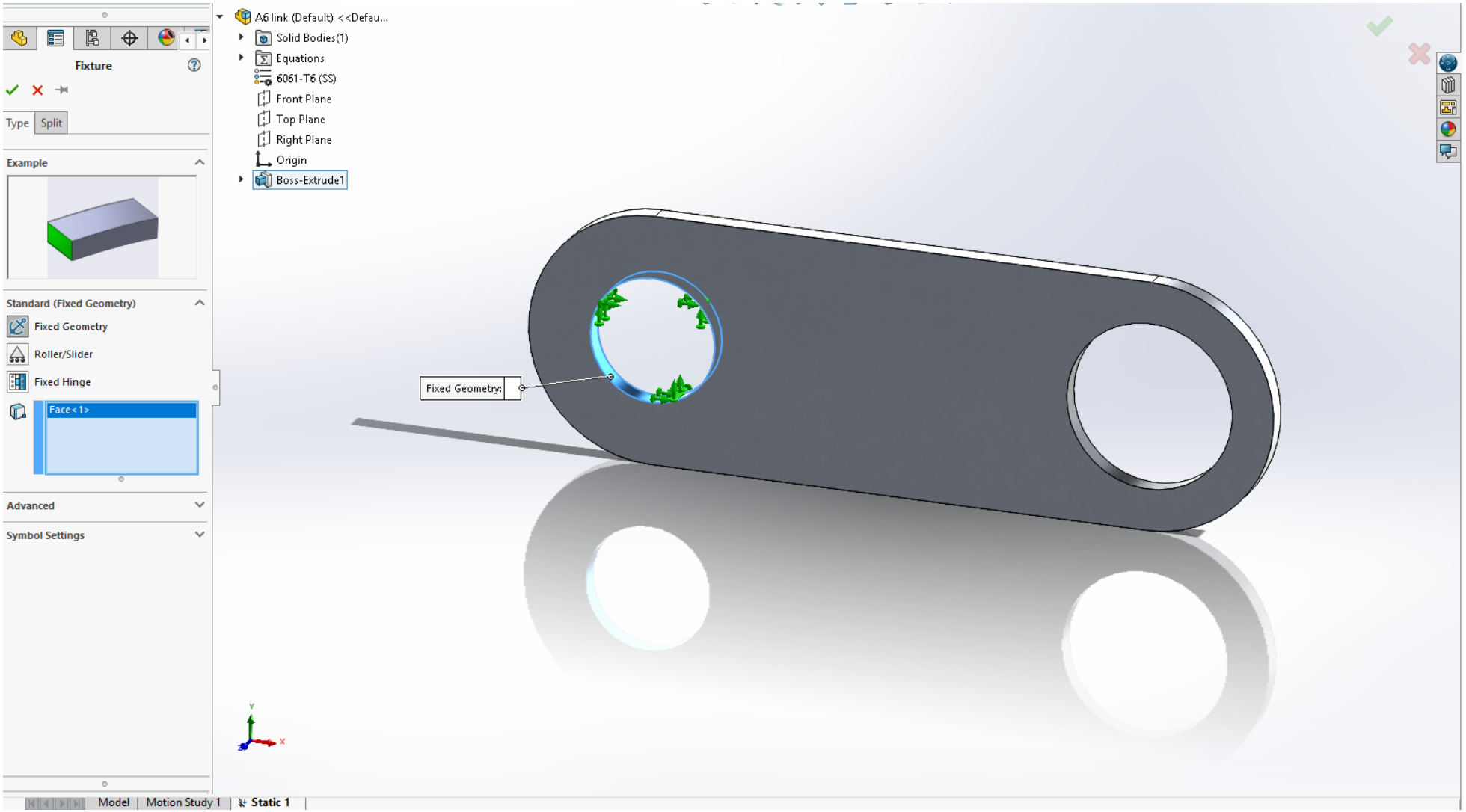
Volume = 5.10 cubic inches

Surface area = 34.35 square inches

Center of mass: (inches)
X = 2.26
Y = 0.48
Z = 0.50

A7 - Figure 50

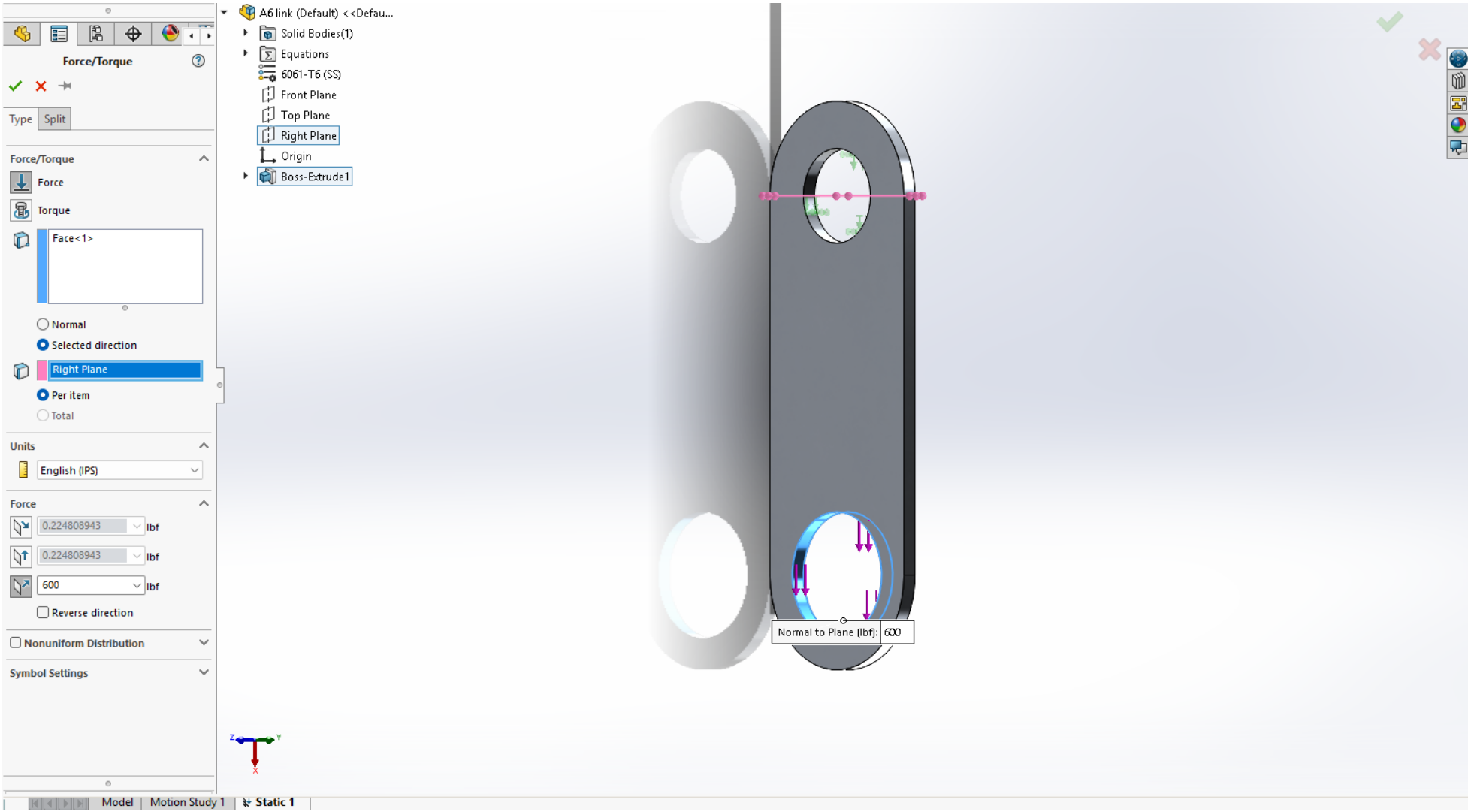
Link fixture on the inner cylindrical face of the smaller hole.



Source: Link/Design 1/Fixed geometry for smaller hole.png

A7 - Figure 51

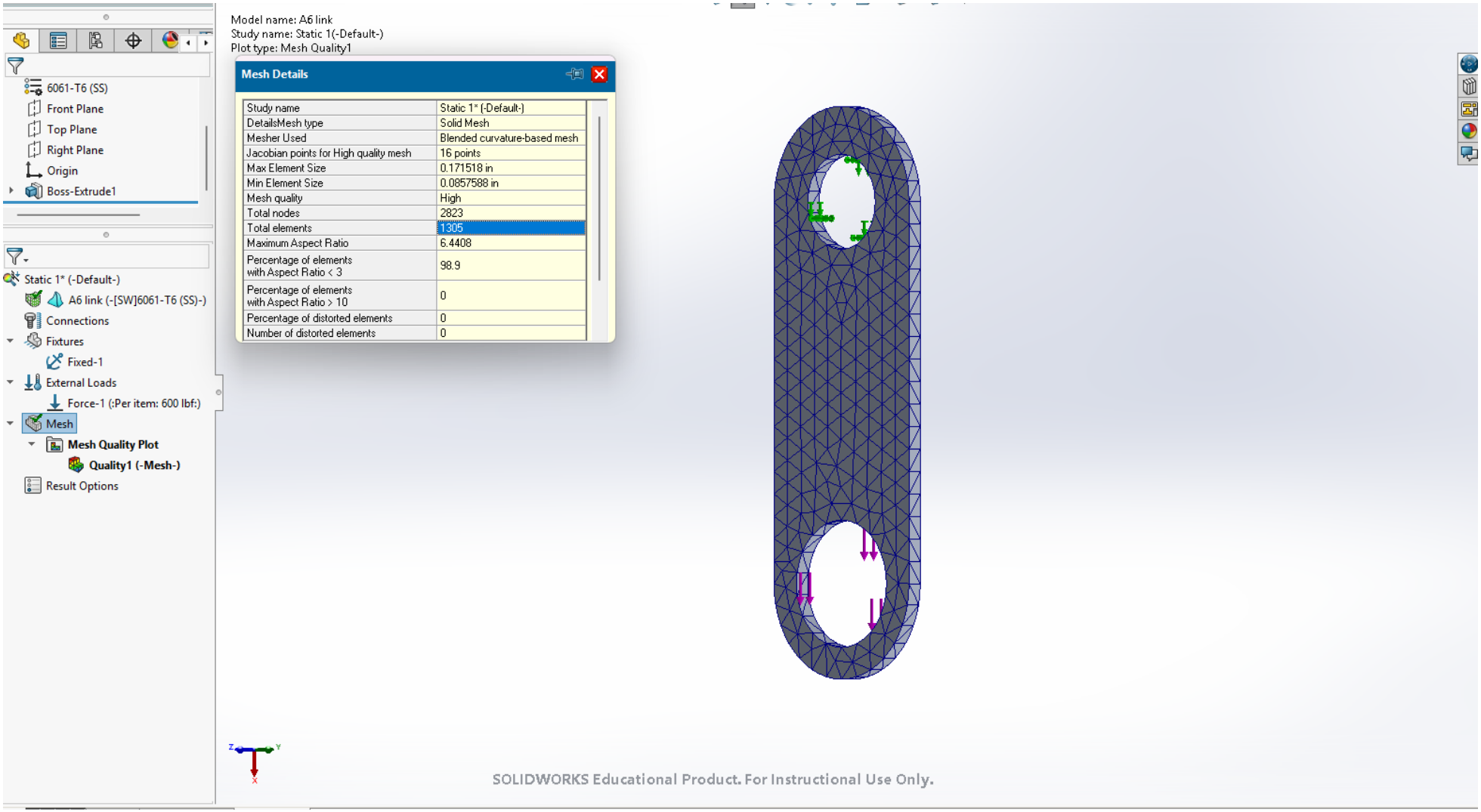
Link force definition: 600 lbf applied to the larger-hole face along the link centerline.



Source: Link/Design 1/Force on larger hole.png

A7 - Figure 52

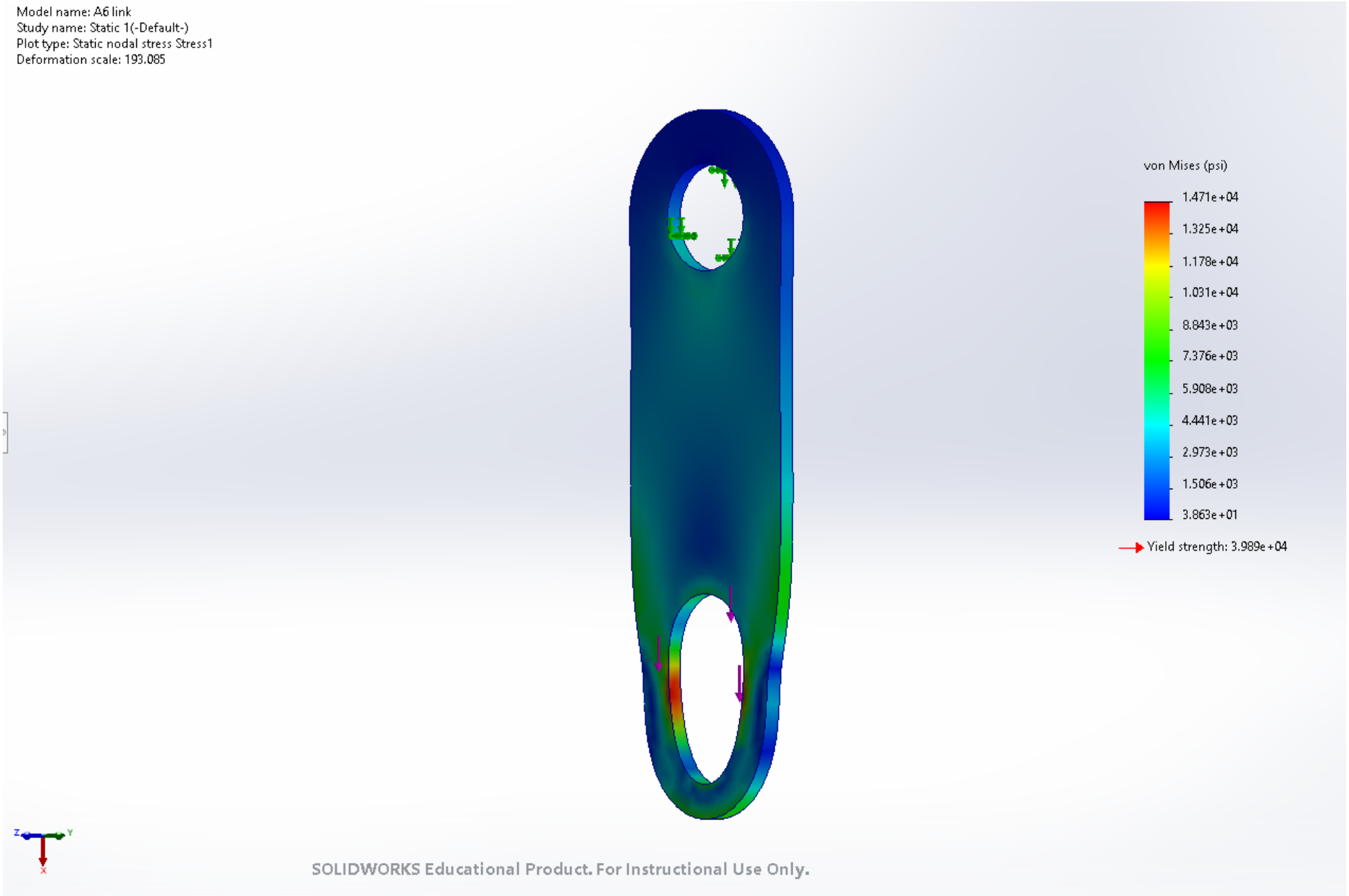
Link Design 1: Coarse Mesh Design 1.



Source: Link/Design 1/Coarse Mesh Design 1.png

A7 - Figure 53

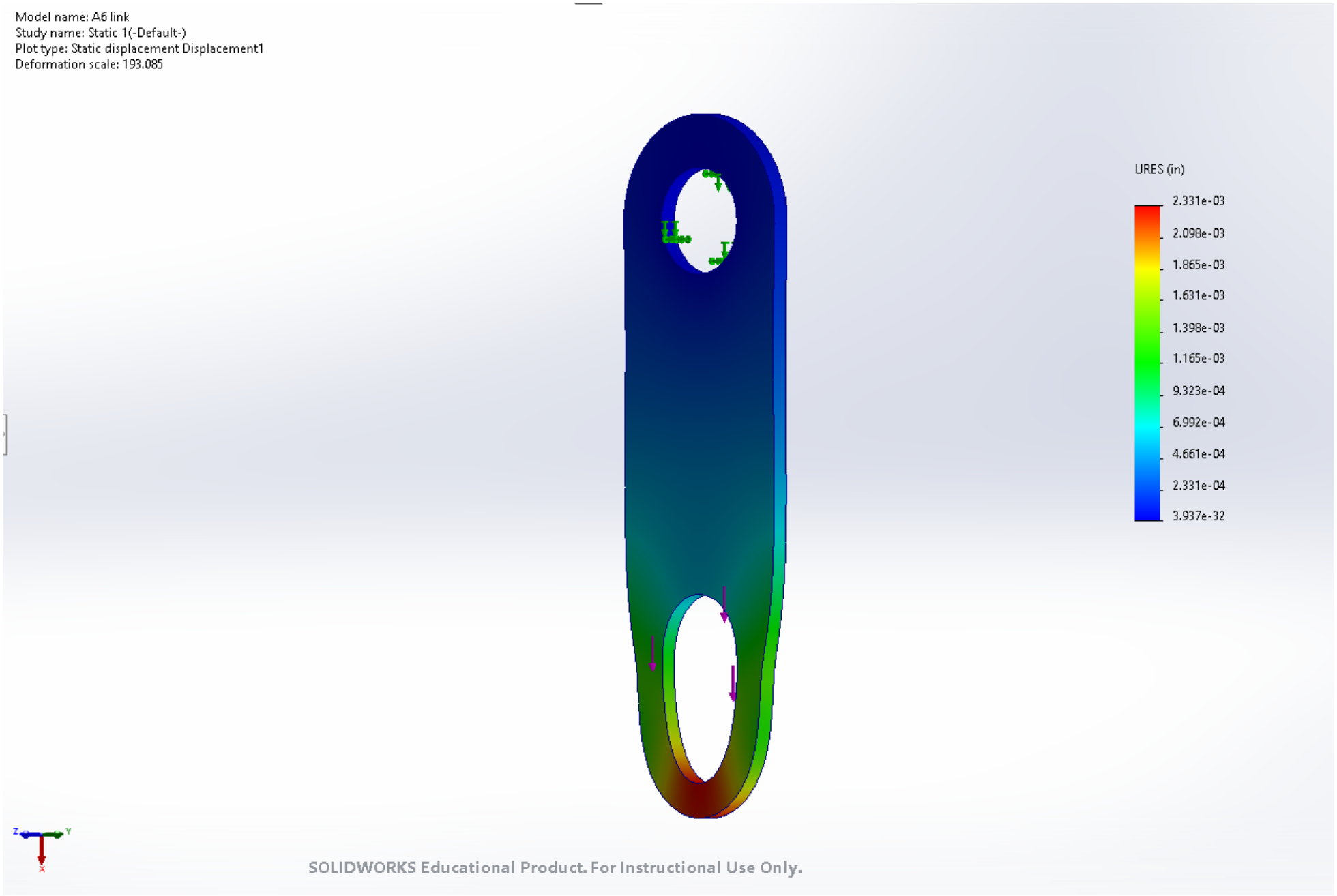
Link Design 1: Design 1 Coarse Stress.



Source: Link/Design 1/Design 1 Coarse Stress.png

A7 - Figure 54

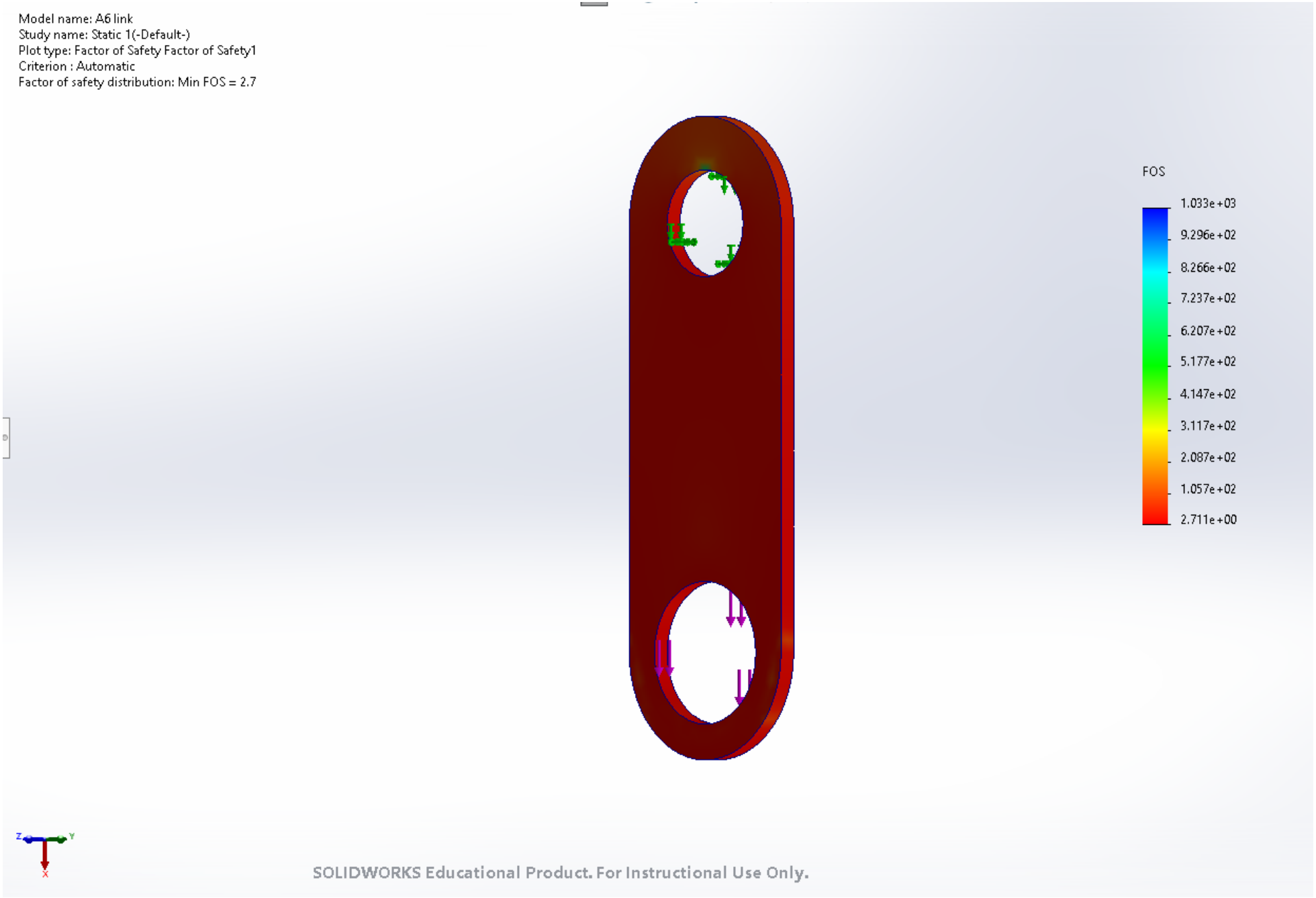
Link Design 1: Design 1 Coarse Displacement.



Source: Link/Design 1/Design 1 Coarse Displacement.png

A7 - Figure 55

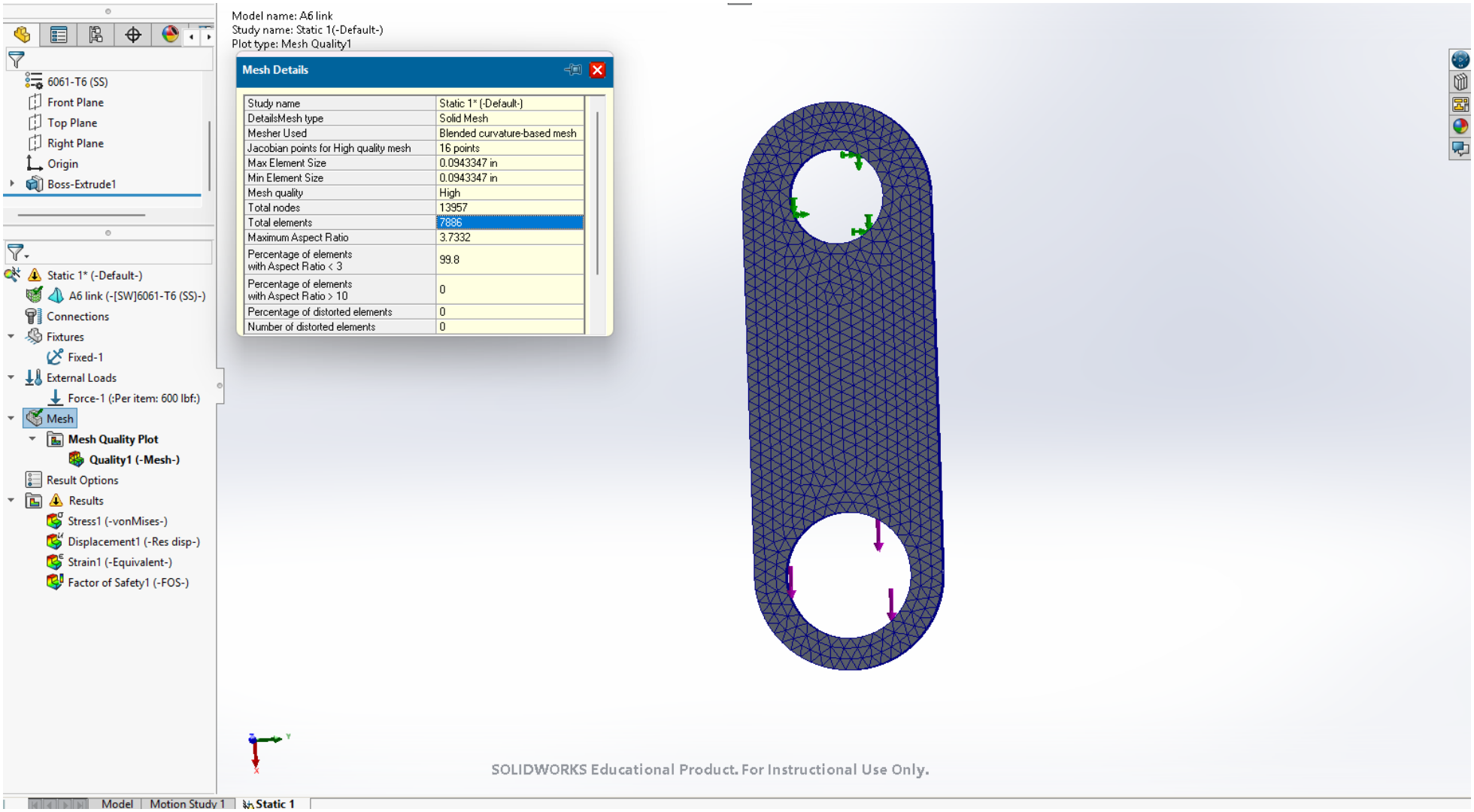
Link Design 1: Design 1 Coarse FOS.



Source: Link/Design 1/Design 1 Coarse FOS.png

A7 - Figure 56

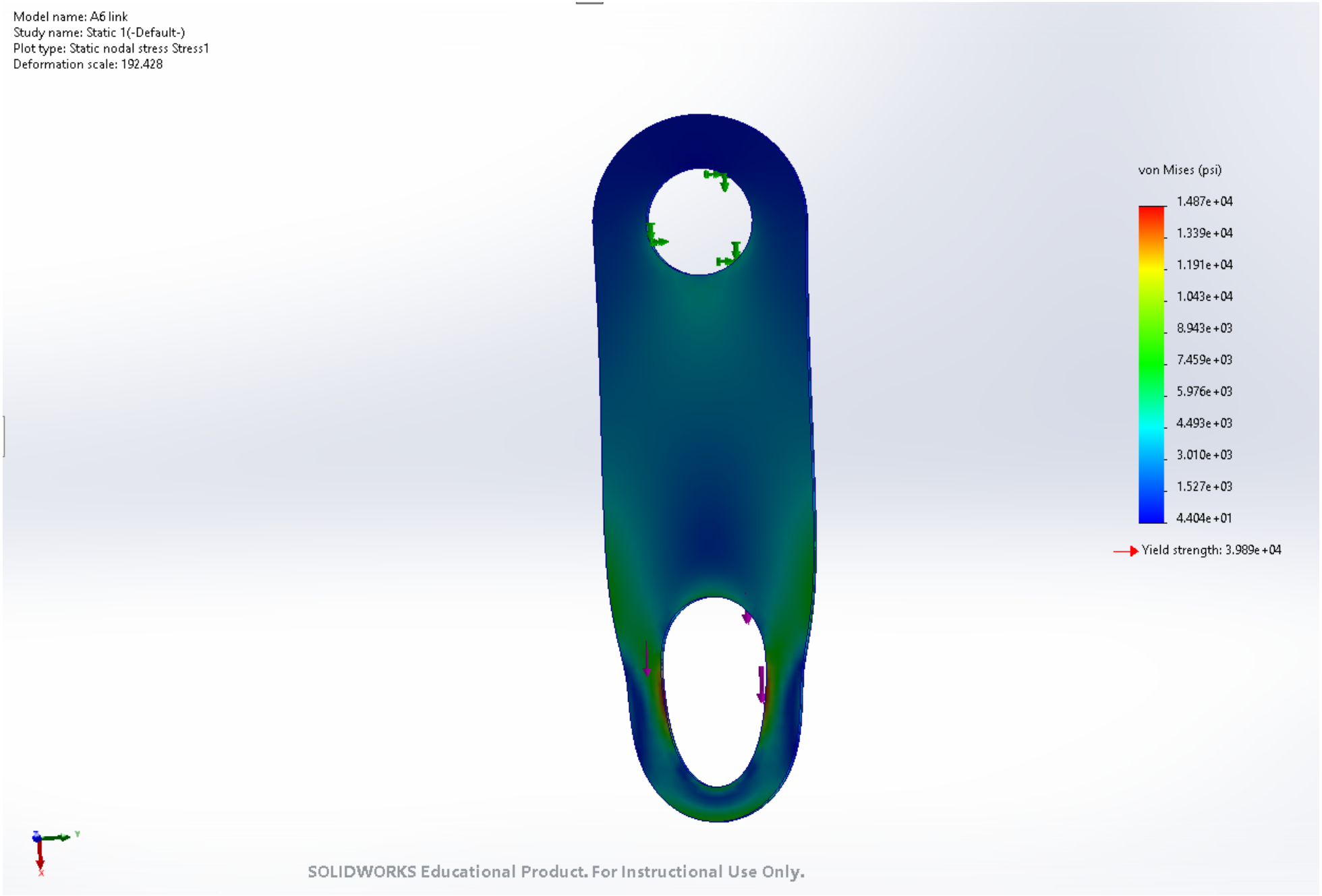
Link Design 1: Design 1 Medium mesh.



Source: Link/Design 1/Design 1 Medium mesh.png

A7 - Figure 57

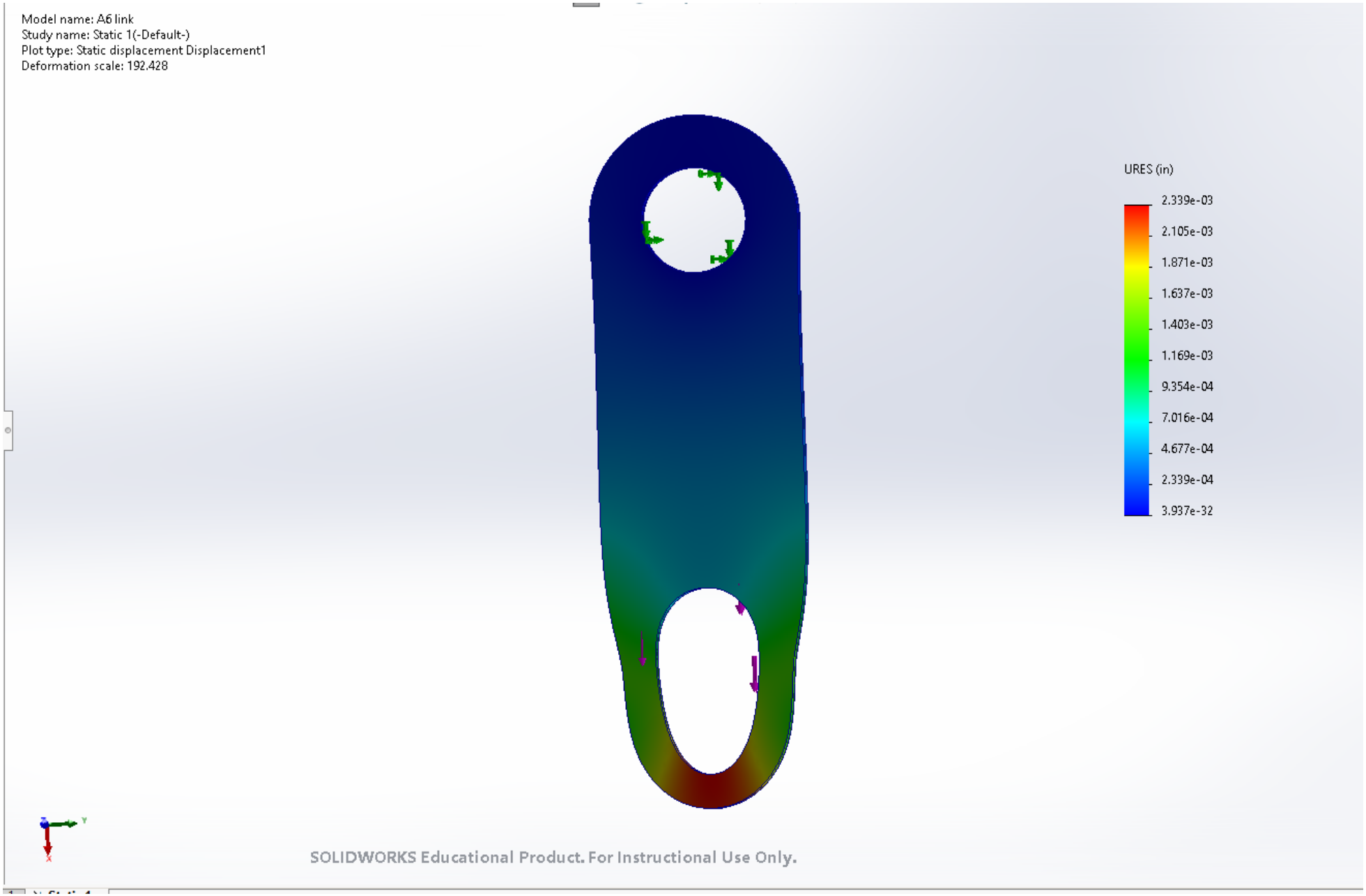
Link Design 1: Design 1 Medium Stress.



Source: Link/Design 1/Design 1 Medium Stress.png

A7 - Figure 58

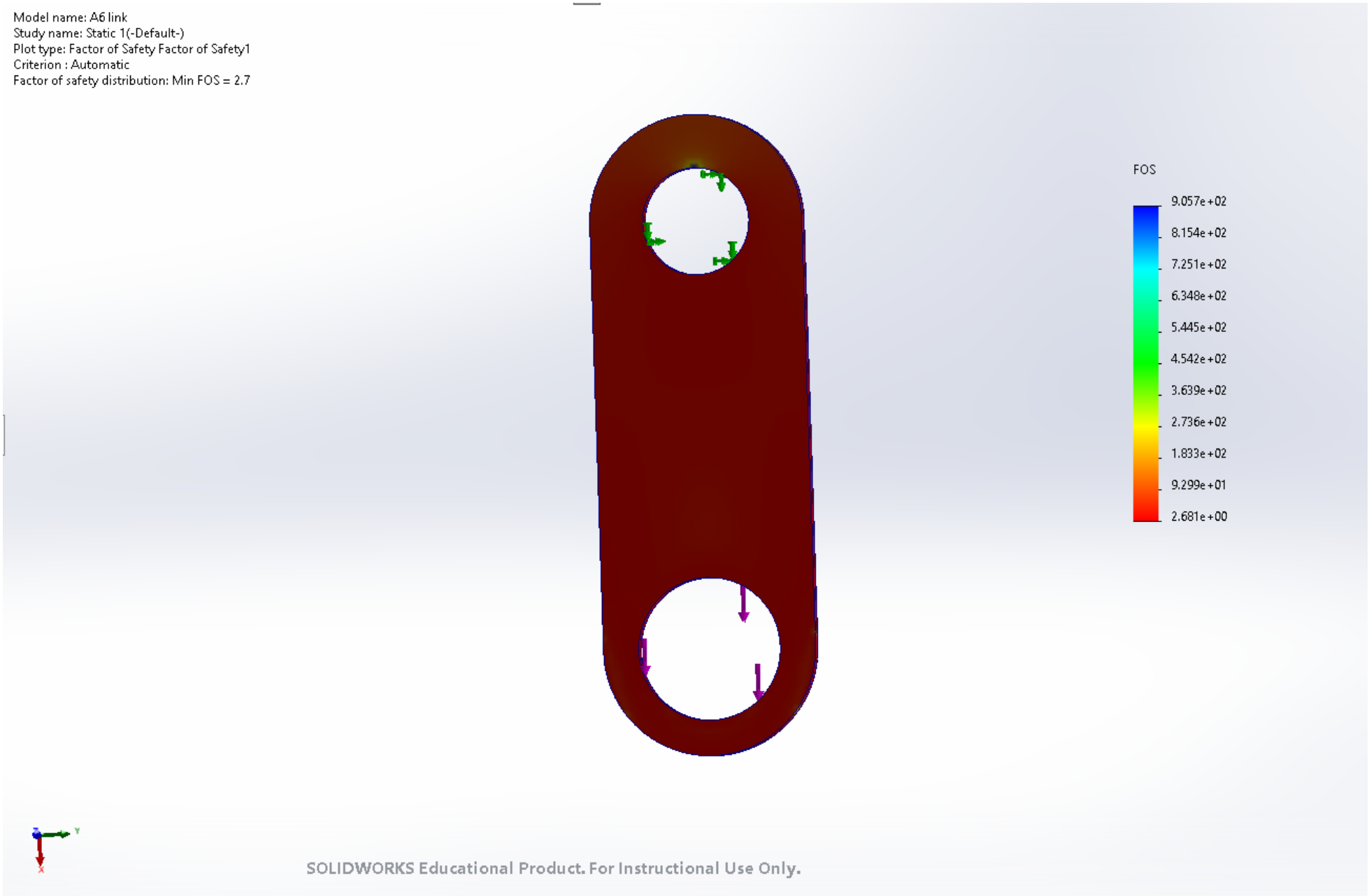
Link Design 1: Design 1 Medium Displacement.



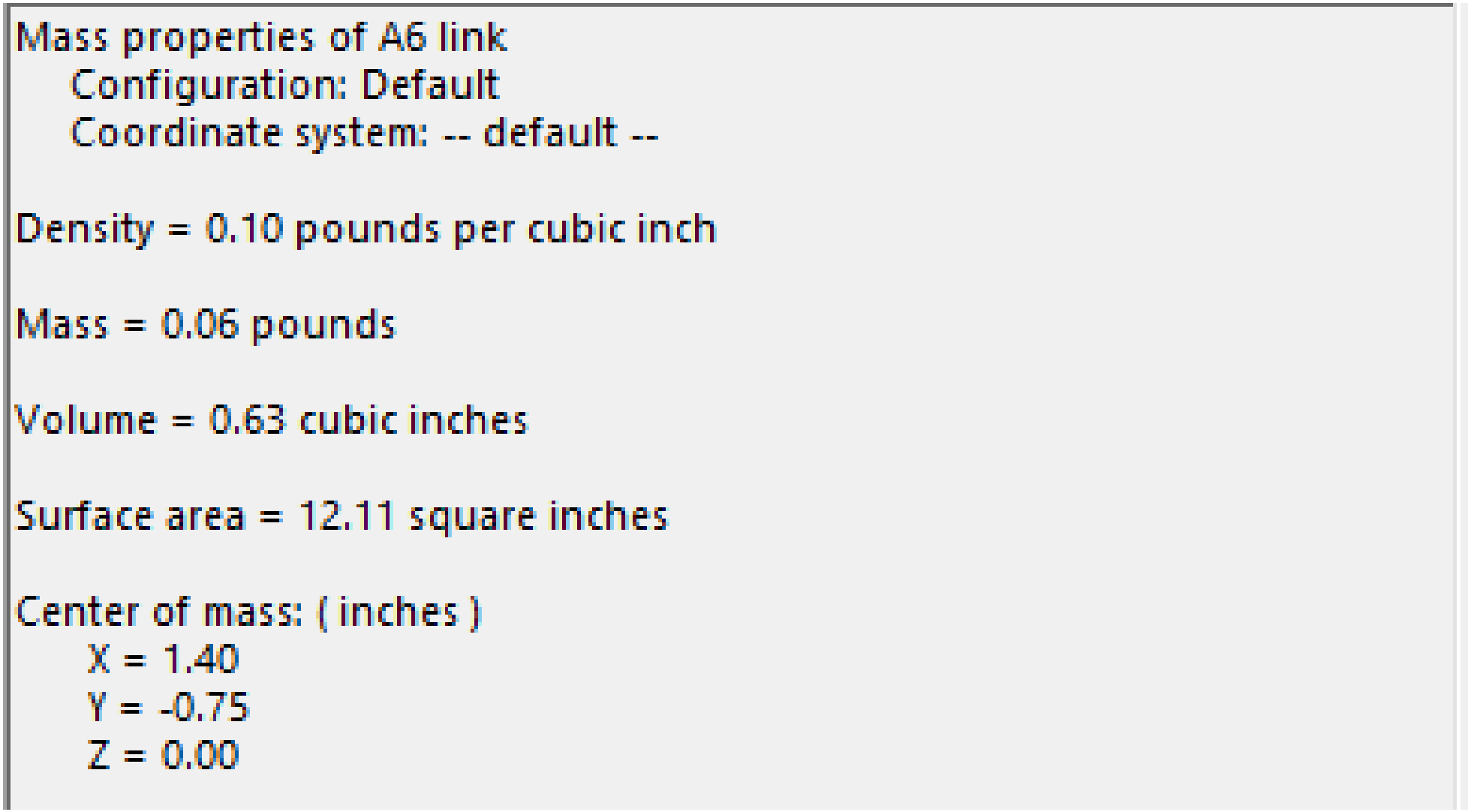
Source: Link/Design 1/Design 1 Medium Displacement.png

A7 - Figure 59

Link Design 1: Design 1 Medium FOS.

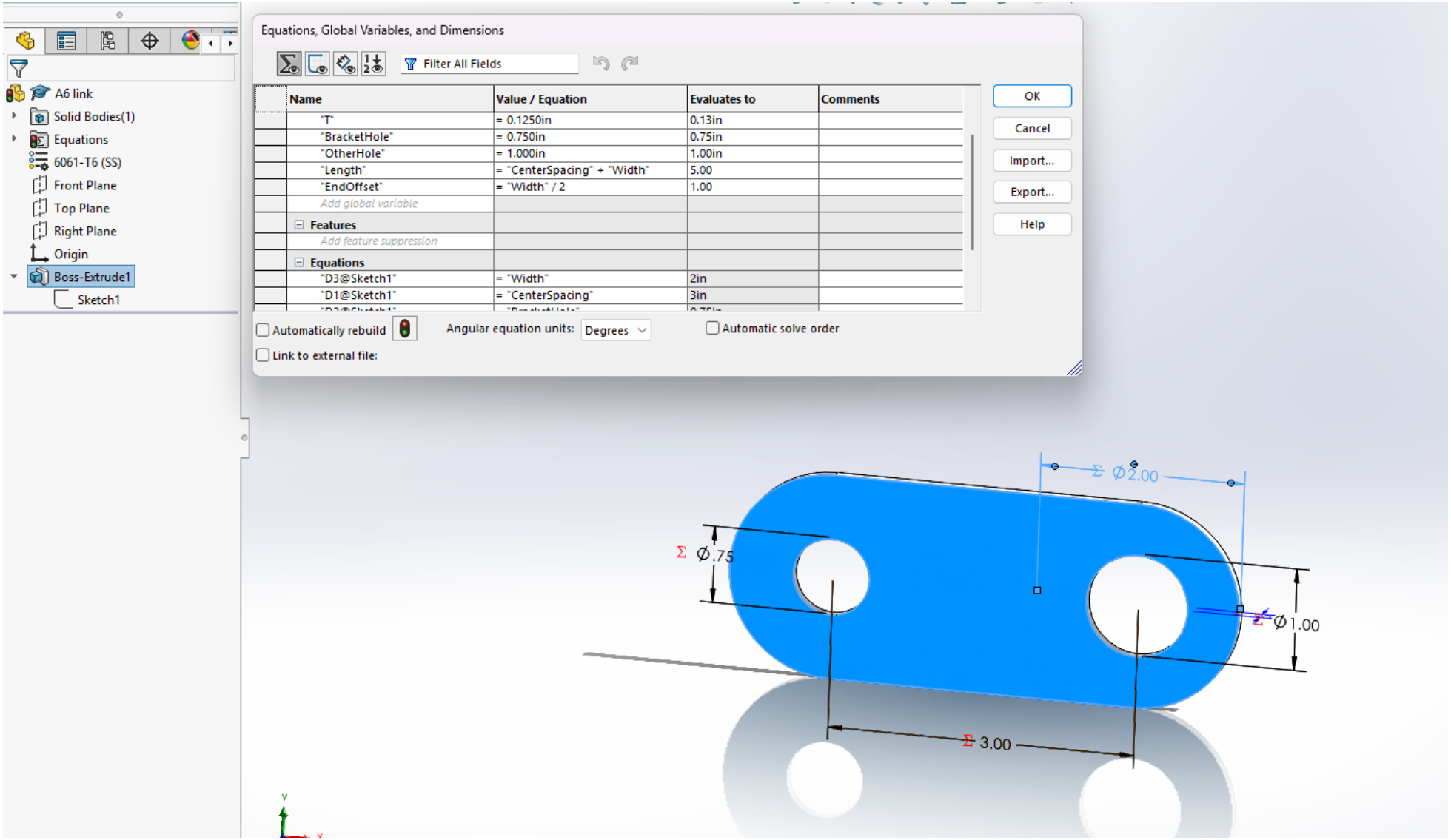


Link Design 1: Mass Design 1.



A7 - Figure 61

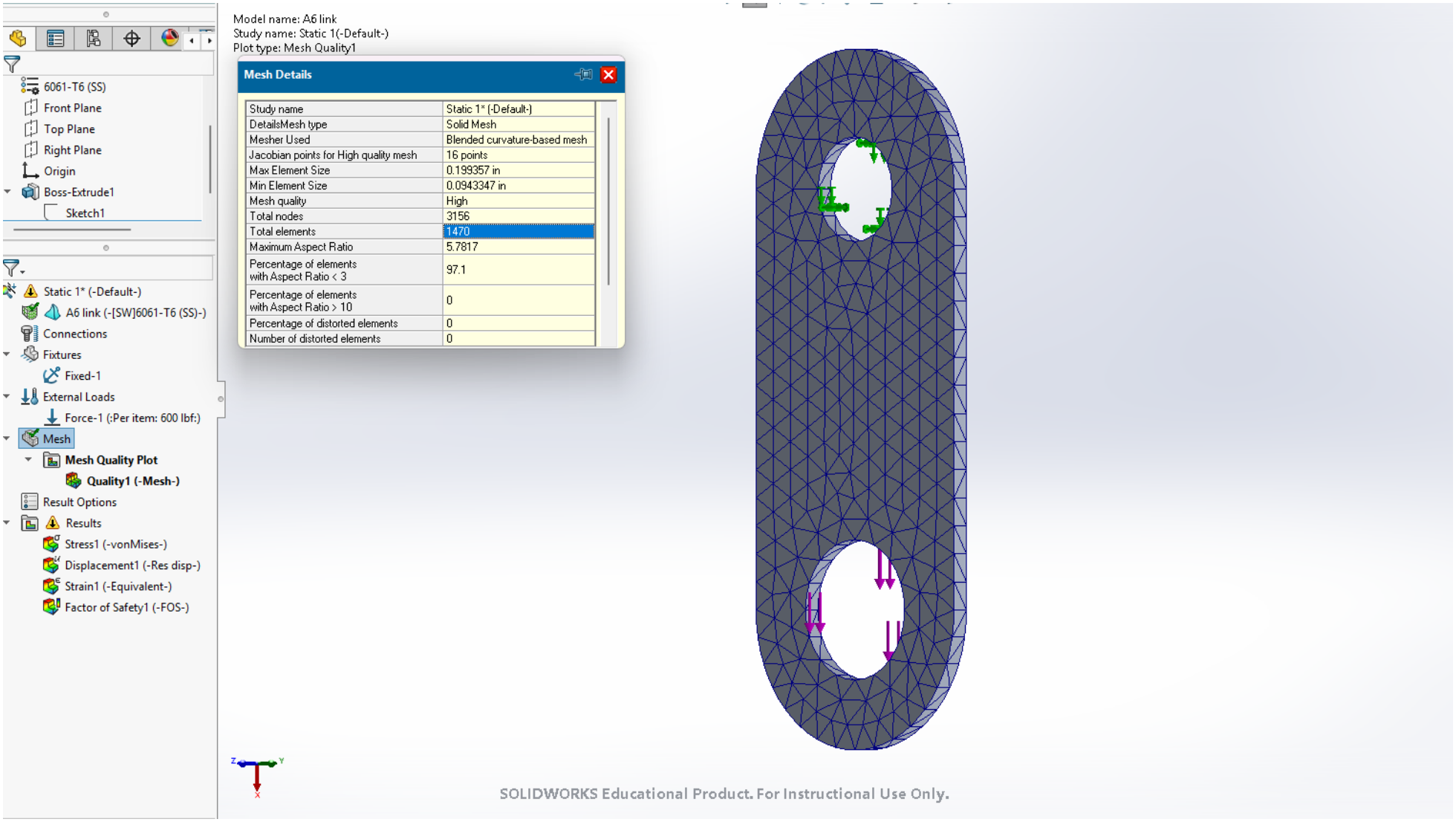
Link width increased from 1.50 to 2.00 in; thickness remains 0.125 in and hole-center spacing remains 3.00 in.



Source: Link/Design 2/Design 2 width Change.png

A7 - Figure 62

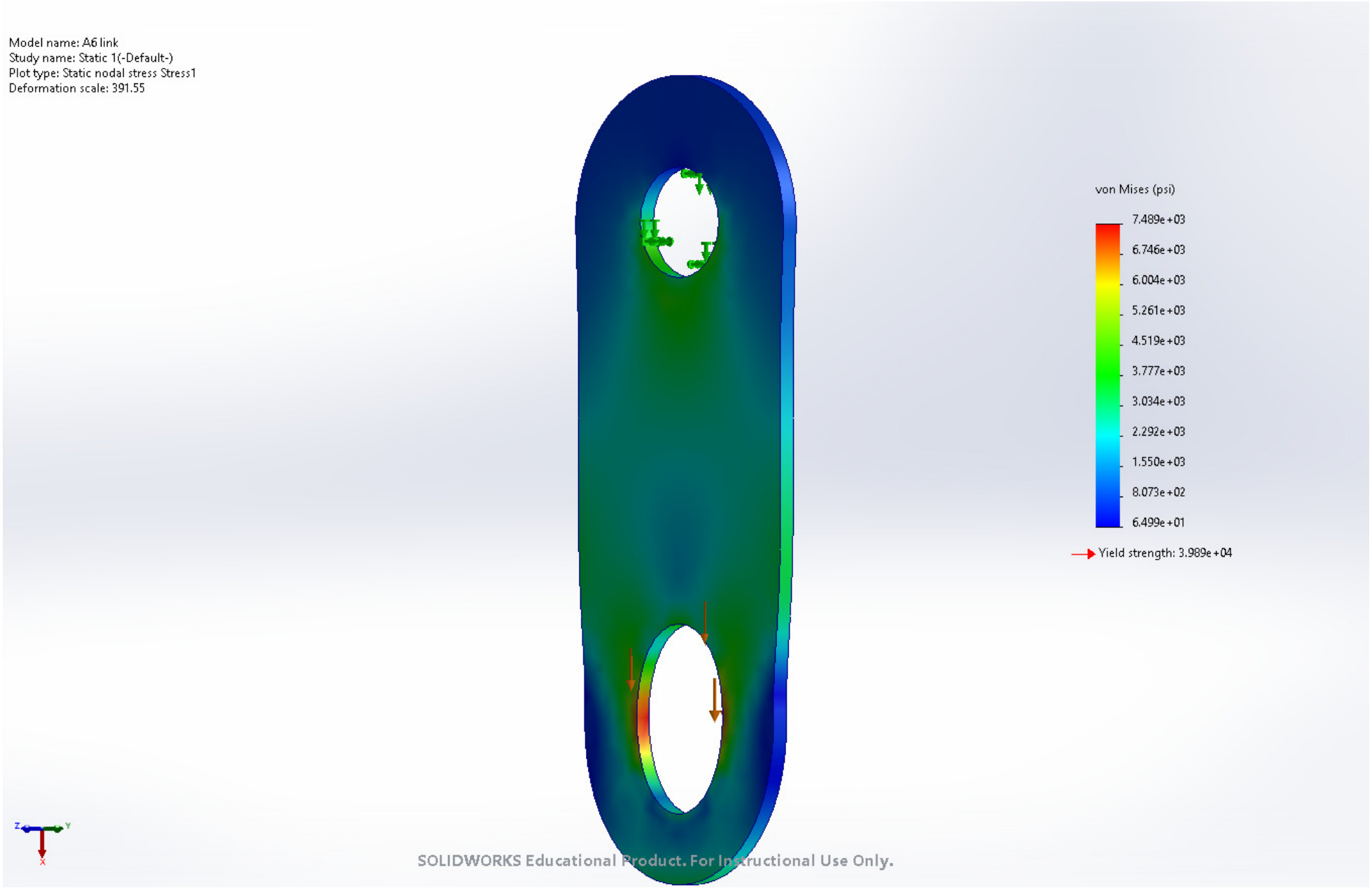
Link Design 2: Design 2 Coarse Mesh.



Source: Link/Design 2/Design 2 Coarse Mesh.png

A7 - Figure 63

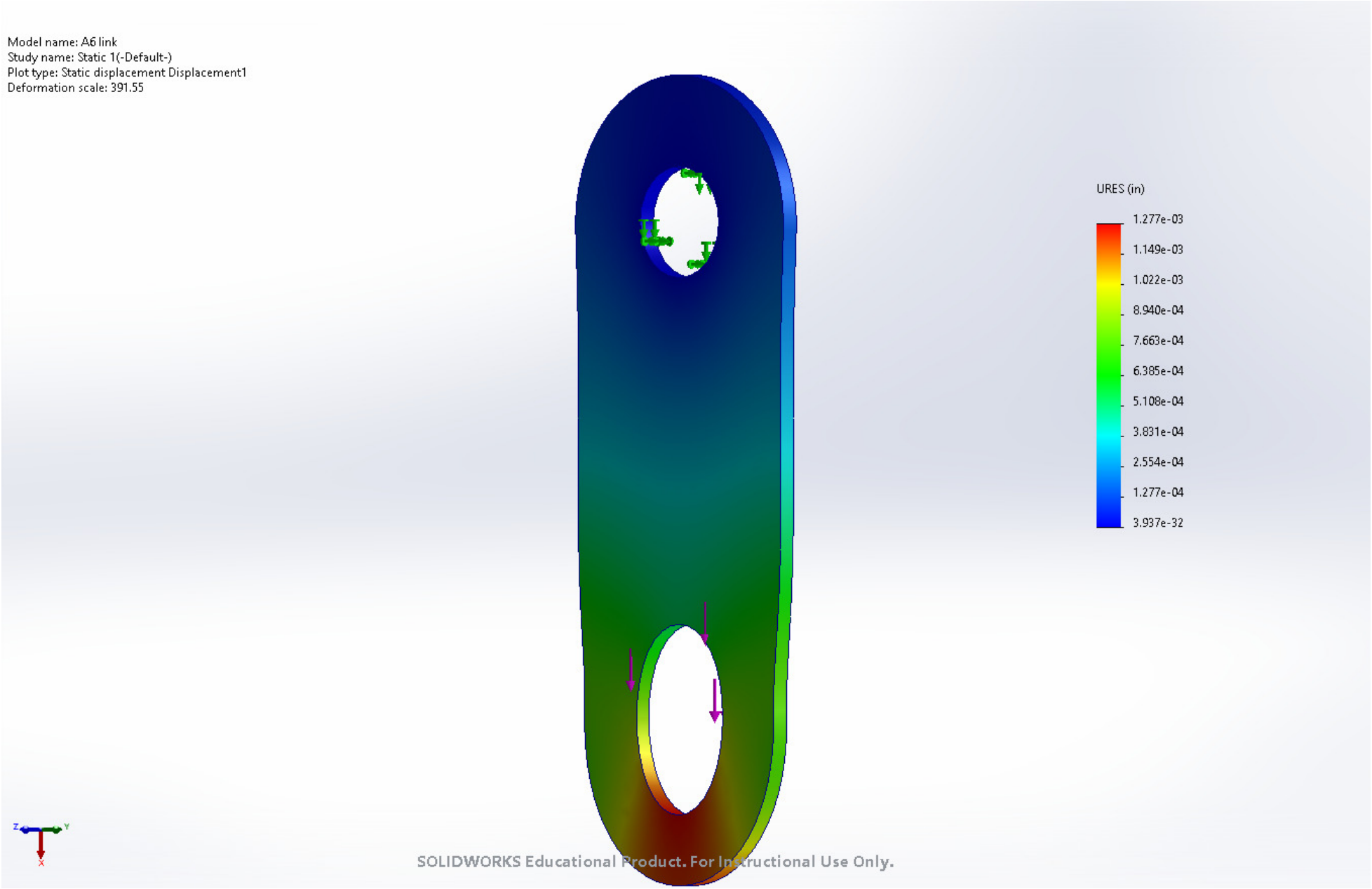
Link Design 2: Design 2 Coarse Stress.



Source: Link/Design 2/Design 2 Coarse Stress.jpg

A7 - Figure 64

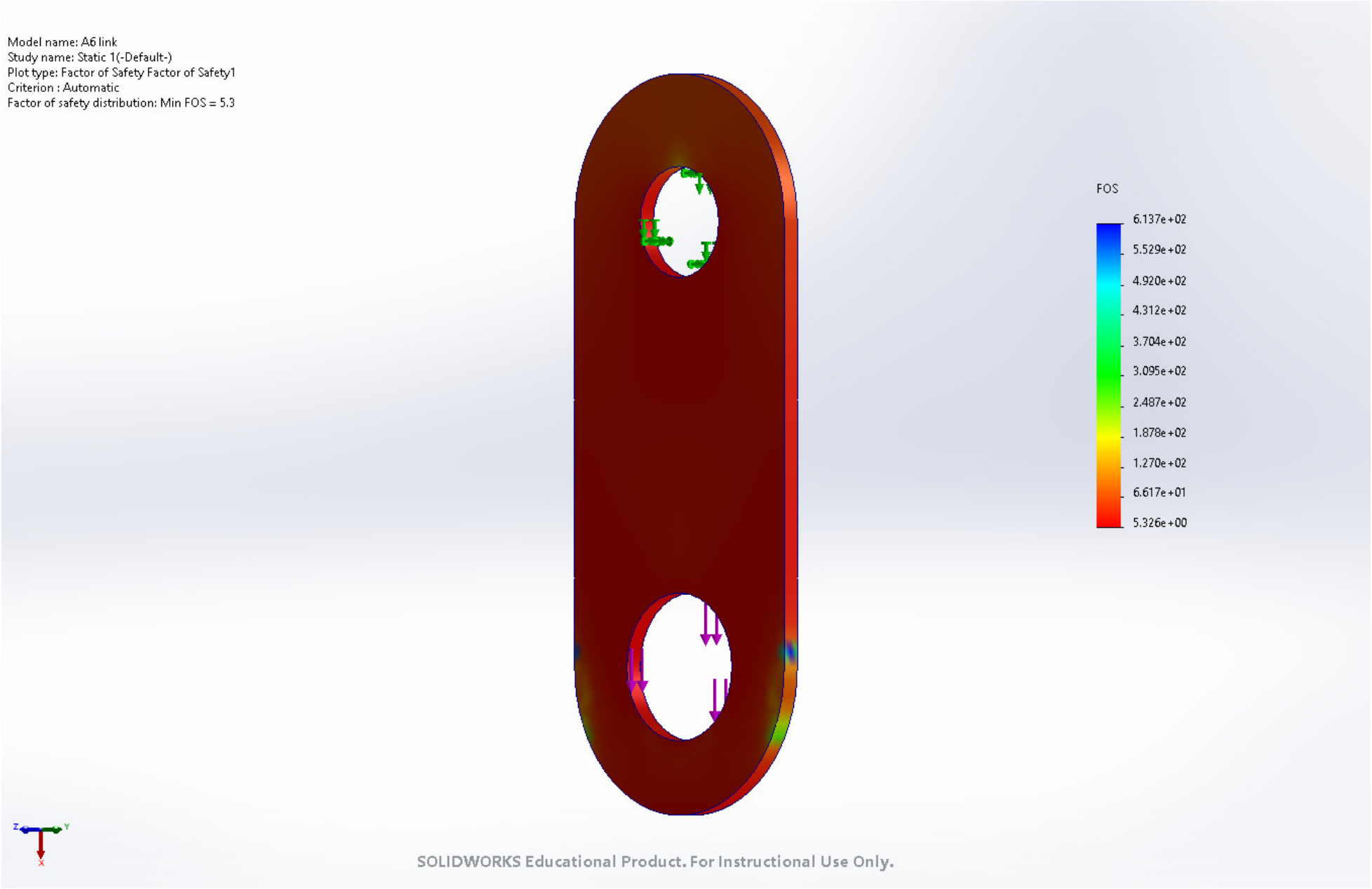
Link Design 2: Design 2 Coarse Displacement.



Source: Link/Design 2/Design 2 Coarse Displacement.jpg

A7 - Figure 65

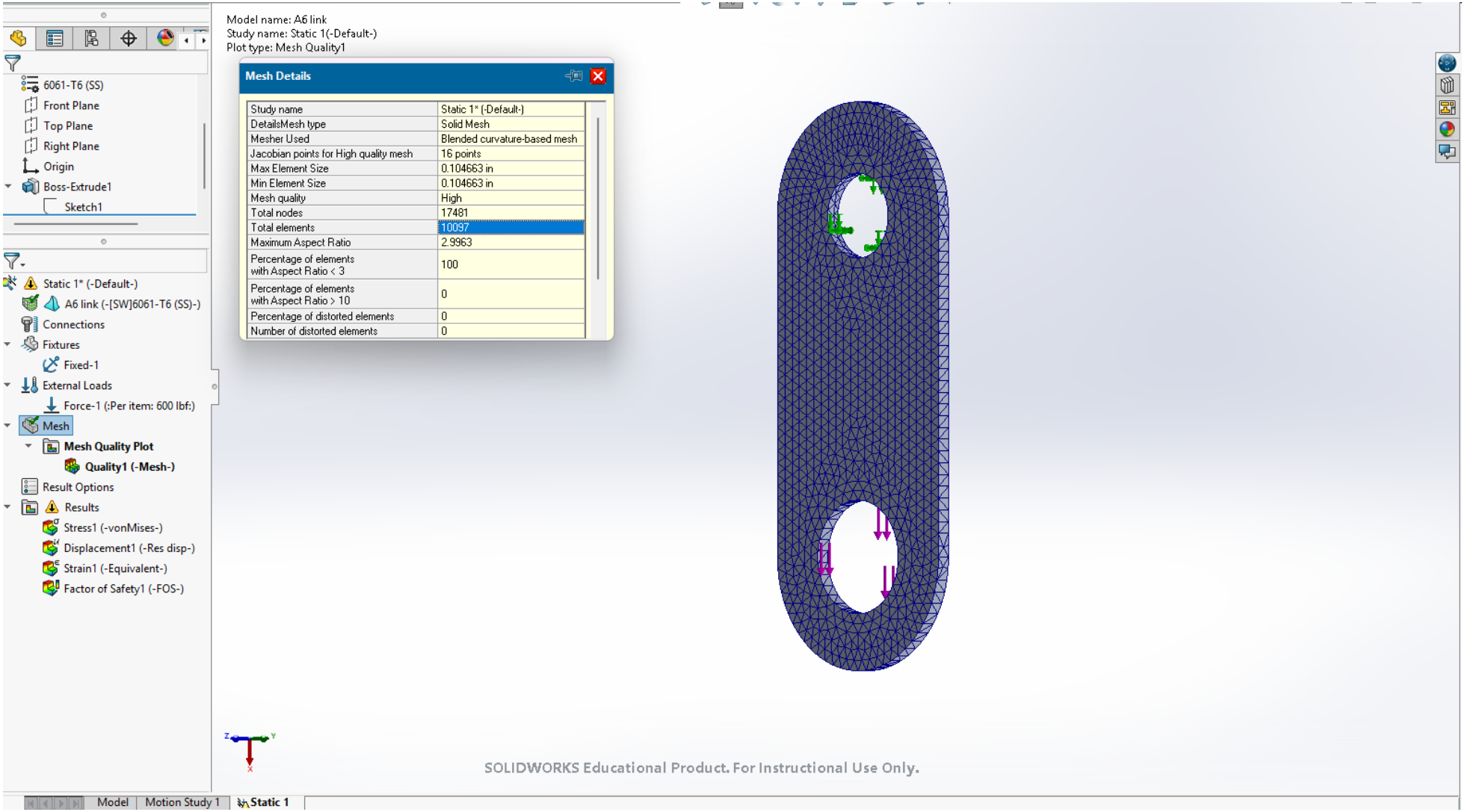
Link Design 2: Design 2 Coarse FOS.



Source: Link/Design 2/Design 2 Coarse FOS.jpg

A7 - Figure 66

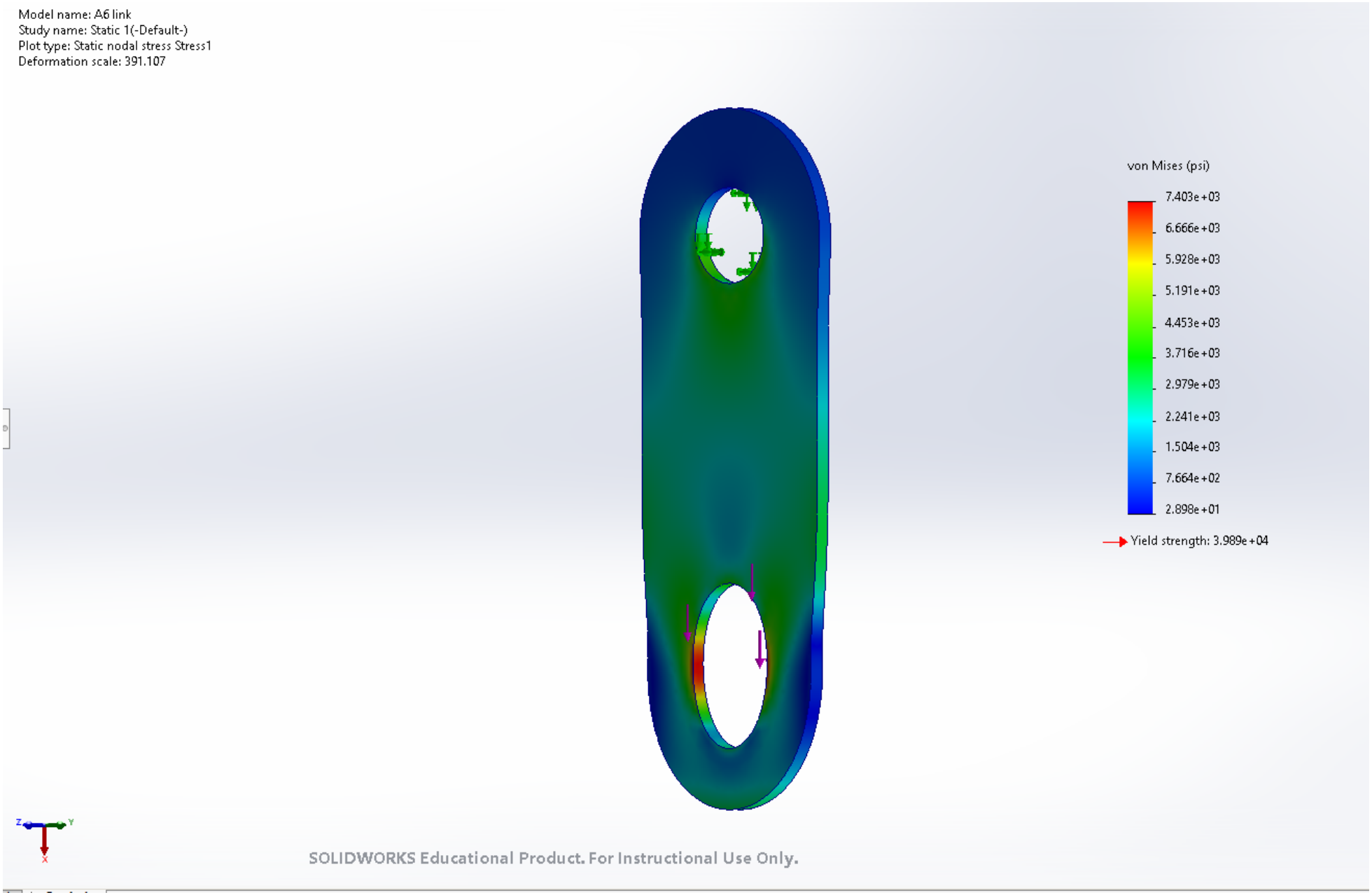
Link Design 2: Design 2 Medium Mesh.



Source: Link/Design 2/Design 2 Medium Mesh.png

A7 - Figure 67

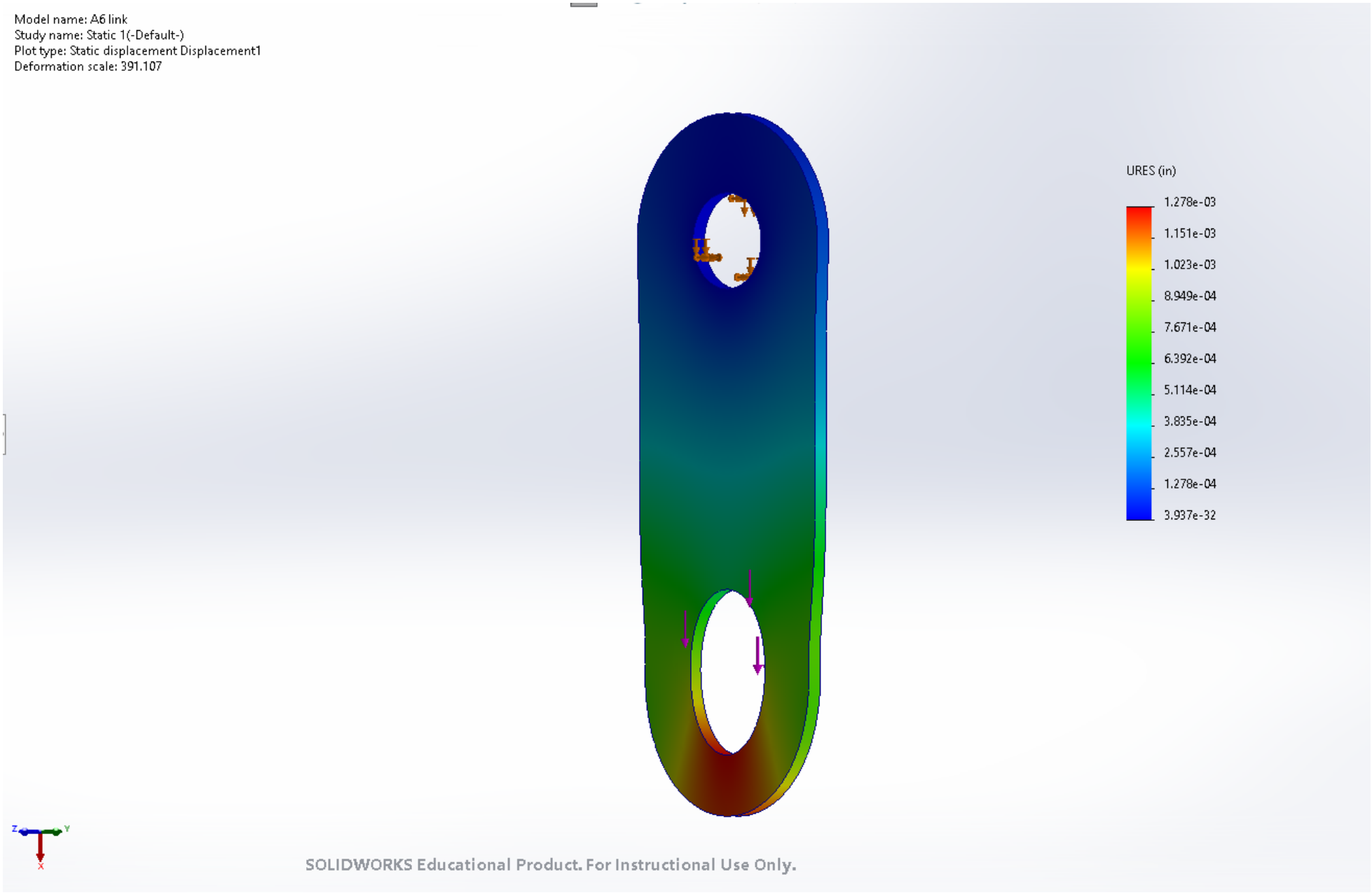
Link Design 2: Design 2 Medium Stress.



Source: Link/Design 2/Design 2 Medium Stress.png

A7 - Figure 68

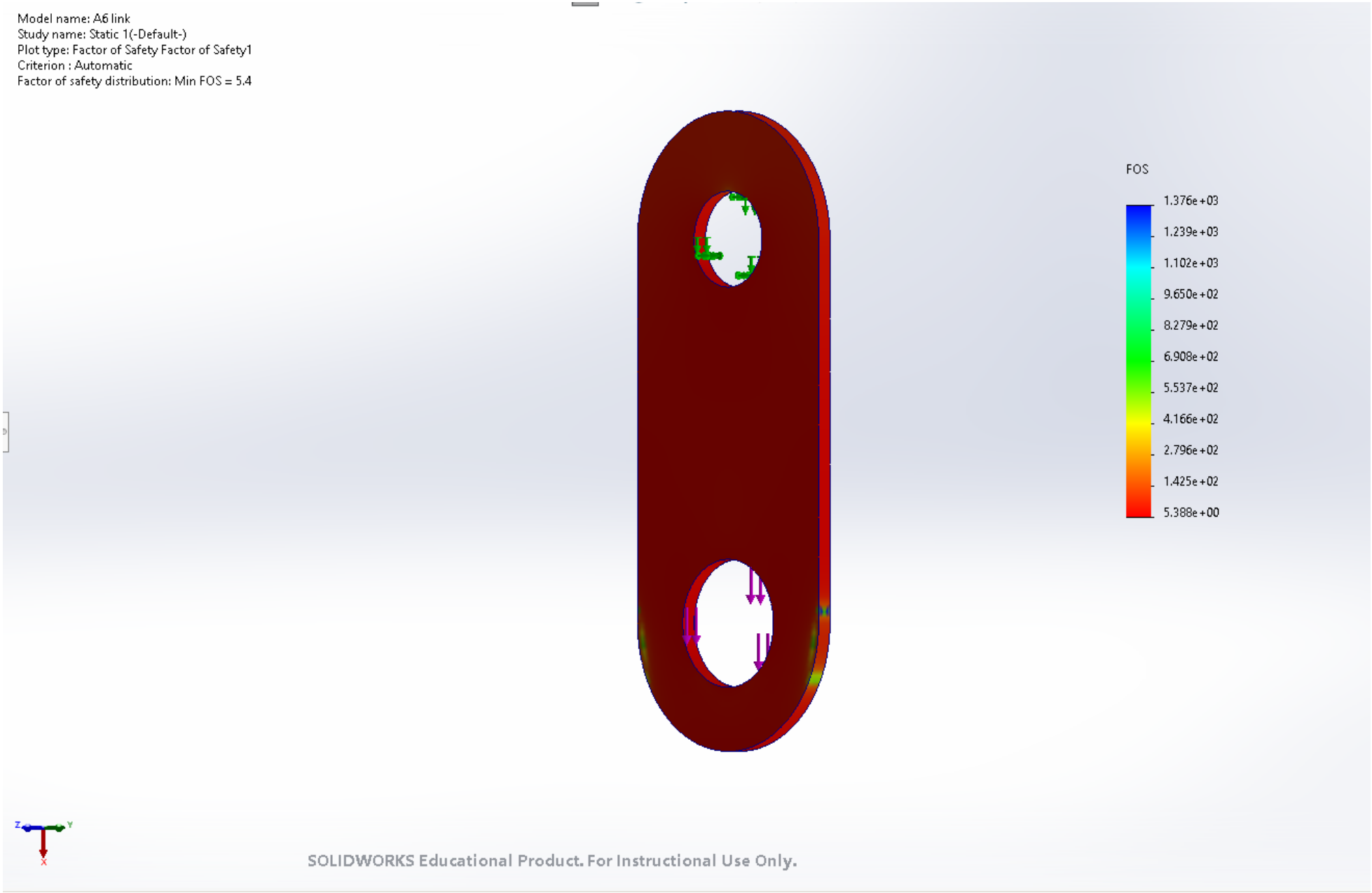
Link Design 2: Design 2 Medium Displacement.



Source: Link/Design 2/Design 2 Medium Displacement.png

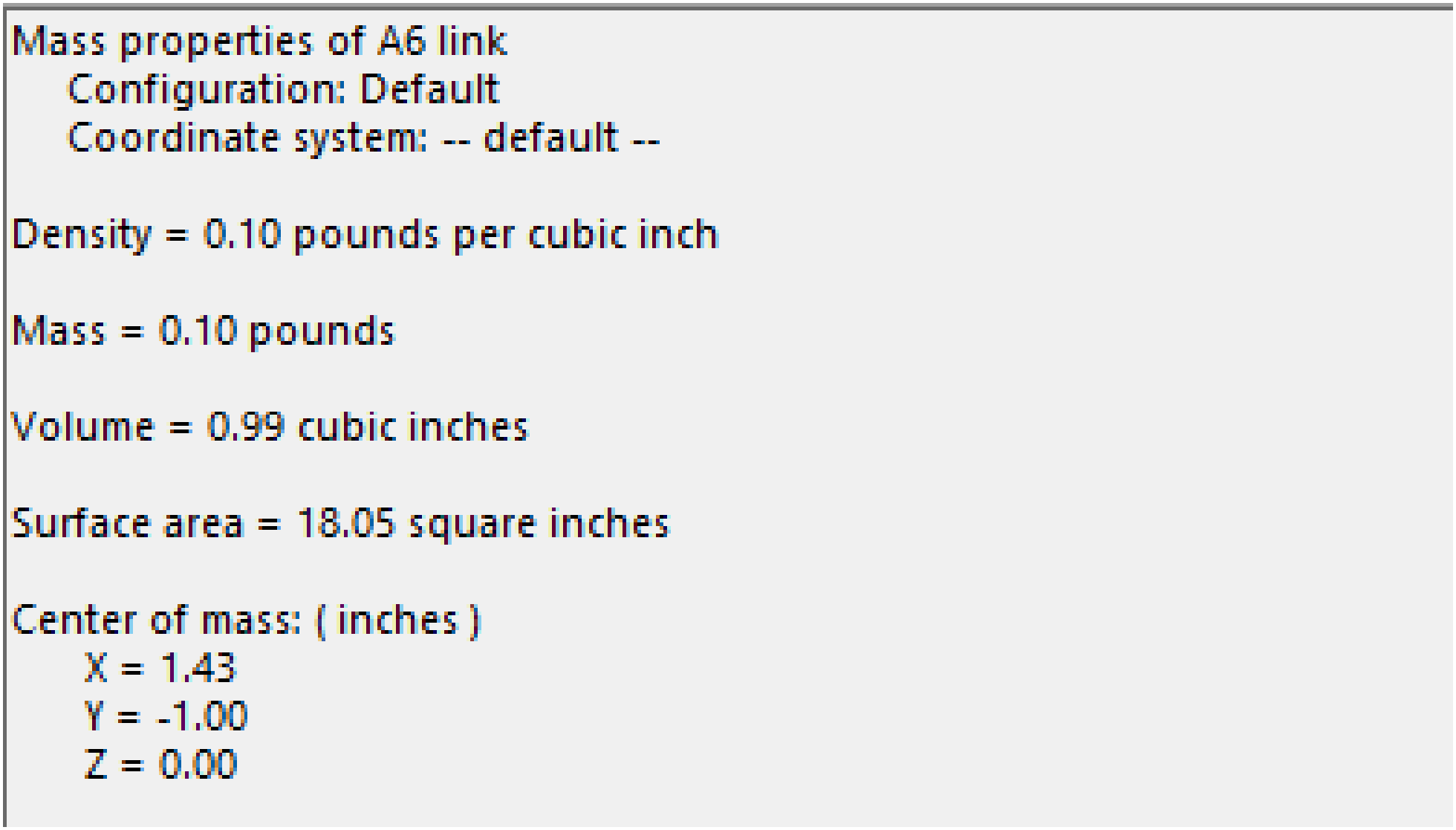
A7 - Figure 69

Link Design 2: Design 2 Medium FOS.



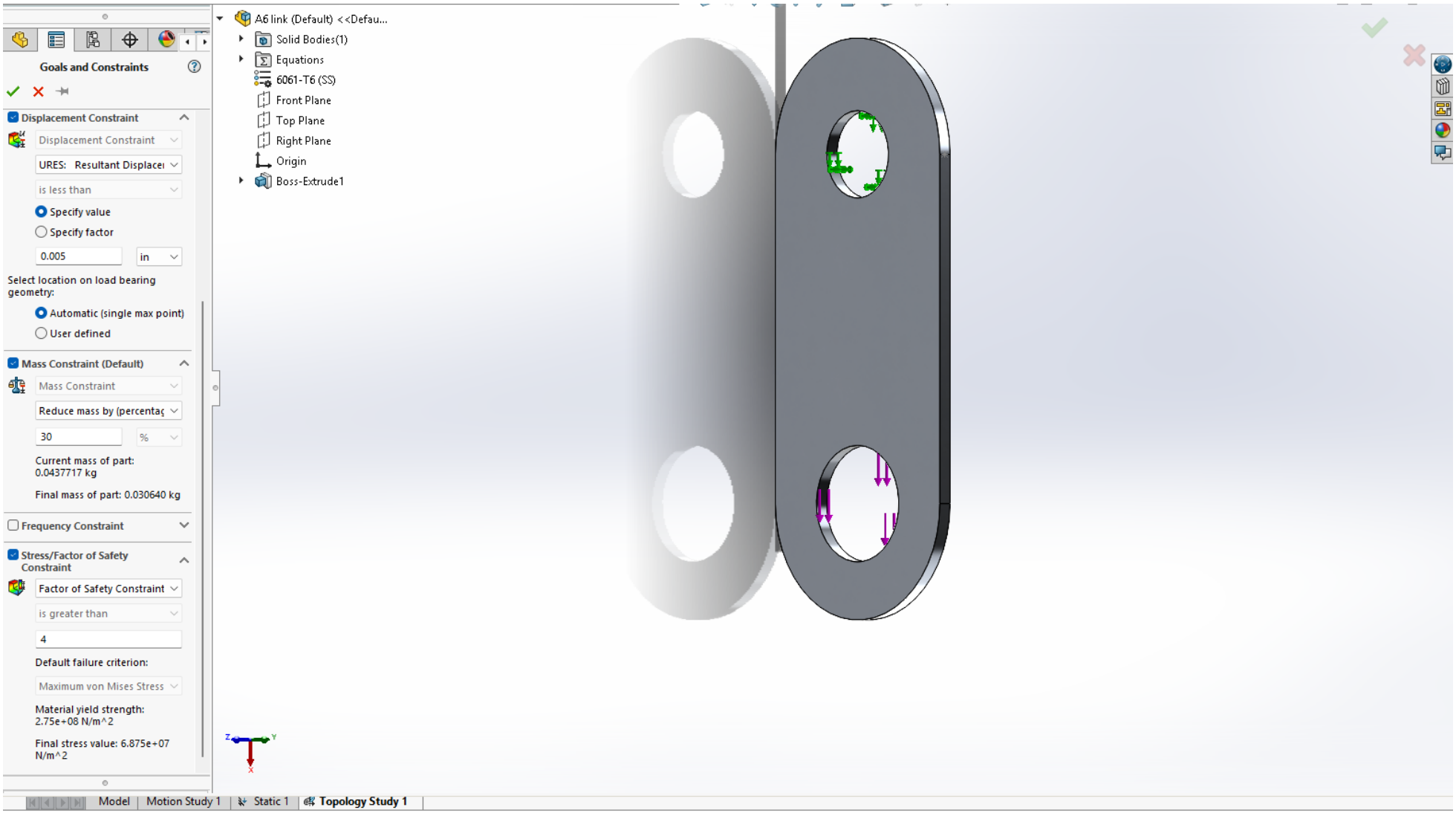
Source: Link/Design 2/Design 2 Medium FOS.png

Link Design 2: mass.



A7 - Figure 71

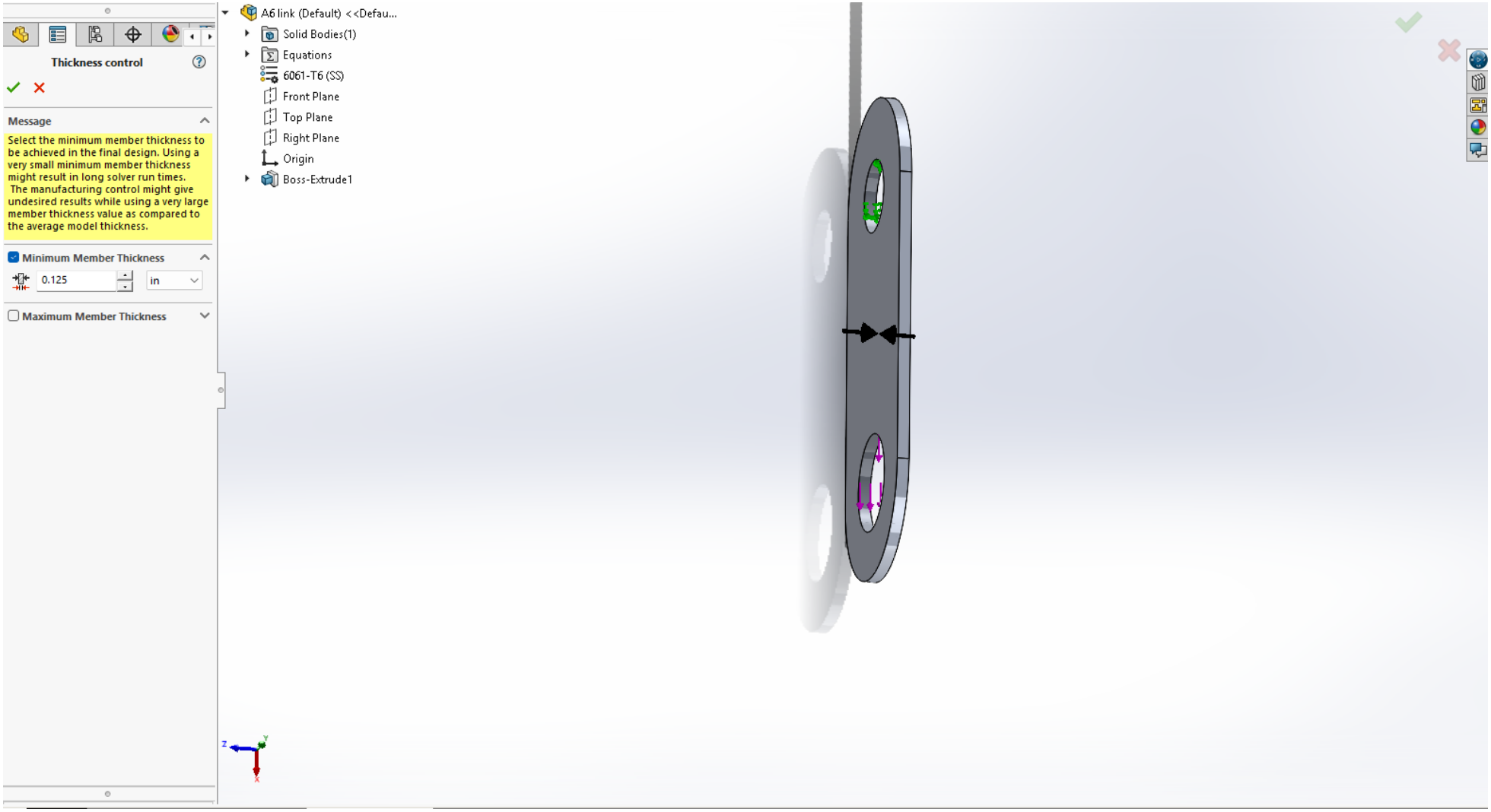
Link topology goals and constraints, including 30% mass reduction and FOS 4.



Source: Link/Design 3/Topology Goals and Conts.png

A7 - Figure 72

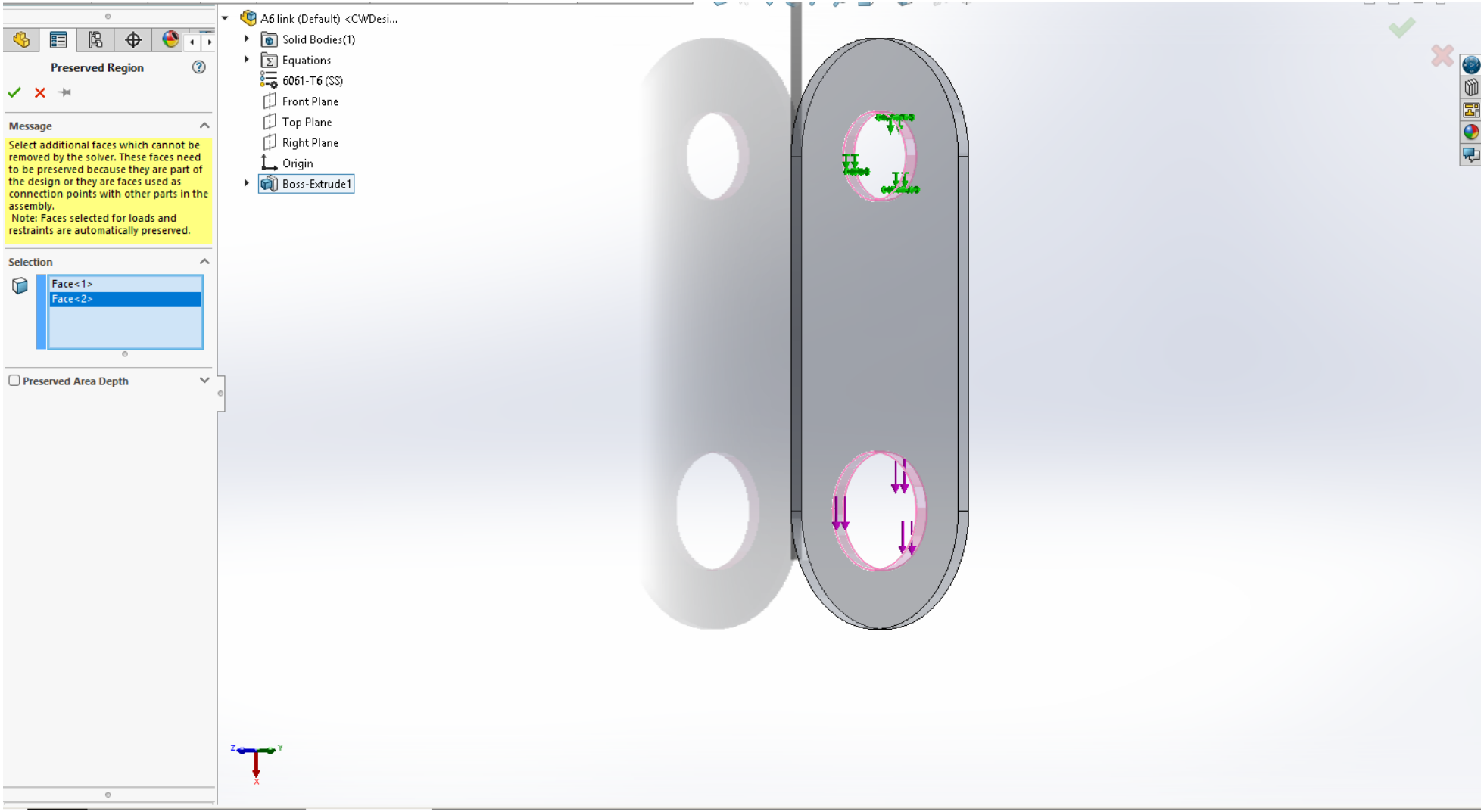
Link topology minimum member-thickness control set to 0.125 in.



Source: Link/Design 3/Thickness.png

A7 - Figure 73

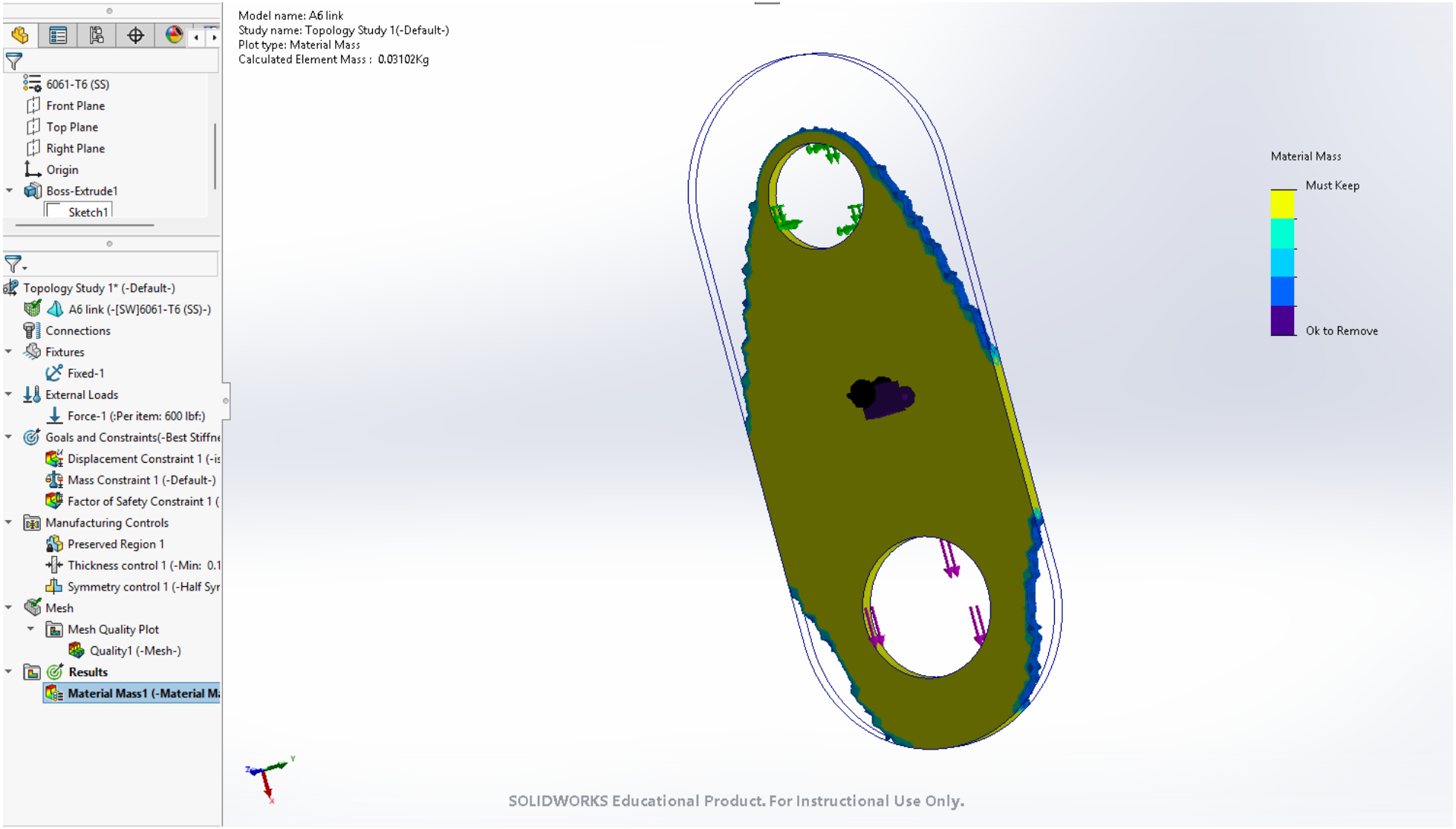
Preserved regions around the two link-hole interfaces.



Source: Link/Design 3/Preserved Regions.png

A7 - Figure 74

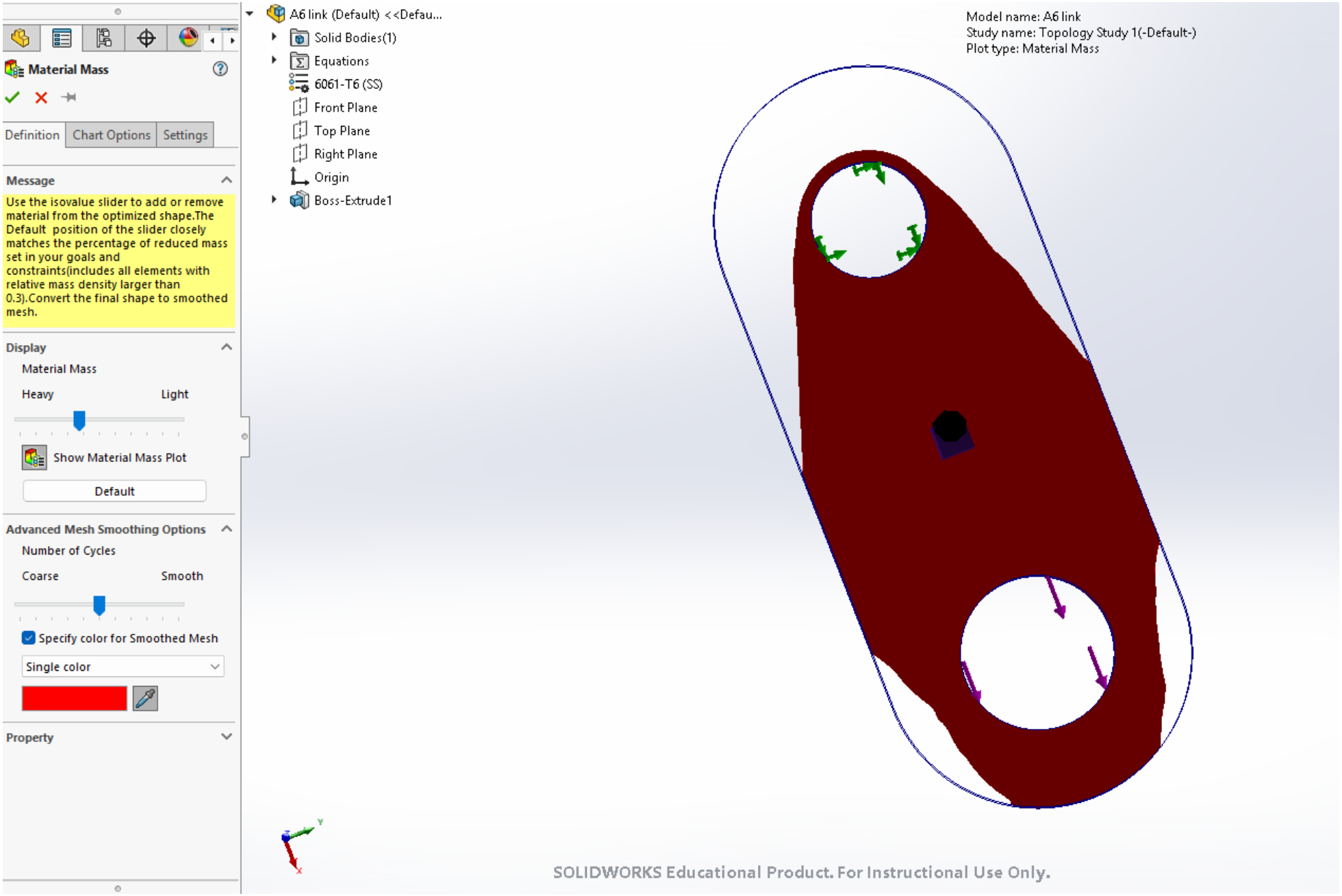
Raw link topology result before smoothing.



Source: Link/Design 3/Coarse topology before smoothining.png

A7 - Figure 75

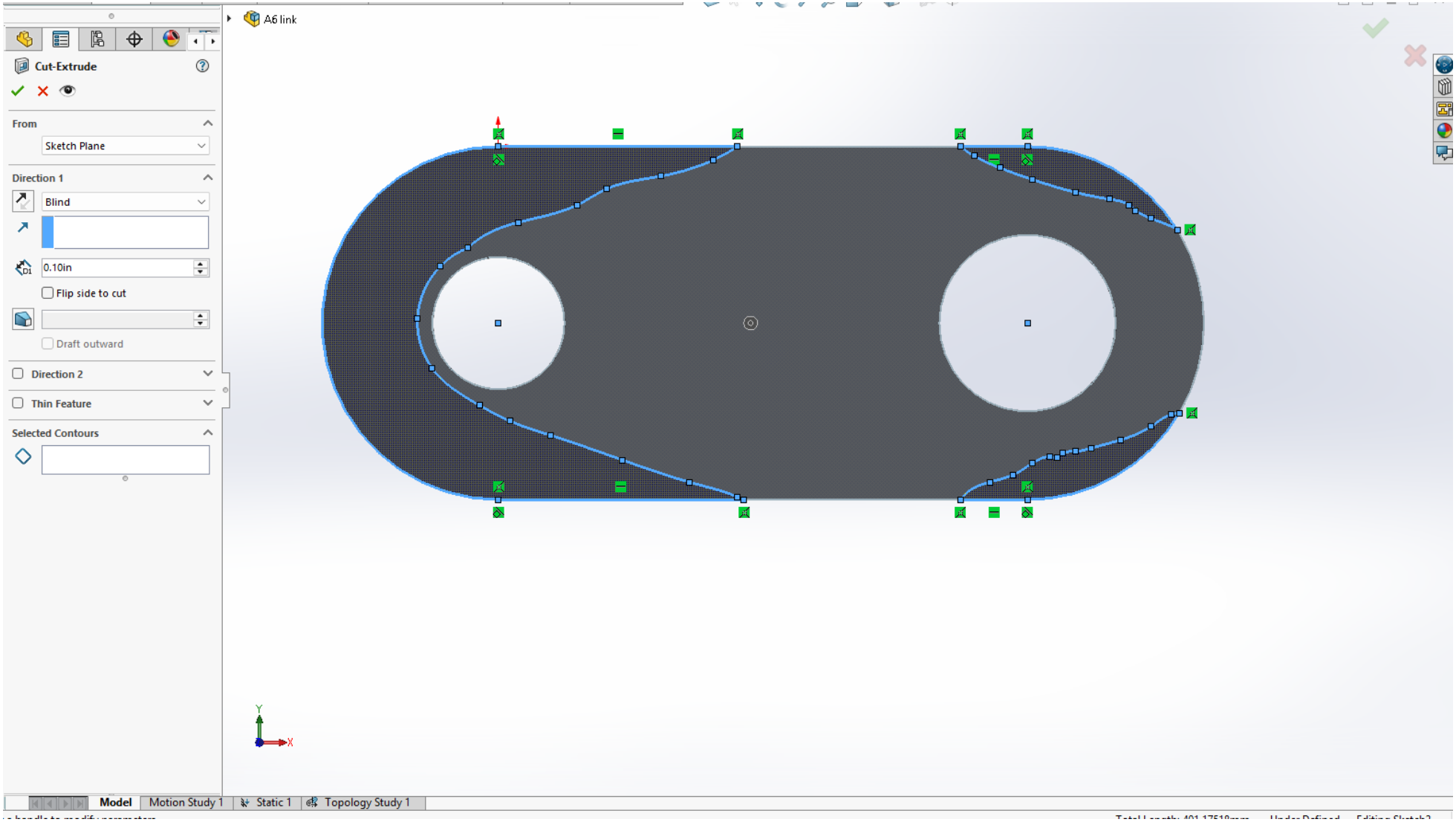
Smoothed link topology result displayed against the original outline.



Source: Link/Design 3/after smoothining.png

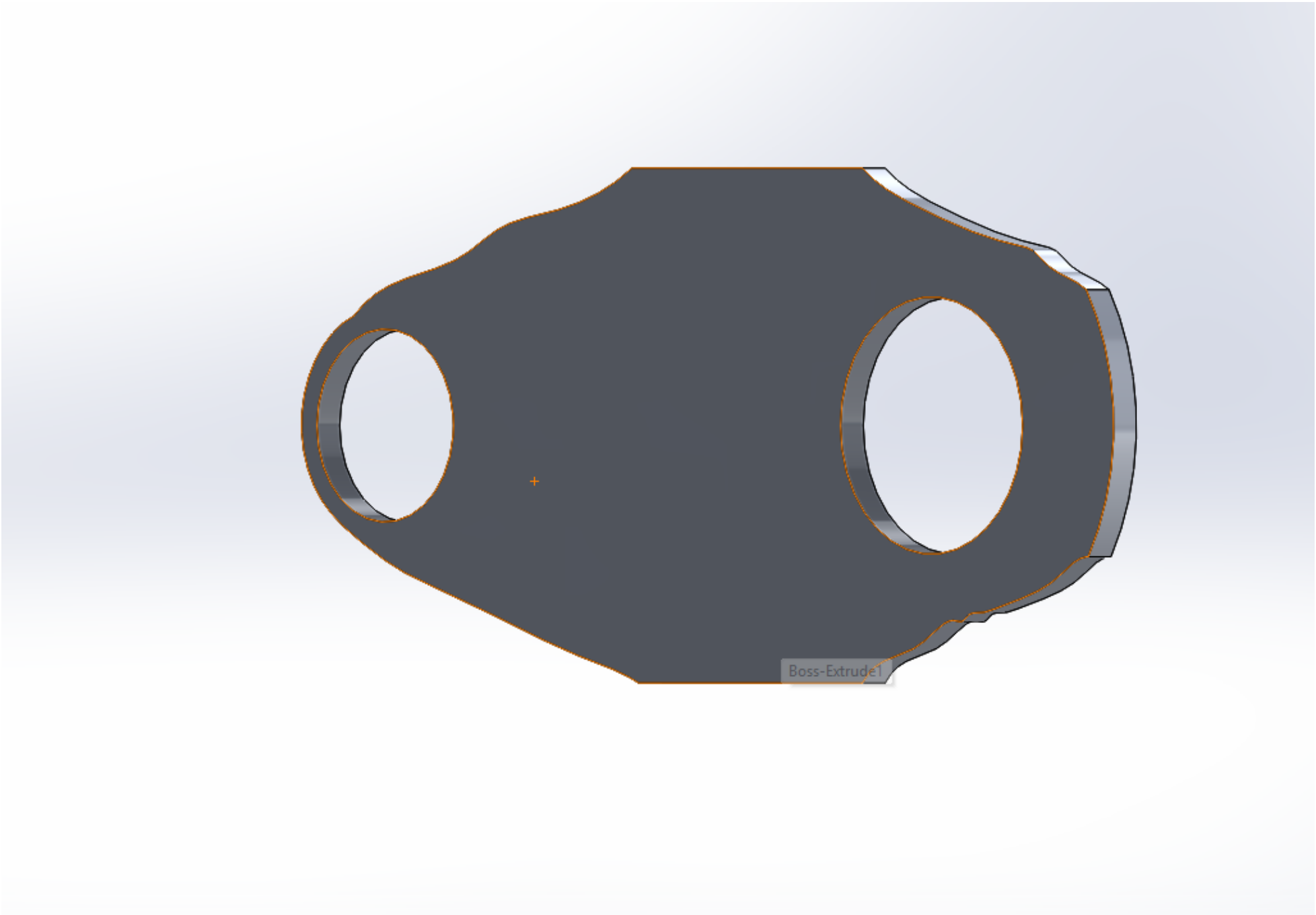
A7 - Figure 76

Manual cut following the topology-inspired link outline, with a 0.10-in blind-cut depth.



A7 - Figure 77

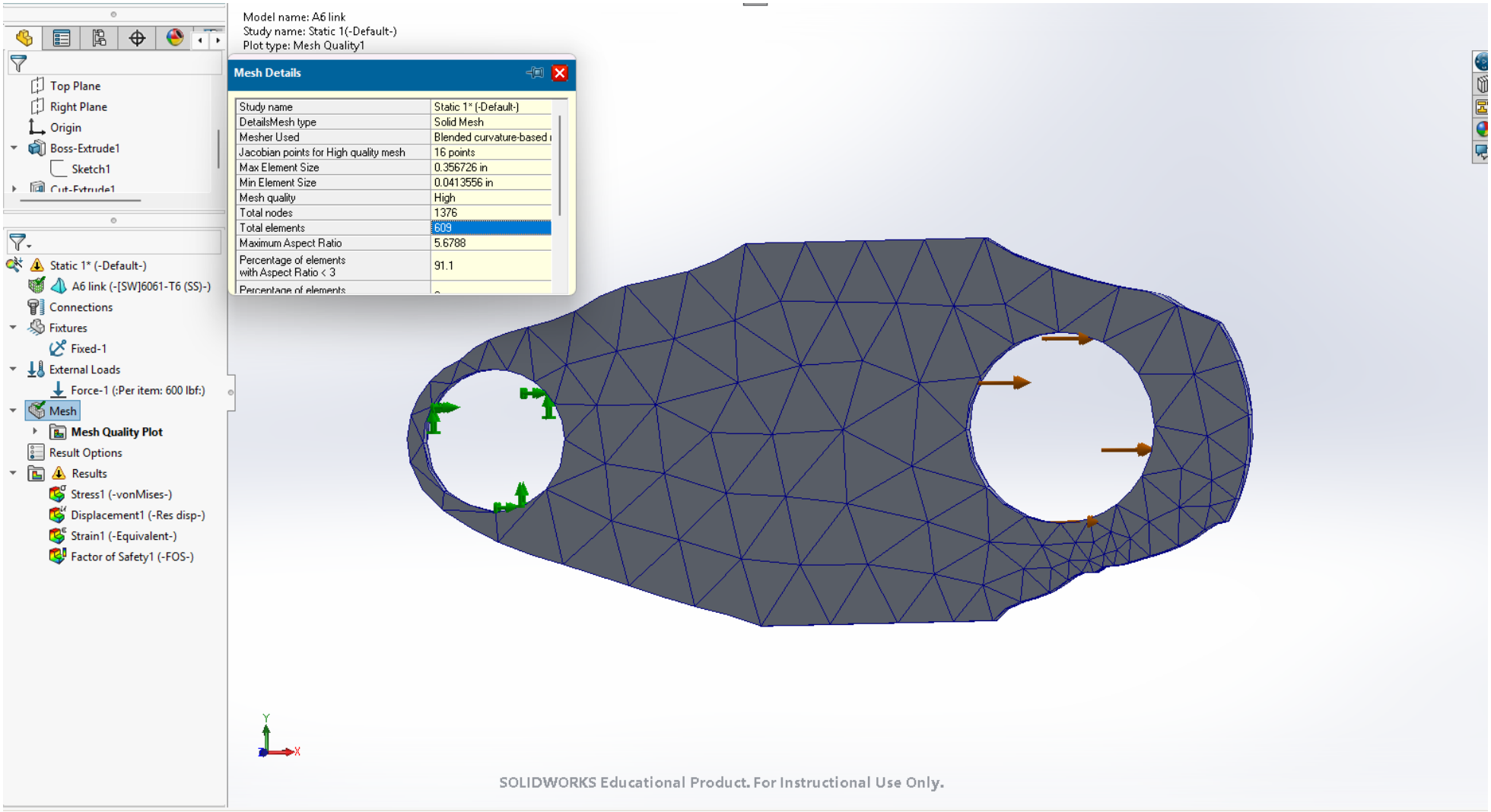
Link after the manual material-removal operation.



Source: Link/Design 3/Topology optimized part.png

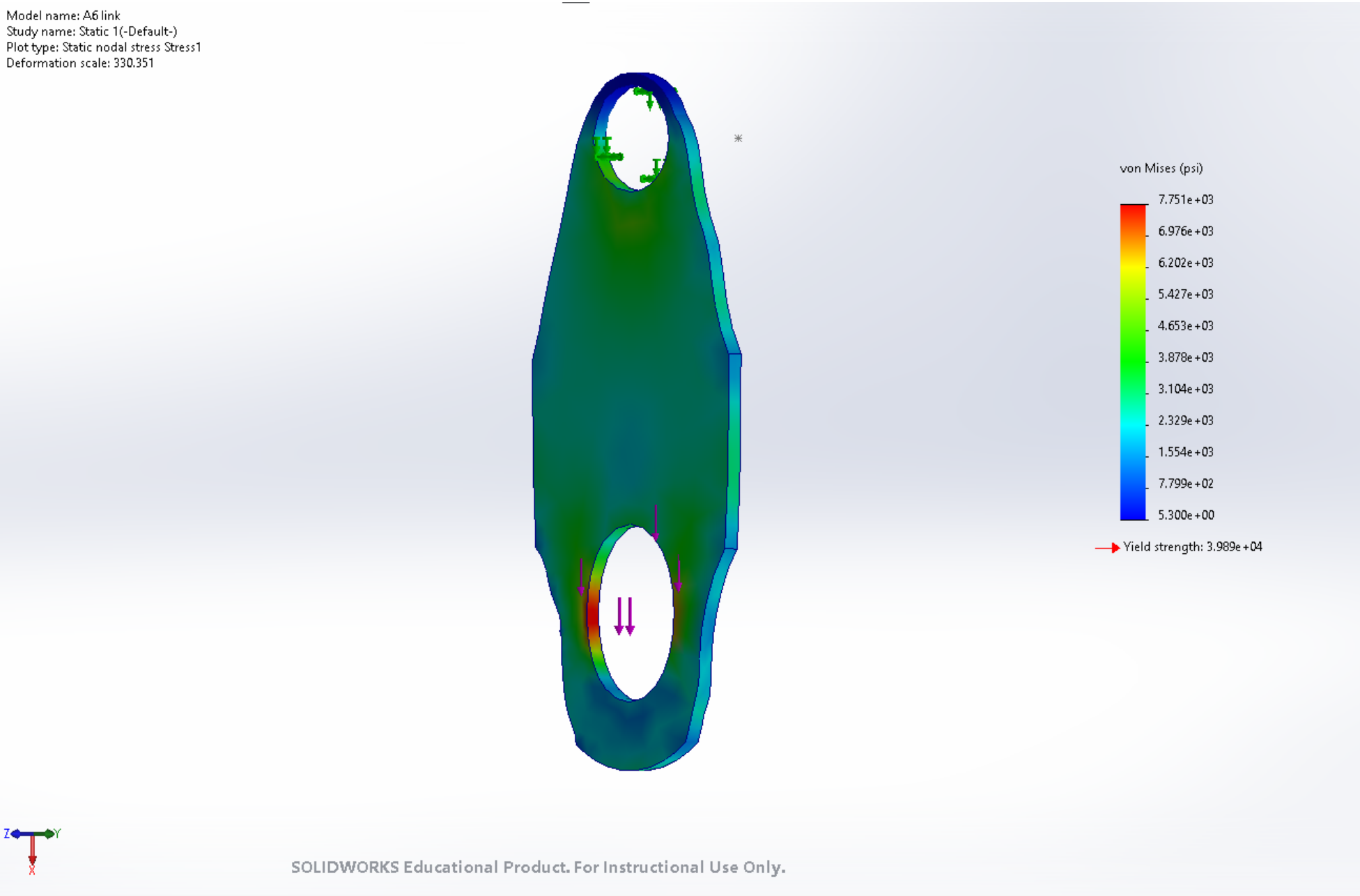
A7 - Figure 78

Link Design 3: Coarse mesh.



A7 - Figure 79

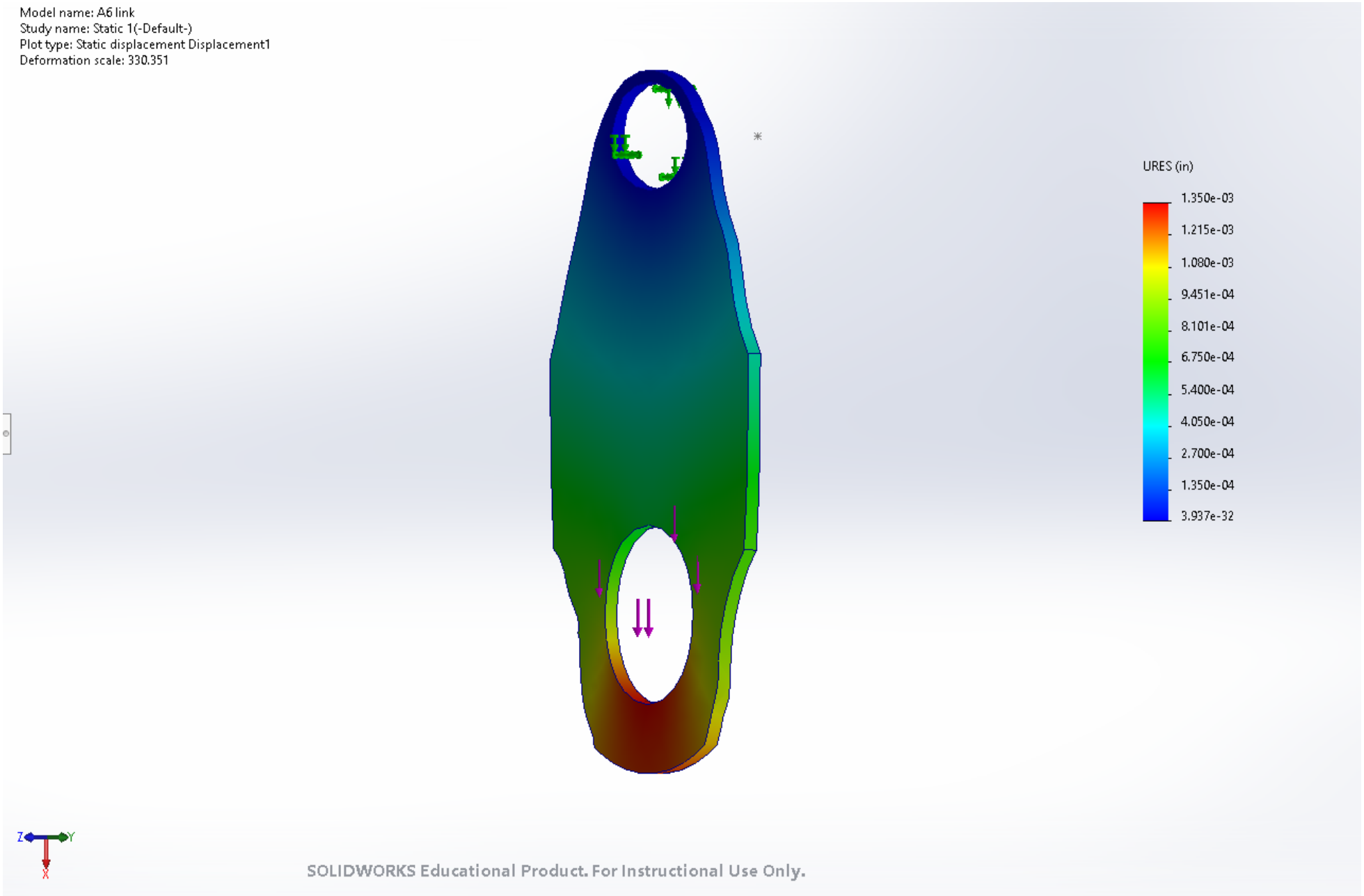
Link Design 3: Coarse stress.



Source: Link/Design 3/Coarse stress.png

A7 - Figure 80

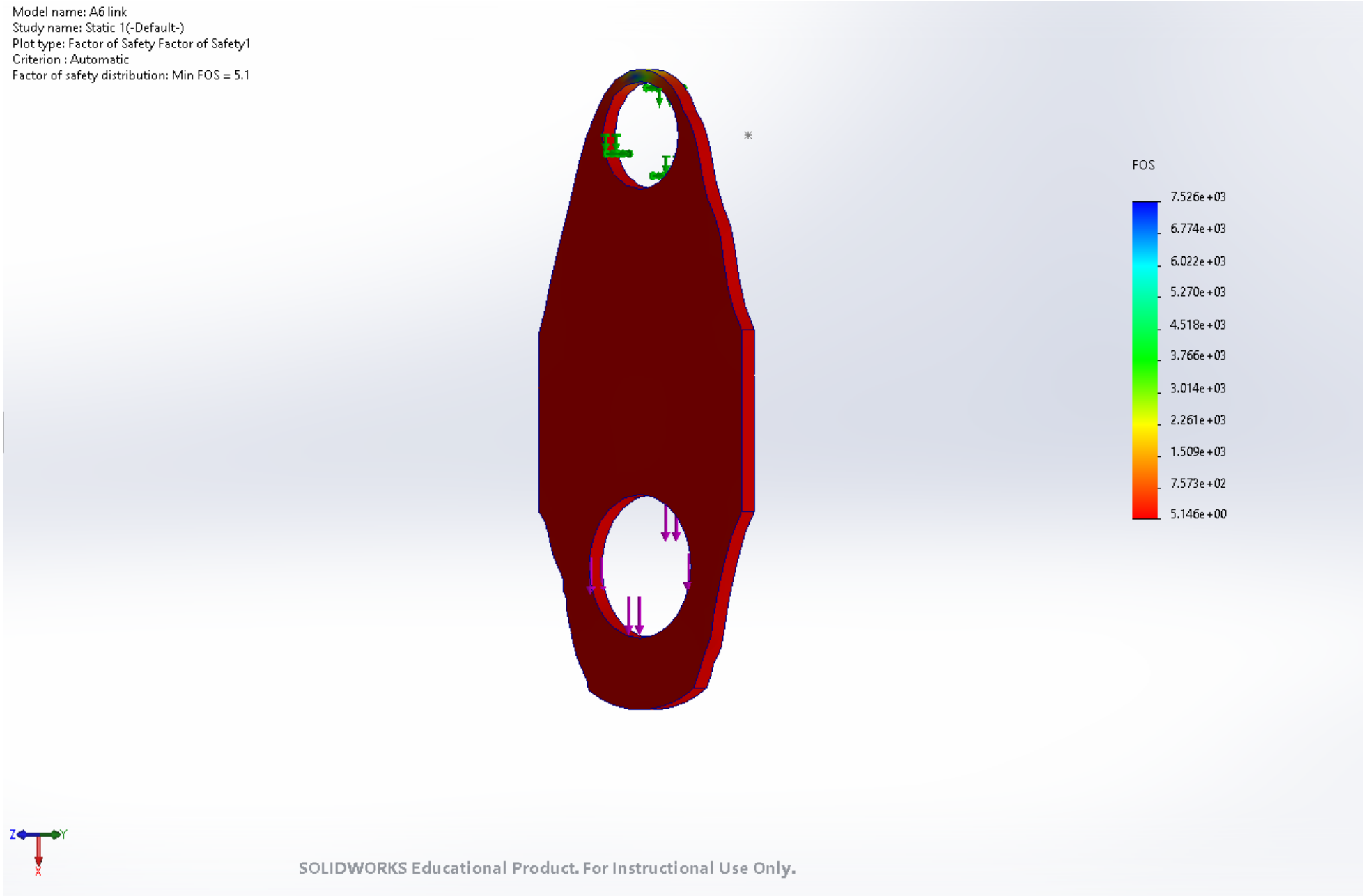
Link Design 3: Coarse Displacement.



Source: Link/Design 3/Coarse Displacement.png

A7 - Figure 81

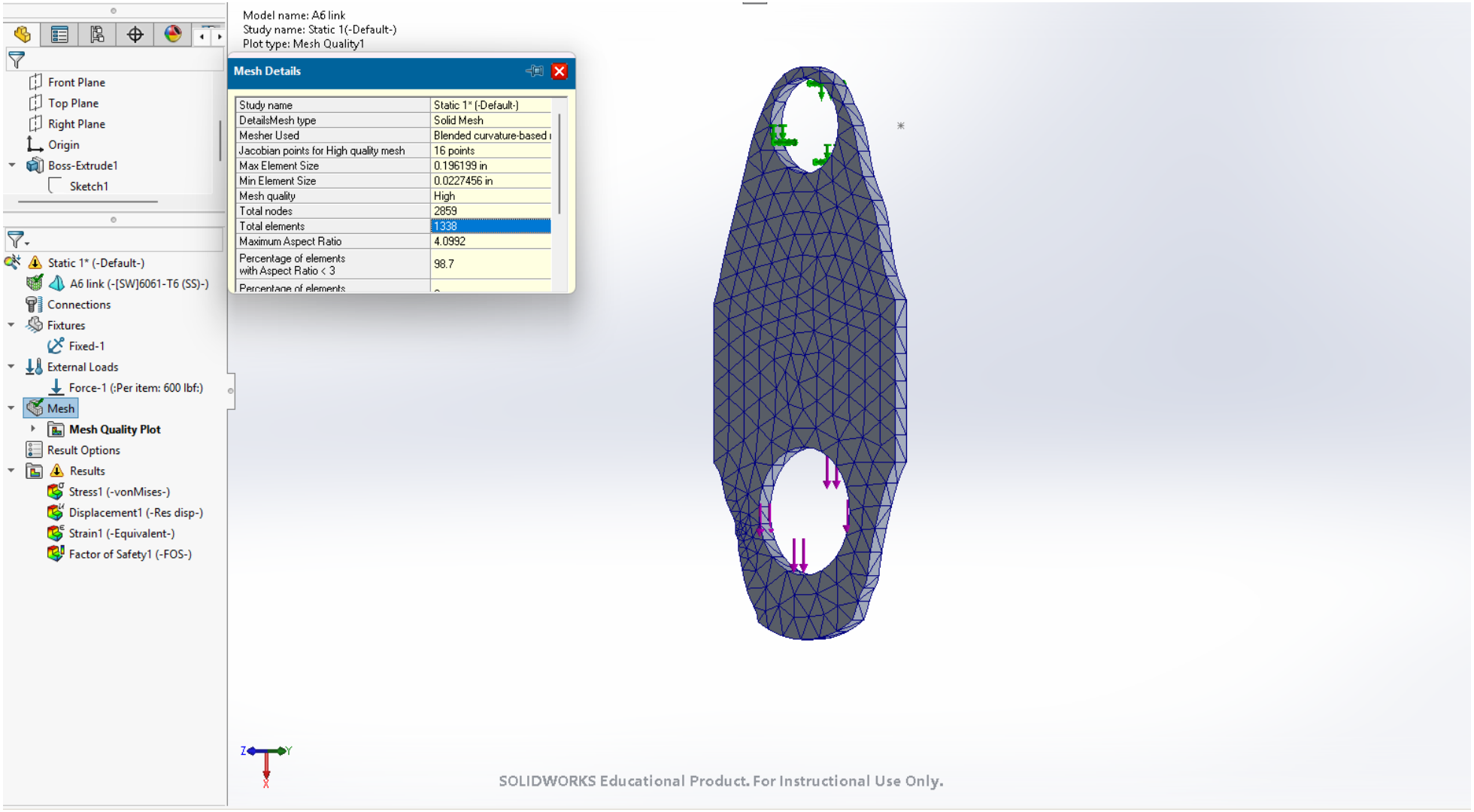
Link Design 3: Coarse FOS.



Source: Link/Design 3/Coarse FOS.png

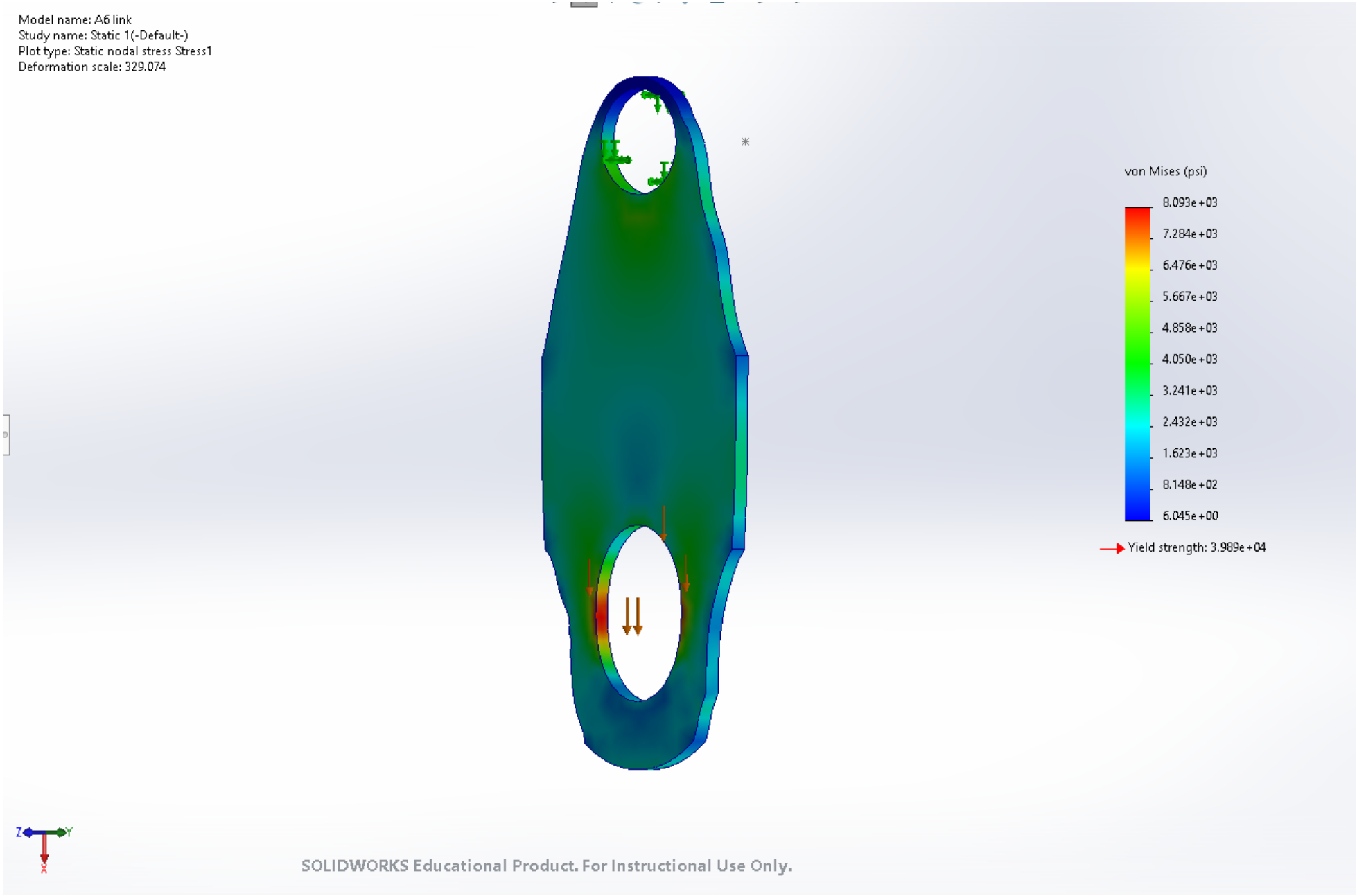
A7 - Figure 82

Link Design 3: Medium Mesh.



A7 - Figure 83

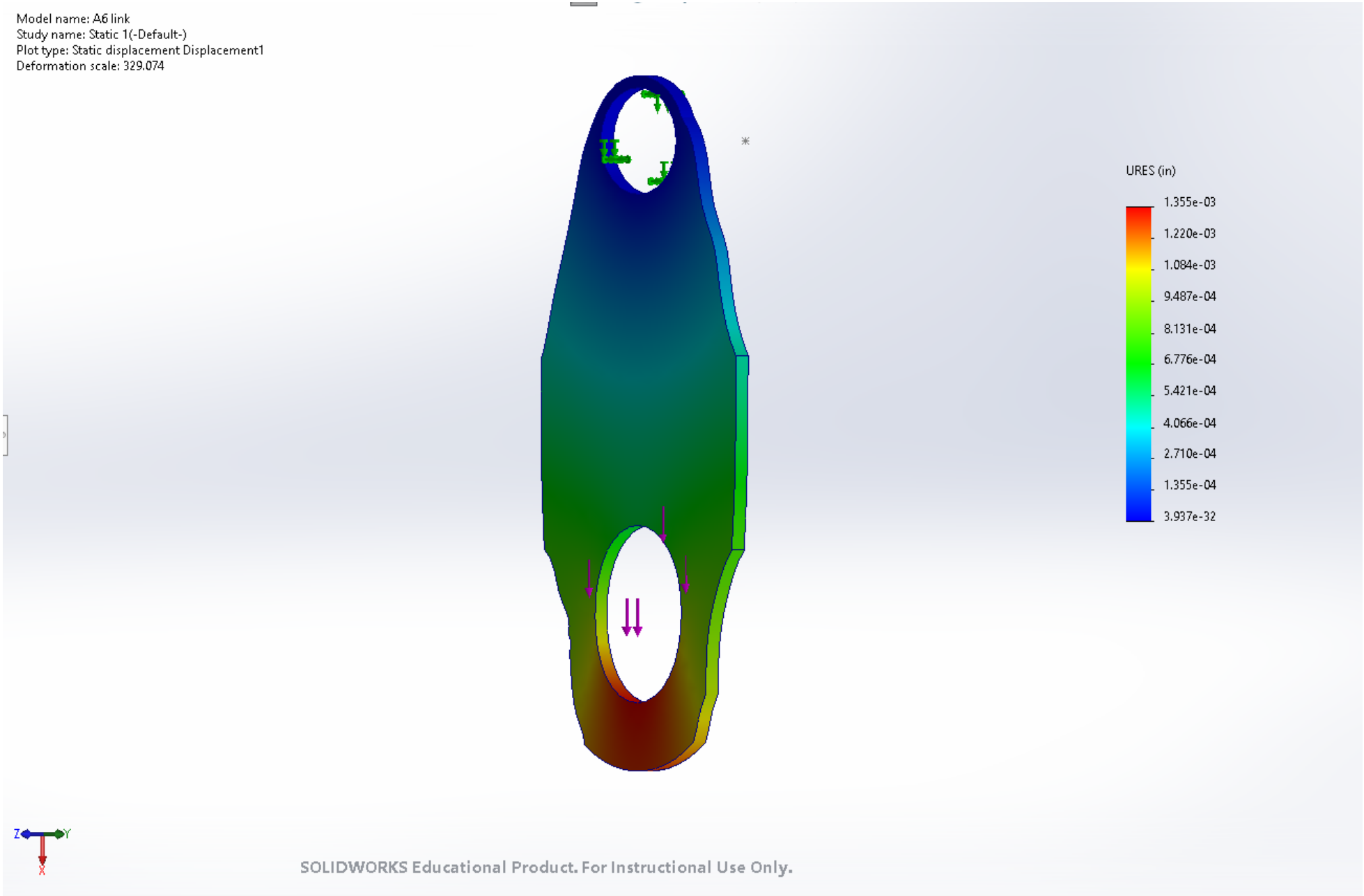
Link Design 3: Medium Stress.



Source: Link/Design 3/Medium Stress.png

A7 - Figure 84

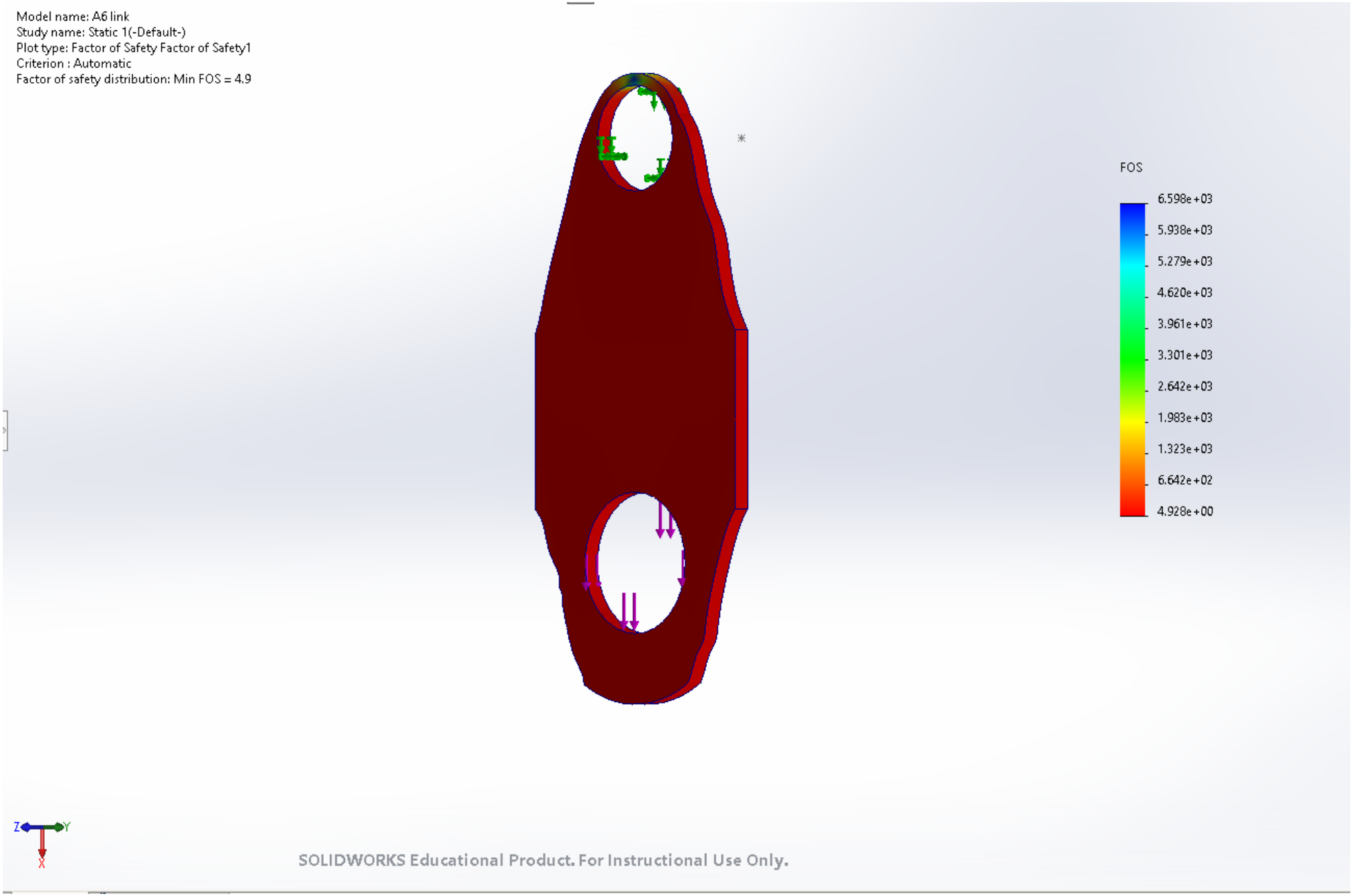
Link Design 3: Medium Displacement.



Source: Link/Design 3/Medium Displacement.png

A7 - Figure 85

Link Design 3: Medium FOS.



Source: Link/Design 3/Medium FOS.png

A7 - Figure 86

Link Design 3: mass.

